PARENTHOOD TIMING AND GENDER INEQUALITY

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Abstract

I develop a new methodology delivering the first causal estimates of the separate effects of parenthood and its timing, using quasi-experimental variation in intrauterine insemination (IUI) success. Women whose first IUI fails often undergo further procedures, generating sequences of fertility shocks that affect whether and when they conceive. In Dutch administrative data, motherhood persistently reduces hours by 12%–19% and earnings by 13%–23% relative to childlessness, causing up to half of post-childbirth gender inequality. A 2.3-year delay—though unintended—not only postpones these losses but reduces them in the four years after childbirth: by 23% for hours and 12% for earnings.

JEL codes: C21, C22, J13, J16,

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1 Introduction

The differential impact of parenthood on women's and men's careers is widely viewed as a key driver of gender inequality in the labor market (Goldin, 2014; Bertrand, 2020; Cortés & Pan, 2023; Olivetti et al., 2024). Assessing this impact is challenging because women choose whether and when to have children, and these decisions may be correlated with career outcomes independently of parenthood. A growing literature addresses this challenge using quasi-experimental variation in fertility, showing substantial and persistent declines in women's employment, hours worked, and earnings following childbirth (Lundborg et al., 2017; Gallen et al., 2024; Bensnes et al., 2025).

Yet our understanding of how motherhood affects careers remains incomplete because existing evidence does not distinguish between two margins: whether women become mothers and when they do so. Studies typically compare women who conceive following a plausibly random fertility shock, such as successful in vitro fertilization or contraceptive failure, with those who do not. The latter group includes both women who remain childless and women who become mothers later. These estimates therefore reflect a weighted average of two effects: the effect of becoming a mother rather than remaining childless and the effect of becoming a mother earlier rather than later.

The direction of the timing effect is theoretically ambiguous. Postponing childbearing mechanically eliminates motherhood-related losses during the delay and may reduce them after childbirth, as women enter motherhood with more experience and better jobs. Yet it may also shift the career interruption to a stage when earnings and promotion potential are greatest, increasing the losses. Quantifying the timing and motherhood effects separately is therefore essential for understanding fertility decisions and designing policies that influence whether and when women have children.

This paper provides the first causal evidence on the separate effects of motherhood and its timing. To do so, I develop a new methodology that disentangles these effects and apply it to a common but previously unstudied source of fertility variation: success in intrauterine insemination (IUI). The approach exploits the fact that women whose first IUI attempt fails often try again. By leveraging variation across the entire sequence of IUI attempts, the method separately quantifies the effect of motherhood relative to childlessness and the effect of later relative to earlier motherhood. I apply the method to Dutch administrative data. I first show that success at both initial and subsequent IUI attempts is uncorrelated with prior labor-market outcomes, conditional on age, supporting the assumption that the resulting fertility variation is plausibly random. Using this variation, I find that motherhood substantially and persistently reduces women's earnings and work hours, while later childbearing—even when unintended—mitigates these losses. My results show that both the incidence and timing of motherhood shape gender inequality. They indicate that women face two distinct career trade-offs: whether to become mothers and when to do so.

The method builds on the idea that success at each IUI attempt contains a random component. Ideally, this repeated variation would help recover average outcomes under motherhood beginning at the first attempt, motherhood beginning later, and childlessness, allowing the incidence and timing effects to be assessed separately. The obstacle is selection, which operates along two margins: women choose how many attempts to pursue, and some eventually become mothers regardless of their IUI outcomes. Both margins may be correlated with career outcomes independently of parenthood. Consequently, directly comparing women with different IUI and conception histories would reintroduce the selection problem that random attempt success is meant to address.

To overcome this challenge, I classify women into latent types along the two margins of selection: how many IUI attempts they would pursue if all failed, and whether they would eventually become mothers regardless of IUI success. I first show that women who remain childless after a given IUI sequence reveal the average childless outcome for their type, while the rates of IUI continuation and conception outside IUI following each failure identify the distribution of types. Together, these objects recover the average childless outcome of women whose motherhood depends on eventual IUI success. I next show that their outcome under motherhood is not point identified without additional assumptions, but can be bounded using the outcomes of women whose first attempt succeeds and the group's population share. This yields bounds on the effect of motherhood relative to childlessness, uncontaminated by timing effects. Finally, applying a symmetric argument to women who conceive outside IUI after their IUI attempts fail yields bounds on the effect of conceiving after the failed sequence rather than at the first attempt—the effect of later relative to earlier motherhood.

The approach requires only that success at each attempt be as good as random, conditional on observables. Crucially, it imposes no restrictions on heterogeneity across women or over time, selection into additional attempts, or conception outside IUI. I show that the resulting bounds are sharp and that intuitive additional assumptions—with testable implications—can progressively narrow them, up to point identification.

Statistical inference for the bounds is complicated by the fact that their construction selects observations from the tails of the outcome distribution, which requires nonparametric estimation of a quantile function. To address this, I propose orthogonal moment functions that eliminate first-order sensitivity to estimation error in this function, justifying standard asymptotic inference.

My analysis focuses on opposite-sex couples who undergo IUI for their first child. The procedure is typically used as a first-line treatment for couples experiencing difficulties conceiving naturally, particularly when fertility issues are male-factor-related or unexplained. By placing sperm directly into the uterus using a flexible catheter, it closely mimics natural conception while remaining minimally invasive, quick, and generally painless.

While IUI users are a selected clinical population, the effects of parenthood in this group are likely informative about the broader population of parents who intended to have children. Most IUI users are otherwise healthy and receive treatment for unexplained or mild fertility impairments, of which they are typically unaware before attempting to conceive. This means that IUI users effectively come from a larger group of families who would have undergone treatment had they experienced sufficient difficulty conceiving. In Section 4.3, I use fecundity rates from the medical literature to obtain a lower bound on the size of this group and estimate that potential IUI users account for at least one quarter of all parents.

The analysis proceeds in five steps. First, I assess the assumption that IUI success is as good as random conditional on age, the main determinant of success, which I control for flexibly throughout the analysis. Since IUI outcomes cannot be manipulated directly, the main concern is systematic health differences that may affect both procedure success and labor-market outcomes. Such differences should also be reflected in pre-procedure outcomes, which therefore provide a direct test. I find that success at both first and subsequent IUI attempts is uncorrelated with a rich set of pre-IUI labor-market outcomes. I complement this with a sensitivity analysis that assigns women with better post-IUI labor-market outcomes higher ex ante success probabilities and tests whether the resulting selection is consistent with pre-IUI outcome balance. I find that selection strong enough to generate meaningful post-IUI imbalance is inconsistent with the observed pre-IUI balance.

Second, I estimate how parenthood affects labor market outcomes and contributes to gender inequality. When motherhood begins at the first IUI attempt, the bounds imply declines of 12%–19% in women's annual work hours and 13%–23% in earnings relative to remaining childless, with effects persisting for at least six years. For men, the bounds are even narrower and centered near zero, ruling out small negative effects. Together, these estimates imply that parenthood accounts for 35%–53% of the gender gap in work hours and 21%–45% of the earnings gap among parents.

Third, I estimate the effects of parenthood timing. Women whose IUI sequence fails conceive, on average, 2.3 years later than under first-procedure success. Over the following six years, they work 8%–14% more hours and earn 5%–14% more than under first-procedure conception. I decompose this effect into two margins. The first operates while later mothers remain childless and earlier mothers have children, so delay avoids motherhood-related labor-market losses entirely. The second operates once both groups have children and captures the effect of having entered motherhood later. Although the second margin fades over time, both contribute similarly to the overall effect.

Fourth, I investigate mechanisms, focusing on whether the gains from postponement can be explained by reduced total fertility or additional time to establish a career before childbearing, which may reduce post-childbirth losses (Adda et al., 2017; Gallen et al., 2024). The average 2.3-year delay leaves women with 0.64 fewer children within six years. To assess the role of the intensive margin, I use couples undergoing treatment for a second child and variation from multiple births at first parity in the IUI and representative samples. These analyses suggest a limited role for the number of children. I also find that the gains from postponement are largest for women with lower pre-IUI earnings, women who earn less than their partner, and younger women. Parenthood effects are also somewhat larger for these groups, though not significantly so. In contrast, fertility responses vary little along the same dimensions. Taken together, these patterns are consistent with the gains arising primarily from additional time to establish a career before motherhood, with reduced fertility mostly a consequence of the delay rather than the main source of these gains.

Lastly, I assess the sensitivity of my estimates to potential selection on unobservables. First, I focus on differences in treatment-related behavior that may be associated with both IUI success and labor-market outcomes. I add detailed procedure-level protocol controls that capture clinical differences and may proxy for other conception-related health behaviors. The estimates are virtually unchanged. Second, I use my estimates of the extent of selection on unobservables consistent with the observed balance to adjust realized outcomes so as to restore balance between success and failure groups, and re-estimate the effects. The estimates remain similar. Finally, I emphasize that if lower success probabilities reflect worse health that also worsens labor-market outcomes, any remaining selection makes the estimated benefits of postponement conservative, because later-conceiving women are negatively selected relative to those who conceive earlier.

I conclude by discussing the broader implications of my results for the role of parenthood in the labor market, especially for women deciding early in their careers whether and when to have children. A central consideration is that, for women whose IUI attempts fail, both remaining childless and becoming mothers later are unplanned outcomes. My estimates may therefore further represent a lower bound on the gains from planned childlessness or delay. This may occur both because women cannot fully optimize their careers in anticipation of these outcomes (see Bensnes et al., 2025, for a discussion) and because unmet childbearing goals may impose a psychological toll (see Bögl et al., 2024, for a discussion). I end with a suggestive calculation, based on my estimates, of the extent to which postponing childbearing can mitigate the career costs of motherhood. I find that a 2.3-year delay not only postpones these losses but also reduces them over the first four years after childbirth by 12% for earnings and 23% for work hours.

My work builds on studies that estimate the effects of parenthood using quasi-experimental variation from infertility, miscarriages, success in in vitro fertilization, or contraceptive failure (Hotz et al., 2005; Agüero & Marks, 2008; Cristia, 2008; Miller, 2011; Lundborg et al., 2017, 2024; Gallen et al., 2024; Bensnes et al., 2025). These studies compare women who give birth at a particular time for plausibly random reasons with women who do not, many of whom become mothers later. They report reduced-form effects that capture a weighted average of parenthood and timing effects, or isolate parenthood effects under restrictions on the role of timing. My approach extends this work by identifying parenthood effects without restricting how they vary with timing, and by quantifying timing effects directly.

There is little causal evidence on the effects of parenthood timing. Bíró et al. (2019) compare women who give birth with women who miscarry and subsequently give birth, addressing selection into later childbirth by balancing observables. Lundborg et al. (2017) and Gallen et al. (2024) compare childbirth effects across groups that differ in the age at which they undergo in vitro fertilization or begin using contraception. These comparisons, however, cannot separate timing effects from heterogeneity in fertility intentions across groups. I provide the first causal estimates of the effect of parenthood timing that address selection on unobservables.

I also contribute evidence from a new population and institutional setting. The closest studies exploit IVF success in Scandinavia (Lundborg et al., 2017, 2024; Gallen et al., 2024; Bensnes et al., 2025). I focus on IUI, a minimally invasive treatment used in cases of mild infertility, and use data from the Netherlands, where parental leave and childcare use are closer to OECD averages. Whereas Scandinavian IVF studies generally find modest earnings losses, I estimate larger and more persistent declines, similar in magnitude to those found by Gallen et al. (2024) following unplanned births after contraceptive failure. This difference is consistent with family-friendly policies mitigating the career costs of motherhood.

My work also contributes to the literature on dynamic treatments. In biostatistics, numerous methods have been developed to evaluate sequential experiments under full compliance (see Hernán & Robins, 2020, for an overview). In economics, Van den Berg & Vikström (2022) study settings where treatment is assigned dynamically among eligible individuals who selectively exit eligibility, while Heckman et al. (2016) and Han (2021) allow treatment to be obtained outside assignment, provided instruments exist for each treatment margin. I contribute a method for settings where individuals selectively undergo additional assignments and may obtain treatment outside assignment without valid instruments—a structure shared by education admission lotteries, job training programs with partially randomized access, and sequential clinical trials with imperfect compliance.

My work also relates to the literature on treatment-effect bounds, beginning with Manski (1989, 1990); Horowitz & Manski (1995), and in particular to the methods of Zhang & Rubin (2003) and Lee (2009). Bounding methods can be applied to virtually any treatment-effect setting, but a central drawback is that the resulting bounds may be economically uninformative. I contribute by showing how bounding techniques can be combined with sequential quasi-experimental variation to obtain bounds that are strictly narrower and valid for a broader population, without additional assumptions. This improvement is crucial in application: existing methods adapted to separate treatment and timing effects yield bounds for my main outcomes that are several times wider and fail to rule out large positive or negative effects.

My work further relates to the literature on dynamic non-compliance with one-time instruments (Cellini et al., 2010; Ferman & Tecchio, 2023; Angrist et al., 2024; Gallen et al., 2024; Bensnes et al., 2025), which exploits initial assignment and repeated outcome measurements to identify treatment effects when treatment duration is central. These approaches require restrictions linking effects for initially- and later-treated individuals, typically equal average effects conditional on treatment duration. Such restrictions may be particularly strong for parenthood, where later mothers may differ systematically from earlier mothers and births at different ages may have different career and fertility consequences (see Gallen et al., 2024; Bensnes et al., 2025, for discussions). My approach instead delivers sharp bounds on parenthood effects under unrestricted heterogeneity across women and conception timings, while separately identifying the timing effect itself.

Finally, I contribute to the broader literature on the consequences of fertility. This includes studies exploiting quasi-experimental variation in the intensive margin, (Rosenzweig & Wolpin, 1980; Angrist & Evans, 1998; Bronars & Grogger, 1994; Black et al., 2005); studies comparing mothers with fathers or with women who have not yet had children (Fitzenberger et al., 2013; Angelov et al., 2016; Chung et al., 2017; Bütikofer et al., 2018; Kleven, Landais, & Søgaard, 2019; Kleven, Landais, Posch, et al., 2019; Eichmeyer & Kent, 2022; Kleven et al., 2024; Melentyeva & Riedel, 2023; Adams et al., 2024); and structural analyses of fertility and career choices (Adda et al., 2017; Jakobsen et al., 2022). The causal evidence I provide on how motherhood timing affects fertility and labor market outcomes helps explain differences among women who enter motherhood at different life-cycle stages.

The remainder of the paper is structured as follows. Section 2 introduces a framework for sequential quasi-experiments. Section 3 discusses existing methods, develops the intuition for my econometric approach, states the formal results, outlines estimation, and introduces a procedure for assessing selection on unobservables. Section 4 describes the institutional setting, intrauterine insemination, and the data, and presents evidence supporting the identifying assumptions. Section 5 presents the results. Section 6 discusses generalizability and life-cycle implications. Section 7 concludes.

2 A Framework for Sequential Quasi-experiments

My framework generalizes the local average treatment effect (LATE) framework (Angrist & Imbens, 1995) by making explicit how a treatment—parenthood—depends not only on an initial treatment assignment, such as the success of the first IUI procedure, but also on subsequent decisions to pursue additional assignments and on their outcomes.

Since the analysis involves timing effects and sequential decisions, the framework could be formulated dynamically. For clarity, I instead use a cross-sectional representation that summarizes the relevant events and decisions within a fixed observation window. Appendix SA1 formally maps the dynamic framework into this representation and shows that there is no loss of generality. Section 2.1 presents the framework, and Section 2.2 covers interpretation details.

2.1 Setup

I observe each woman over a fixed window beginning at her first IUI attempt for a first child; in the main analysis, the window spans seven years. All events considered below—IUI attempts, their outcomes, and the onset of motherhood—occur within this period, and all latent variables are defined with respect to it. All variables are woman-specific, and I suppress individual subscripts throughout.

Each woman is described by two latent variables. First, $W \in \{1, \ldots, \overline{w}\}$ is the number of IUIs she would have undergone for her first child within the window had every attempt failed, with upper bound $\overline{w}$; I refer to W as *willingness* to undergo IUI. Second, $R \in \{0, 1\}$ indicates whether she would have remained childless throughout the window had all W IUIs failed; I refer to R as *reliance* on IUI. Women with $R = 1$ (*reliers*) rely on IUI to become mothers, while women with $R = 0$ (*non-reliers*) would have become mothers even if all their IUIs had failed. The timing of IUI attempts and non-IUI motherhood may vary across women. Non-IUI motherhood may include adoption, but it is almost nonexistent in the data.

Potential outcomes describe a woman's earnings or work hours over the window. I define three: Y^E is her outcome if the first IUI succeeds (the *early motherhood* outcome), Y^L is her outcome if the first IUI fails but she becomes a mother within the window (the *later motherhood* outcome), and Y^N is her outcome if she remains childless throughout the window (the *childless* outcome). Depending on the analysis, I consider either outcomes cumulated over the entire window, such as total earnings over the period, or contemporaneous outcomes measured at a given horizon relative to the first IUI attempt, such as earnings in the third year after the attempt.

Y^L is not explicitly indexed by the timing of motherhood, although different conception times may correspond to different values of Y^L; the framework places no restrictions on how Y^L varies with timing. Throughout the analysis, I focus on the case in which motherhood occurs through non-IUI conception after the full IUI sequence has failed. When Y^L refers to a different conception time, I make this explicit. I discuss the timing interpretation in more detail in the next section.

The first parameter of interest is the *effect of parenthood*, defined as the difference between conceiving earlier and remaining childless: $\tau = Y^E - Y^N$. The second is the *effect of parenthood timing*, the effect of later relative to earlier motherhood: $\delta = Y^L - Y^E$. Both τ and δ are defined at the individual level and may vary arbitrarily across women. The two main effects I focus on are the average treatment effect among reliers, $\tau_{ATR} = \mathbb{E}[\tau \mid R = 1]$, and the average timing effect among non-reliers, $\delta_{ANR} = \mathbb{E}[\delta \mid R = 0]$. The two estimands reflect the two margins along which IUI success induces variation. For reliers, who remain childless after the IUI sequence fails, success affects the incidence of motherhood. For non-reliers, who conceive even when the IUI sequence fails, success shifts the timing of motherhood.

The observed indicator for the success of the jth IUI for the first child is Z_j, which equals 1 if the procedure succeeded and 0 if it failed or was not undertaken. The first key variable linking observed and latent variables is the realized number of IUIs, $A = \min(\{j : Z_j = 1\} \cup \{W\})$: a woman undergoes IUIs until one succeeds or until she reaches her willingness. The outcome of the last attempt is Z_A. Willingness is thus observed exactly when the sequence ends in failure, in which case $A = W$; when an attempt succeeds, W is only known to satisfy $W \geq A$. This censoring structure is central to the identification argument in Section 3.

The second key variable is the parenthood (*treatment*) indicator $D = Z_A + (1 - Z_A)(1 - R)$: a woman becomes a mother if an IUI succeeds or if she is a non-relier, who conceives independently of IUI success. Similar to willingness, reliance is only observed when the sequence ends in failure, in which case $R = 1 - D$. A woman's realized labor market outcome is $Y = Z_1 Y^E + (1 - Z_1)[D\, Y^L + (1 - D)\, Y^N]$, and it depends on whether she conceived at the first IUI, later, or not at all.

After introducing the core idea, I also leverage information on non-IUI births among women who conceive their first child via IUI. For these women, $R^+ \in \{0, 1\}$ is a latent indicator of whether a woman relies on IUI for all additional children after conceiving her first child via IUI. Women with $R^+ = 1$ (*subsequent reliers*) would have only IUI-conceived children after conceiving their first via IUI, whereas those with $R^+ = 0$ would also have one or more non-IUI children.

The indicator for having at least one non-IUI child is $D^+ = Z_A(1 - R^+) + (1 - Z_A)D$: a woman has at least one such child if an IUI succeeded and she is not a subsequent relier, or if all IUIs failed and she became a mother regardless.

2.2 Framework Interpretation and Discussion

In this subsection, I discuss the interpretation of reliance and willingness, as well as the treatment and timing effects, and connect my framework to the LATE framework.

2.2.1 Interpreting Willingness and Reliance

Willingness and reliance are counterfactual summary statistics that capture all endogeneity in the realized number of IUIs, and in fertility outside observed IUI, that is relevant for identification. Willingness reflects how many IUIs a woman would undergo if all attempts failed, while reliance reflects whether she would remain childless under that same counterfactual, regardless of how these outcomes arise. Because this counterfactual path may end through non-IUI conception, willingness counts the IUIs completed along the path rather than a fixed maximum number of procedures a woman is prepared to undergo.

Willingness and reliance do not describe every dimension of fertility behavior. For example, two women with the same willingness and reliance may start the procedure sequence at different ages or progress through subsequent procedures at different paces. The framework leaves these dimensions implicit and unrestricted, but they matter for identification because IUI success probabilities depend on age. Section 3.1 introduces the identifying assumption and explains how the analysis accounts for age throughout the procedure sequence. Women may likewise conceive outside IUI at different ages if the sequence fails, which affects the interpretation of timing effects; I discuss this in Section 2.2.2. Appendix SA1 makes timing explicit in a dynamic framework and shows that the results carry over.

Willingness and reliance can be given an economic interpretation, but they may also be treated as statistical constructs. Reliance, for instance, may reflect a woman's decision to attempt natural conception alongside IUI and hence her effort to conceive naturally. Alternatively, all women undergoing IUI may attempt natural conception simultaneously, with success largely random. The two variables also impose no particular information structure. Willingness and reliance may reflect plans made under perfect foresight, or emerge from a dynamic process in which women update their behavior in response to earlier procedures, medical advice, career changes, or other information. The identification results accommodate all of these possibilities.

The interpretation of reliance matters for the parenthood and timing effects. If reliance mainly reflects effort to conceive naturally, the estimated parenthood effect pertains to women less likely to conceive outside IUI, whereas the timing effect pertains to women more likely to do so. If reliance is instead largely driven by random success in non-IUI conception, reliers and non-reliers need not differ systematically, making the two sets of estimates more directly comparable. Willingness, by contrast, is averaged over in the main estimands, so heterogeneity in willingness is less consequential for their interpretation.

Most analyses in this paper estimate effects separately for reliers and non-reliers and therefore do not rely on comparisons between the two groups; where such comparisons are made, I state the assumptions explicitly. This matters because the groups do differ on observables: Section 4.4 characterizes their pre-IUI characteristics and shows that non-reliers are younger, more educated, and have higher employment and earnings.

Two measurement details are important for interpreting willingness and reliance. The first concerns the measurement of IUI procedures. The design exploits variation in IUI success that is plausibly exogenous only once sperm has been injected into the uterus. Willingness is defined accordingly: it counts only the counterfactual injections that would have occurred had all previous attempts failed, rather than procedures that would have been intended or scheduled but never carried out. Accordingly, the data measure completed procedures.

The second concerns the definition of IUI births. Some IUIs are unobserved, for example when performed abroad. Births that cannot be linked to an observed IUI are therefore treated as non-IUI births, and the corresponding women as non-reliers. This does not undermine identification, since the method relies only on randomness in observed IUI success and allows arbitrary selection into all other conceptions; it only widens the treatment effect bounds relative to a setting with complete information on IUI use.

2.2.2 Interpreting Treatment and Timing Effects

In line with existing work, motherhood is an absorbing state, so subsequent births are included in both the early- and later-motherhood states. Their number may differ across states, making the intensive fertility margin one potential mechanism behind timing effects. Section 5.1 presents effects on total fertility, while Section 5.5 assesses the role of subsequent births. These analyses indicate that the intensive margin contributes little to the estimated labor market effects.

The timing effect is the total effect of postponing entry into motherhood, not the partial effect of maternal age holding the rest of the life course fixed. When childbirth occurs later, a woman enters motherhood at a later career stage and, at any subsequent age, has a younger first child. These changes cannot be separated: age at first birth, child age, and the mother's current age are linked by an identity. There is therefore no feasible intervention that varies birth timing while holding fixed both career stage and child age. The estimand captures this joint change, which is also the object relevant for women deciding whether to postpone motherhood and for policies that shift the age at which women have children.

The conception dates associated with a woman's potential outcomes are shaped partly by endogenous decisions. Whether and when she becomes a mother depends on IUI success, treatment continuation decisions, and conception outside IUI. The framework exploits plausibly exogenous variation in IUI success while placing no restrictions on the other two. For example, if a woman undergoes her first IUI at age 25, conception at 25 corresponds to her early motherhood outcome. If it fails, she may continue treatment, conceive outside IUI, or remain childless. Suppose continued IUI failure eventually leads to conception outside IUI at age 30. The potential conception dates themselves may be selective; IUI success only shifts women between them, in this case between conception at ages 25 and 30.

The delays induced by IUI failure may also differ across women. For example, two women may both undergo their first IUI at age 25 and a series of unsuccessful IUIs at identical ages, yet one may conceive outside IUI at age 27 and the other at age 30. This difference may itself be related to their potential outcomes. The timing effect averages the effects of these heterogeneous delays, with each delay weighted by the share of non-reliers who experience it. These weights are identified: among women whose sequence fails, the realized delay is the difference between the first IUI and subsequent non-IUI conception, which recovers its full distribution. The effects, by contrast, cannot be separated by delay length. Because potential delays are heterogeneous and may be selective, outcomes following first-IUI success cannot be linked to the counterfactual delay each woman would have experienced had the procedure failed.[1]

I consider three dimensions of timing effects. The first is cumulative: how does postponement affect total earnings or hours worked over the observation window or, when that window is sufficiently long, over the life cycle? The second asks whether the effects arise only while motherhood is delayed or persist after the delayed birth. For example, when comparing motherhood at ages 25 and 30, one can compare outcomes between ages 25 and 30, when only the early-birth woman is a mother, and after age 30, when both are mothers. Heterogeneous delays complicate this comparison: at a given time since the first IUI, some later births have already occurred and others have not, so no single point in time separates the two periods for every woman. I explain how I address this when presenting the results in Section 5.3. The third dimension compares effects over a fixed number of years after each woman's own first birth, for example, the first four years after childbirth. This requires comparing reliers with non-reliers; Section 6 discusses the additional assumptions required and presents the results.

2.2.3 Relation to the Local Average Treatment Effects Framework

The willingness–reliance framework is closely related to the local average treatment effects framework, which separates treatment take-up into selective and quasi-random components. In standard potential-outcome notation, let $Y_z(d)$ denote the outcome under first-IUI success status z and parenthood status d, so that $Y_1(1) = Y^E$, $Y_0(1) = Y^L$, and $Y_0(0) = Y^N$. The remaining cell $Y_1(0)$ is empty, since a woman whose first IUI succeeds necessarily becomes a mother; the IUI setting therefore contains only compliers—who would remain childless if the first IUI failed—and always-takers—who would become mothers even if the first IUI failed.

Connecting the two frameworks requires specifying not only how many subsequent attempts a woman would undergo (willingness) and whether she would conceive outside IUI (reliance), but also the counterfactual outcomes of those subsequent attempts. Let $Z_j(0)$ denote the success of the jth IUI in the counterfactual in which all previous attempts fail. Compliance with respect to the first IUI then satisfies $C = R\prod_{j=2}^{W}(1 - Z_j(0))$. A woman is therefore a complier if she is a relier and every subsequent IUI she would undertake after failure of the first also fails. Compliers are thus a subset of reliers; reliers additionally include women who would remain childless if all IUIs failed but would conceive through a subsequent IUI. The parenthood effects identified by the willingness–reliance framework therefore cover a broader population than the standard LATE.

A simple example clarifies the distinction. Suppose all women would conceive only through IUI and would undergo exactly one additional procedure if the first failed. All women are then reliers, but only those whose second IUI would fail are compliers with respect to first-IUI success;

[1] This is also why the framework does not index potential outcomes by conception age; Appendix SA1 shows that doing so yields no additional identifiable parameters. Identifying effects of specific conception delays would require exogenous shifts of a given length, which quasi-experimental fertility designs generally do not provide because subsequent conception remains partly endogenous.

those whose second IUI would succeed are always-takers. Compliance status therefore combines potentially selective components—willingness and conception outside IUI—with later IUI outcomes that may themselves be quasi-random. The willingness–reliance framework keeps these components separate, allowing variation in later IUI success to contribute to identification.

There are two natural ways to define compliance in a dynamic setting, corresponding to different treatment definitions and identifying assumptions. The first defines compliance over the full data window and is suited to cumulative outcomes when the effect of motherhood does not depend on when within the window it begins. The second defines compliance at a particular horizon and is suited to contemporaneous outcomes under the assumptions that earlier births have the same effect regardless of their exact timing and later births have no anticipatory effects; I discuss these assumptions in Section 3.2. Reliance can be defined analogously at a particular horizon, which I use to separate timing effects before and after the later birth.

3 Econometric Approach

Section 3.1 introduces the central identification assumption. Section 3.2 describes the limitations of conventional methods. Section 3.3 presents the intuition behind my approach. Section 3.4 formalizes the procedure. Section 3.5 outlines estimation. Section 3.6 introduces a procedure to assess selection on unobservables.

3.1 Local sequential unconfoundedness

To formalize the intuition underlying the approach, I introduce the following assumption.

Assumption 1 (Local sequential unconfoundedness). $(Y^m, R, W) \perp\!\!\!\perp Z_j \mid A \geq j$*, for all* m, j*.*

This assumption states that, among women who undergo IUI attempt j—that is, who actually have sperm inserted into the uterus—the success of that attempt is effectively random: independent of potential outcomes and individual type. This aligns with the standard unconfoundedness assumption used in studies exploiting IVF, where pregnancy conditional on embryo transfer is assumed to be random. Unlike in those settings, however, the assumption here applies not only to the first procedure but also to all subsequent attempts.

To simplify exposition, I abstract from covariates. The main method in Section 3.4 allows IUI success rates to vary flexibly with observables, most importantly age at each procedure, the key medical determinant of success. It can also accommodate additional covariates, such as calendar time, time since the previous attempt, education, and current labor market outcomes, either at the time of the particular procedure or over their full histories across previous attempts. These covariates may themselves be endogenous. For instance, women may choose the age at which they initiate each IUI. Success rates may also vary across attempts for medical or behavioral reasons.

Crucially, each quasi-experiment is defined among realized inseminations. Assumption 1 concerns only the success of procedure j (Z_j), conditional on that procedure—i.e., insertion of sperm into the uterus—occurring ($A \geq j$). Everything leading up to it—whether a woman seeks subsequent procedures, how long she waits after a failure, and what she does to conceive in between—may depend on observed and unobserved determinants of outcomes. Consequently, women who reach later attempts may form a selected sample; for example, career potential may be systematically related to treatment continuation. The assumption requires only that, among women who undergo attempt j, whether that insemination succeeds is unrelated to their potential outcomes.

Decisions to continue IUI or pursue conception outside IUI may also respond to observable and unobservable shocks realized between attempts that affect labor-market outcomes. Women may postpone or cancel planned treatment following a health shock, a career change, adverse experiences with earlier procedures, or planned or unplanned conception outside IUI. Importantly, the assumption requires only that, among women who selectively undergo a given procedure at a given age, success at that IUI attempt is independent of potential outcomes and shocks to future success rates.

In line with the literature, Assumption 1 is stated as joint independence for expositional simplicity, but identification allows correlation between type and the factors determining IUI success, provided those factors are unrelated to potential labor market outcomes. Sterility is a leading example. It mechanically affects conception probabilities and is therefore correlated with reliance, while learning about it may also affect treatment continuation. Yet sterility need not otherwise affect women's health or labor market outcomes, and women are often unaware of it before attempting to conceive. In that case, these correlations do not threaten identification. This feature is inherent in quasi-experimental fertility designs that exploit variation in conception, where women with underlying sterility are disproportionately represented among those who remain childless.

In contrast, many factors may jointly shape continuation decisions, conception outside IUI, and labor market outcomes without affecting the success of individual IUI attempts. Work-family preferences provide a clear example. They may influence how many attempts a woman is prepared to undergo, how intensively she tries to conceive outside IUI, and her labor market outcomes, while remaining unrelated to whether a given insemination succeeds. The willingness–reliance framework accommodates precisely this form of selection, allowing fertility choices to be correlated with potential labor market outcomes.

Identification is thus threatened only by factors related to both per-attempt success probabilities and potential labor-market outcomes, or to both per-attempt success probabilities and continuation decisions insofar as those decisions are themselves correlated with potential labor-market outcomes. Although the assumption is not directly testable, medical evidence suggests that many natural candidates are only weakly associated with IUI success. For instance, studies of psychological factors, such as stress and anxiety (Syam et al., 2017; Guan et al., 2021), and lifestyle factors, such as body mass index and obesity (Kan et al., 2020; Kim et al., 2023), generally find little association with IUI success. Even for smoking, a well-established predictor of fertility, the evidence is mixed, with studies often finding only small negative associations or no effect on IUI outcomes (see ASRM Practice Committee, 2024, for an overview). The main remaining concern is therefore that success may be correlated with latent health differences that also shape labor market outcomes.

I provide three pieces of evidence in support of the assumption. First, if IUI success probabilities were systematically related to labor market potential, these differences might also appear in pre-IUI outcomes. In Section 4.6, I find little imbalance between successes and failures at either the first or later procedures, suggesting that success remains unrelated to labor market potential throughout the sequence. This also speaks directly to sterility: women with lower success probabilities are mechanically overrepresented among failures, yet their pre-IUI labor market outcomes are similar, suggesting that underlying sterility is not systematically associated with labor market performance.

Second, Section 3.6 develops an approach to quantify the extent of selection on unobservables that may remain despite these balance results. Applying it in Section 4.6, I find little scope for such selection. Section 5.6 then uses these results to assess the sensitivity of the main estimates to violations of the assumption and finds that the estimates change little.

Third, I investigate whether women who continue to subsequent procedures differ systematically in their likelihood of success. If women who are generally healthier are both more likely to succeed and more likely to continue treatment, success rates may change across attempts as the sample becomes increasingly selected. Instead, Section 4.6 shows that age-conditional per-attempt success rates remain stable across the sequence, providing little evidence of such sorting.

In sensitivity analysis in Section 5.6, I further relax the assumption by conditioning on clinical protocol details at each procedure, including semen testing and preparation, ovulation monitoring, and hormone monitoring and use. Success then need only be as good as random among women receiving the same protocol at the same attempt. The results are virtually unchanged.

Finally, even if some unobserved selection due to latent health differences remains, the direction of the resulting bias is predictable. If women whose IUIs succeed are positively selected on latent health, those whose IUIs fail are negatively selected, and if latent health is positively correlated with labor market outcomes, estimates will understate both the cost of parenthood and the benefit of delaying motherhood. This is particularly relevant for the timing effects: under selection on unobserved health, the estimated gains from delay would provide a lower bound on the true effects.

3.2 Limitations of Standard Methods

The standard IV approach uses the success of a woman's first IUI as an instrument for parenthood. First, consider a setup in which parenthood status is measured at the end of the observation window, as in my analysis, and outcomes capture cumulative earnings over the full window. The reduced form is then the difference in average outcomes between women whose first IUI succeeds and those whose first IUI fails:

$$\lambda_{RF} = \mathbb{E}[Y \mid Z_1 = 1] - \mathbb{E}[Y \mid Z_1 = 0].$$

The first group conceives at the first IUI, whereas the second contains both women who remain childless (compliers) and women who become mothers later (always-takers). Following Angrist & Imbens (1995), under unconfoundedness the reduced form identifies a weighted combination of the average parenthood effect for compliers and the average timing effect for always-takers:

$$\lambda_{RF} = \mathbb{E}[Y^E - Y^N \mid C=1]\Pr(C=1) + \mathbb{E}[Y^E - Y^L \mid C=0]\Pr(C=0) \tag{1}$$
$$= \mathbb{E}[\tau \mid C=1]\Pr(C=1) - \mathbb{E}[\delta \mid C=0]\Pr(C=0). \tag{2}$$

Scaling by the first stage—the difference in the share of mothers between the two groups, which identifies the complier share—gives

$$\frac{\mathbb{E}[Y \mid Z_1=1] - \mathbb{E}[Y \mid Z_1=0]}{\mathbb{E}[D \mid Z_1=1] - \mathbb{E}[D \mid Z_1=0]} = \mathbb{E}[\tau \mid C=1] - \mathbb{E}[\delta \mid C=0]\frac{\Pr(C=0)}{\Pr(C=1)}. \tag{3}$$

The standard IV exclusion restriction requires that the instrument have no effect among always-takers. In this setting, this means that outcomes do not depend on the timing of motherhood, so that $\delta = 0$. Otherwise, the second term in (3) contaminates the IV estimand.

The direction of this bias is ambiguous. Career costs may be underestimated when younger children require more care and delayed motherhood coincides with women's peak career years ($Y^L < Y^E$), and overestimated if early motherhood generates lasting career setbacks ($Y^L > Y^E$). Its magnitude also depends on the scaling factor $\Pr(C=0)/\Pr(C=1)$. In the context of IUI (and IVF; Lundborg et al., 2017), around 75% of women whose first procedure fails eventually have children, implying $\Pr(C=0) = 0.75$ and $\Pr(C=0)/\Pr(C=1) = 3$. Even relatively small timing effects can therefore generate substantial bias.[2]

An alternative is to define parenthood status and compliance at a particular horizon within the observation window. Suppose the econometrician observes two periods, indexed by j, with Y_j and D_j the outcome and motherhood status in period j. Assume that no woman whose first IUI fails becomes a mother in the first period, so that $D_1 = Z_1$, and let C_2 indicate whether a woman would remain childless through the second period if her first IUI failed. The approach also imposes no anticipation: before becoming a mother, later-motherhood and childless potential outcomes are equal, $Y_1^L = Y_1^N$.

In the first period every woman with $Z_1 = 0$ is childless, so the reduced form identifies the effect of one period of motherhood. In the second period women with $Z_1 = 0$ are a mixture—those with $C_2 = 1$ remain childless, those with $C_2 = 0$ have become mothers—so the reduced form again combines the parenthood effect for women who would otherwise remain childless with the timing effect for those who become mothers later. The exclusion restriction now requires $Y_2^E = Y_2^L$, so that once motherhood begins, outcomes do not depend on when it began.

One way to address this contamination is to account for parenthood duration. Gallen et al. (2024) and Bensnes et al. (2025) use a recursive IV approach that first identifies short-run effects and then uses them to adjust longer-run estimates. In the two-period example, identification requires

$$\mathbb{E}\left[Y_1^E - Y_1^N\right] = \mathbb{E}\left[Y_2^L - Y_2^N \mid C_2 = 0\right], \tag{4}$$

[2]The same bias can be expressed in terms of negative weights (see Bensnes et al., 2025). In this example, weight 4 is assigned to the average parenthood effect and weight −3 to the average later-motherhood effect among always-takers, $\mathbb{E}[Y^L - Y^N \mid C=0]$.

equating the mean effect of one period of motherhood beginning at the first IUI with the effect of one period of motherhood in the second period for always-takers who become mothers in that period. The former is identified by the first-period reduced form and $\Pr(C_2 = 0) = \mathbb{E}[D_2 \mid Z_1 = 0]$ is observed, so the second-period parenthood effect follows by substitution.

The two approaches therefore impose different restrictions, but both ultimately restrict the relationship between early- and later-motherhood outcomes, Y^E and Y^L. Conventional IV allows effects to vary over the life cycle, but requires that, holding life-cycle stage fixed, they do not depend on when motherhood began. Recursive IV allows effects to vary with the duration of motherhood, but requires that, holding duration fixed, they do not depend on the current life-cycle stage. Crucially, both approaches rule out effects that depend on the age or career stage at which motherhood begins, as well as differences arising from subsequent fertility paths induced by earlier or later motherhood.[3]

Bensnes et al. (2025) show that recursive IV estimates can differ markedly from conventional IV estimates, illustrating the importance of assumptions about the role of timing. In Section 5.7, I replicate this divergence in my setting and compare both approaches with my estimates.

Before turning to the method, I discuss two intuitive ways to use the IUI sequence for identification and why neither resolves the timing challenge. The first is to treat each attempt as a separate observation. One can construct a stacked dataset in which each block contains women who reached a given attempt and estimate the blocks separately or pool them. This does not solve the identification problem: within each block, women whose attempt fails include both those who remain childless and those who conceive later than women whose attempt succeeds. Comparisons between successful and failed attempts therefore still combine parenthood and timing effects, and averaging across attempts does not separate them.

The second is to redefine the instrument as success of the last attempt a woman undergoes, Z_A. This may seem appealing because reliers are the compliers with respect to Z_A, since $D = Z_A + (1 - Z_A)(1 - R)$. However, Z_A need not be as good as random because the number of attempts is selected. Suppose maximum willingness is two and attempt j succeeds with probability p_j. For women with $W = 1$, $\Pr(Z_A = 1 \mid W = 1) = p_1$. For women with $W = 2$, $\Pr(Z_A = 1 \mid W = 2) = p_1 + (1 - p_1)p_2$. Thus, the probability of $Z_A = 1$ varies with willingness, generating bias whenever willingness is correlated with potential outcomes. This concern is empirically relevant: Section 4.5 shows that willingness is systematically related to pre-IUI labor market outcomes.

Lastly, while it may seem helpful to control for the number of procedures undergone, this would introduce a bad-control problem because the realized number of attempts is affected by earlier successes. Conditioning on willingness would avoid this selection, but willingness is unobserved when the sequence ends in success.

[3] The two estimates may deviate from the true effect in either the same or opposite directions. In the two-period example, the recursive-IV discrepancy depends on the difference between $\mathbb{E}\left[Y_1^E - Y_1^N\right]$ and $\mathbb{E}\left[Y_2^L - Y_2^N \mid C_2 = 0\right]$, while the conventional-IV discrepancy depends on $\mathbb{E}\left[Y_2^L - Y_2^E \mid C_2 = 0\right]$; there need not be any systematic relationship between the two.

3.3 Intuition for the Econometric Approach

In this section, I illustrate the intuition behind the method in a simplified setting. The main simplification, reflected in Assumption 1, is to abstract from the role of covariates—in particular, age at each IUI—in shaping procedure success. This makes it easier to see how variation across the procedure sequence is used to identify the effects without also tracking when procedures occur. The formal results in Section 3.4 account for age throughout the sequence.

The identification argument has two main steps. The first uses women who experience no IUI success within the window. The core intuition is that each woman has several possible fertility paths, characterized by her type—how many attempts she would undergo under continued failure and whether she would conceive outside IUI if the sequence failed—and by which attempts succeed or fail. On the path where every procedure fails, the conditions defining type are realized, so her type is directly observed: the number of completed procedures gives her willingness, and whether she subsequently conceives outside IUI gives her reliance. Moreover, among women of the same type, reaching this path is as good as random: continuation decisions are fixed by type, so whether a woman reaches it depends only on procedure outcomes, each of which is as good as random. The outcomes of women reaching this path therefore identify the relevant average counterfactual outcome at each willingness level—childlessness for reliers and later motherhood for non-reliers. The frequencies of failed sequences identify the corresponding type shares, allowing these counterfactual outcomes to be aggregated across willingness levels.

The second step bounds the average early-motherhood outcome for reliers and non-reliers using women whose first IUI succeeds. For these women, the early-motherhood outcome is observed and, under random first-attempt success, they form an as-good-as-random sample of the overall sample. While their latent types are unknown, I can use the type shares identified in the first step to bound each type's average early-motherhood outcome by assuming that the type fully occupies either the lower or upper tail of the distribution. Combining these bounds with the identified childless and later-motherhood outcomes yields bounds on the average parenthood and timing effects for reliers and non-reliers, respectively.

To illustrate the first step, I focus on the case in which willingness is capped at two attempts, $W \leq 2$; the argument extends straightforwardly to an arbitrary cap. Throughout this section I write $Z_2 = \varnothing$ when no second attempt takes place and $Z_2 = 0$ when it takes place but fails. I write p_1 and p_2 for the success rates at the first and second procedures, the latter among women who actually undergo a second attempt.

Figure 1 separates each latent type into a column, indexed by willingness and reliance, while each row within a column describes a state that the type can reach. The resulting rectangles group individuals by observed fertility histories, summarized by IUI outcomes and parenthood status. When only one type can reach a given state, the corresponding rectangle is contained within that type's column, so the observed group is known to contain only that type; when several types can reach it, the rectangle spans the relevant columns, so the type of individuals in the observed group is not known. Within each rectangle, the first row shows the realized and observed fertility path,

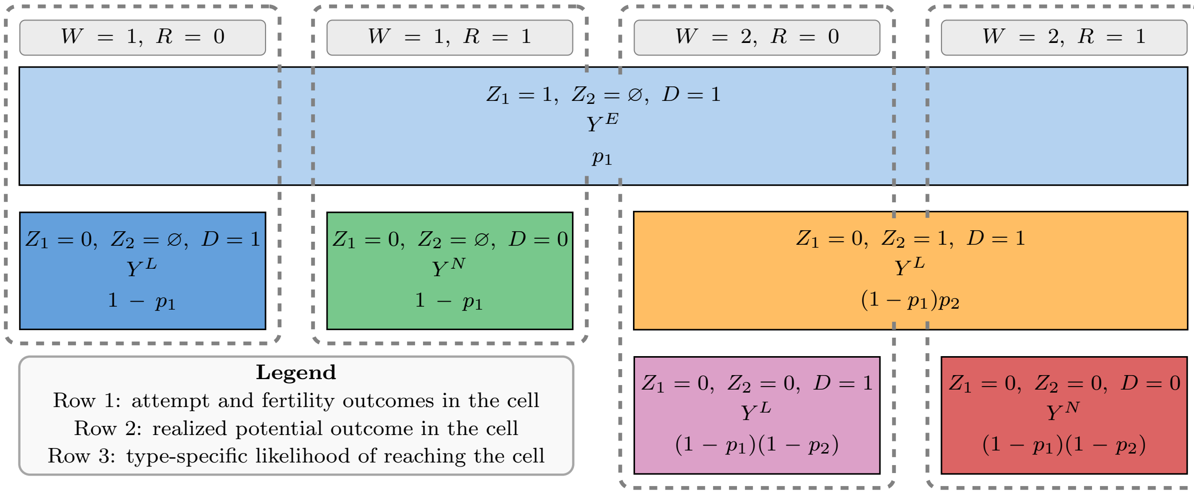

Figure 1: Illustration of the Identification Argument

Note: Column headers denote latent types. Cells show the possible states in which each type may be observed, as determined by Z_1, Z_2, and D. The notation $Z_2 = \varnothing$ indicates that no second attempt took place. The likelihoods use p_1 and p_2 for the success probabilities at the first and second procedures. Dashed outlines group states belonging to the same latent type and indicate that, conditional on type, the realized cell is determined by as-good-as-random variation in attempt success. Researchers observe Z_1, Z_2, D, and Y, but not the latent type. Horizontally merged cells represent observationally identical states in which researchers cannot distinguish among the latent types.

the second shows the realized potential outcome corresponding to that path—early motherhood, later motherhood, or childlessness—and the third shows the probability that an individual of that type reaches the cell.

Consider non-reliers willing to undergo one attempt, $(W = 1, R = 0)$. Women who belong to this type may end up in one of two fertility paths. If the first IUI succeeds, they reach the state $(Z_1 = 1, Z_2 = \varnothing, D = 1)$, where the early-motherhood outcome Y^E is realized; the probability of reaching that state is p_1. Women observed in this state cannot be distinguished from other types, since every type reaches it under first-attempt success, which is why the rectangle in the first row spans all four columns. Alternatively, if the first IUI fails, no second attempt takes place, and the woman nonetheless has a child, she reaches the state $(Z_1 = 0, Z_2 = \varnothing, D = 1)$, where the later-motherhood outcome Y^L is realized. Only the $(W = 1, R = 0)$ type can reach this state: reliers would remain childless, while women willing to undergo two attempts would proceed to a second procedure. Women observed in this state are therefore known to be of this type.

The dashed outlines around each column represent the key identifying assumption. Because success rates are independent of potential outcomes and type, conditional on type, the state, or row, in which a woman ends up within her column is as good as random. For example, the $(W = 1,\ R = 0)$ type can end up only in the first row or in the first column of the second row. Since the latter cell contains only this type, and assignment between these two cells is as good as random, women observed in the state $(Z_1 = 0,\ Z_2 = \varnothing,\ D = 1)$ are a random sample of the $(W = 1,\ R = 0)$ type. Since their later-motherhood outcome is realized, the type-specific average is identified:

$$\mathbb{E}[Y \mid Z_1 = 0,\ Z_2 = \varnothing,\ D = 1] = \mathbb{E}[Y^L \mid W = 1,\ R = 0].$$

The same argument applied to the second column identifies the average childless outcome among reliers willing to undergo one attempt:

$$\mathbb{E}[Y \mid Z_1 = 0,\ Z_2 = \varnothing,\ D = 0] = \mathbb{E}[Y^N \mid W = 1,\ R = 1].$$

The argument extends directly to types willing to undergo two attempts. Only $(W = 2,\ R = 0)$ can reach the state in which both attempts fail and they remain childless, so

$$\mathbb{E}[Y \mid Z_1 = 0,\ Z_2 = 0,\ D = 1] = \mathbb{E}[Y^L \mid W = 2,\ R = 0].$$

Symmetrically, for $(W = 2,\ R = 1)$ the childless outcome is identified from the state with $D = 0$. Extending the argument to an arbitrary number of willingness types simply requires adding the corresponding columns and proceeding to the terminal failure state within each column.

A similar argument identifies the share of each type. The probability that a $(W = 2,\ R = 1)$ woman reaches the state $(Z_1 = 0,\ Z_2 = 0,\ D = 0)$ is the probability that both attempts fail, $(1 - p_1)(1 - p_2)$, so her type's share solves:

$$\Pr(Z_1 = 0,\ Z_2 = 0,\ D = 0) = \Pr(W = 2,\ R = 1)\,(1 - p_1)(1 - p_2).$$

Because the probability of reaching a fully failed sequence declines with willingness, women willing to undergo more attempts are underrepresented among those observed there; dividing by $(1 - p_1)(1 - p_2)$ is what corrects for this.

Together, these arguments identify the relier and non-relier shares, $\Pr(R = 1)$ and $\Pr(R = 0)$, the average later motherhood outcome for non-reliers, $\mathbb{E}[Y^L \mid R = 0]$, and the average childless outcome for reliers, $\mathbb{E}[Y^N \mid R = 1]$, by averaging the type-specific quantities over willingness.

The figure also shows a state the analysis does not use. Women willing to undergo at least two attempts who conceive at the second, $(Z_1 = 0,\ Z_2 = 1,\ D = 1)$, occupy a cell spanning the third and fourth columns, so the two types cannot be separated there. Their realized outcome could in principle be used to bound timing effects between the first and second attempts for women willing to undergo at least two procedures. However, these attempts typically occur only a month apart, so such comparisons may reveal little about timing effects. Focusing on non-reliers instead isolates the group for whom IUI failure generates the largest timing shift: conception at the first attempt versus only after the entire IUI sequence fails.

When outcomes have bounded support, having identified the average childless outcome for reliers and the average later-motherhood outcome for non-reliers, one could bound the remaining unobserved outcomes using the support of the outcome distribution.[4] These bounds may, however, be very wide. I instead bound the average early-motherhood outcome separately for reliers and non-reliers, which yields tighter bounds on the corresponding average parenthood and timing effects by exploiting observed outcome variation rather than only the limits of outcome support.

I start from the observation that, since IUI outcomes are as good as random, the outcome distribution among women whose first IUI succeeded represents the full distribution of early motherhood outcomes in the IUI sample. Using the relier share identified in the previous step, I construct worst-case bounds for average relier early motherhood outcome by assuming that reliers are those with the highest or lowest early motherhood outcomes. For instance, suppose 100 women had a successful first IUI and the relier share is 80%. It is not known which 80 are reliers, but I can obtain the upper (lower) bound on their average outcome by taking the average of the top (bottom) 80 outcomes.[5] A symmetric argument applies to bounding the non-relier early motherhood outcome.

I then refine the bounds by incorporating pre-IUI covariates. Suppose that after splitting the sample by pre-IUI earnings, each group is estimated to have an 80% relier share. Selecting the bottom 80% of outcomes among women whose first IUI succeeded, without accounting for pre-IUI earnings, may yield a set of potential reliers whose pre-IUI earnings are inconsistent with the group-specific shares. Because this selection produces the most conservative lower bound, any alternative selection can only raise it. To construct the refined bounds, I identify the relier share within each pre-IUI earnings group and select the corresponding share of the lowest and highest early motherhood outcomes in that group. The argument for non-reliers is symmetric.

I can further narrow the bounds only by imposing assumptions on which women are (not) reliers. For example, it may be reasonable to assume that women who have a second or third child without IUI after having their first through IUI would have had at least one child even if all IUIs had failed—or, equivalently, that women who are reliant on IUI for their first child are also reliant for subsequent children. To see how this helps narrow the bounds, consider the previous example. If 10 of the 100 women whose first IUI succeeded subsequently have a non-IUI child, they can be excluded from the pool of potential reliers, as they are certainly not reliant on IUI. Selecting the 80 lowest (highest) outcomes from the remaining 90 can only yield a higher (lower) average than selecting from the full set of 100.

Assumption 2 (Subsequent reliance monotonicity). *$R^+ \geq R$.*

The monotonicity condition ensures that women whose first IUI succeeded and who have any non-IUI children, and are therefore not subsequent reliers, are also not reliers. This allows them to be

[4]The childless outcome for non-reliers and the later-motherhood outcome for reliers can be assigned their lowest or highest feasible values. Combined with the identified type-specific outcomes and the overall early-motherhood outcome, which is identified from first-IUI successes, this yields bounds on the average parenthood and timing effects.

[5]Trimming outcome distribution tails to bound subgroup averages follows the logic of Zhang & Rubin (2003) and Lee (2009) for addressing sample selection. The innovation lies in identifying and overcoming the challenges involved in appropriately leveraging sequential assignment with bounding, yielding narrower bounds that apply to a broader subpopulation (see Section 5.7).

excluded from the pool of potential reliers used to construct the bounds.

One way to interpret the monotonicity condition is that families are more determined to have a first child than additional ones. From a fertility-choice perspective, it rules out couples who prefer multiple children but would rather have none than only one. This restriction seems reasonable, as couples who initiate IUI intend to have at least one child and are likely aware that they may not be able to have multiple.

However, fertility may not be entirely determined by choice. For example, having a first child may improve relationship stability or mental health relative to failing to conceive, leading to more natural conception attempts. This may lead to non-IUI births that would not have occurred had the first IUI failed, thereby violating monotonicity. I address such concerns in Appendix SA4 and relax the assumption in Appendix SA5.

The monotonicity assumption also yields testable implications, namely that the subsequent relier share at each covariate value exceeds the relier share. Appendix SA5 presents the testing procedure and results, and also reports estimates based on a relaxed specification that allows the direction of monotonicity to vary with covariates. Monotonicity is not rejected, and the estimates under the alternative assumption remain similar.[6]

Finally, I discuss what drives the remaining uncertainty in the bounds and when they collapse to a point. Under monotonicity, all reliers—women who would remain childless if the IUI sequence failed—are subsequent reliers—women who would not have subsequent children outside IUI if the first IUI succeeded. Some subsequent reliers, however, are not reliers: they would have no subsequent children outside IUI if the first IUI succeeded, but would conceive outside IUI if the sequence failed. Because it is not known which subsequent reliers are reliers, the bounds assign reliers either the highest or the lowest potential outcomes among subsequent reliers. Since all women in the IUI sample are actively trying to conceive, however, it may be reasonable to assume that, among subsequent reliers, whether a woman would remain childless after sequence failure is largely random or, more weakly, that potential outcomes are the same on average for those who would and would not remain childless:

Assumption 3 (Subsequent relier mean independence). $\mathbb{E}[Y^E \mid R = 1] = \mathbb{E}[Y^E \mid R^+ = 1]$.

This assumption is a restriction on selection into non-IUI fertility. The formal identification approach weakens it by imposing the restriction only conditional on pre-IUI covariates, including prior labor market histories. Crucially, the bounds remain valid even if Assumption 3 fails.

Under Assumption 3, the average early-motherhood outcome for reliers is point identified by the observed average among women whose first IUI succeeds and who subsequently have no non-IUI children—the subsequent reliers. The corresponding non-relier outcome can then be recovered from the overall average early-motherhood outcome and the identified type shares. Since the average

[6]The remaining results assume monotonicity. In settings without variables that can credibly help identify reliers, sharp bounds without monotonicity (i.e., under trivial monotonicity) can be obtained by redefining $R^+ = 1$ and $D^+ = (1 - Z_A)D$, thus treating everyone as subsequent reliers.

childless outcome for reliers and the average later-motherhood outcome for non-reliers are already point identified, the parenthood and timing effects are consequently point identified.

Both the monotonicity and random reliance assumptions imply further testable restrictions when the approach is applied to predetermined outcomes. Because outcomes realized before the first IUI cannot depend on subsequent fertility, all three potential outcomes coincide for every woman, so the parenthood and timing effects are zero. The procedure itself does not impose this. If monotonicity fails, excluding women with non-IUI children from the pool of potential reliers may exclude some true reliers, so the resulting bounds on average relier pre-IUI outcomes need not contain the relier mean identified using women who actually remain childless after IUI failure. Similarly, if subsequent reliers differ systematically from reliers, the point estimate of the relier pre-IUI outcome obtained from subsequent reliers may differ from the mean identified using women who actually remain childless after IUI failure. Section 5.2 shows that the placebo bounds contain zero and that the point estimates are close to zero, providing support for both assumptions.

3.3.1 Treatment Effect Identification in the Absence of Timing Effects

Before turning to the formal results, I consider an alternative way to identify the average treatment effect for reliers under an exclusion restriction analogous to standard IV. This approach targets the same population as my main estimand—reliers rather than compliers—while imposing assumptions similar to existing IV approaches. It therefore helps distinguish differences arising from identifying assumptions from those arising from the populations covered by the estimands.

The previous section identifies the overall average early-motherhood outcome, the average childless outcome for reliers, the average later-motherhood outcome for non-reliers, and the corresponding type shares. These quantities can be combined into a *sequential reduced form* comparing average outcomes when the first IUI succeeds with those when all IUIs fail. Let $Y^* = Y^N R + Y^L(1-R)$ denote the outcome when all IUIs fail. For reliers, it is the childless outcome, for non-reliers it is the later-motherhood outcome. Then $\mathbb{E}[Y^*] = \mathbb{E}[Y^N \mid R=1]\Pr(R=1) + \mathbb{E}[Y^L \mid R=0]\Pr(R=0)$, where all terms are identified in the previous section. It follows that $\mathbb{E}[Y^E - Y^*] = \tau_{ATR}\Pr(R=1) - \delta_{ANR}\Pr(R=0)$. Scaling by the relier share yields

$$\frac{\mathbb{E}[Y^E - Y^*]}{\Pr(R=1)} = \tau_{ATR} - \delta_{ANR}\frac{\Pr(R=0)}{\Pr(R=1)}. \tag{5}$$

I refer to this as the *sequential IV* estimand because it uses the sequence of IUI attempts to construct an estimand analogous to conventional IV in Equation (3), but targets reliers rather than compliers. Under the standard exclusion restriction that the timing of motherhood does not affect outcomes, $\delta_{ANR} = 0$, and sequential IV identifies τ_{ATR}. When timing matters, the estimand is contaminated by δ_{ANR}, but the timing term receives less weight than under conventional IV because non-reliers are a subset of always-takers: $\Pr(R=0)/\Pr(R=1) \leq \Pr(C=0)/\Pr(C=1)$.

Importantly, the sequential IV estimand need not lie within the bounds on τ_{ATR}. If it falls outside the bounds, the exclusion restriction can therefore be rejected, providing a direct test for the presence of timing effects and quantifying their magnitude.

3.4 Formal Identification Statement

This section provides a concise formalization of the intuition underlying the identification argument developed in the previous section. Whereas the previous section established identification using conditional expectations, this section establishes it using inverse probability weighting. The two approaches are combined in the estimation section, analogously to how doubly robust estimators combine outcome regression and inverse probability weighting under a single treatment assignment (e.g., Abadie, 2003; Chernozhukov et al., 2018).

As in previous sections, the outcome (Y) is measured either over the entire data window or at a particular horizon from the first IUI, whereas parenthood status (D), the IUI attempt history (A), and attempt outcomes (Z_j) summarize events occurring between the first IUI and the end of the data window. I begin by introducing the conditional local sequential unconfoundedness assumption:

Assumption 4 (Conditional local sequential unconfoundedness).
$(Y^m, R^+, R, W) \perp\!\!\!\perp Z_j \mid X_j,\ A \geq j$ *for all* m, j, *and* $X_j \in \mathcal{X}_j$.

where $X_j \in \mathcal{X}_j$ are covariates measured at assignment j, such as age at the time of the procedure, time since the last procedure, and other observable characteristics. In words, the success of IUI j is independent of potential outcomes and type, conditional on undergoing at least j IUIs and on these covariates. Note that the measurement timing of X_j may vary across women and be correlated with potential outcomes: it is determined by when they choose to undergo each procedure. The next assumption provides regularity conditions. Let $e_j(x) = \Pr(Z_j = 1 \mid X_j = x, A \geq j)$.

Assumption 5 (Regularity).

1. $0 < \underline{e} < e_j(x) < \overline{e} < 1$ *for all* j *and* $x \in \mathcal{X}_j$, *for some fixed* $\underline{e}$ *and* $\overline{e}$.
2. Y *has a probability density function for* $Z_1 = 1$, $D^+ = 0$, *and all* $x \in \mathcal{X}_1$.

It contains two parts. First, the probability of IUI success conditional on undergoing the procedure and covariates at the time differs from 0 and 1. Second, Y is a continuous random variable conditional on the first IUI succeeding, having only IUI children, and any value of X_1. In practice, adding a negligible amount of continuously distributed noise to Y is sufficient to avoid ties in trimming without meaningful bias.

The bounding procedure begins with identifying several nuisance functions involved in the trimming step. First, the covariate-conditional relier share is identified using the share of women without children among those whose IUIs failed:

$$r(x) = \mathbb{E}\left[(1 - D^+)\Pi_{j=1}^{A} \frac{(1 - Z_j)}{(1 - e_j(X_j))} \middle| X_1 = x\right]. \tag{6}$$

Because the denominator decreases in A, women who undergo more IUIs receive larger weights, correcting for their lower likelihood of never experiencing IUI success and thus their underrepresentation in this group. Next, the covariate-conditional share of subsequent reliers is identified from the share of women having only IUI children among those whose first IUI succeeded

$r^+(x) = \mathbb{E}[1 - D^+ \mid Z_1 = 1, X_1 = x]$. Under monotonicity, the covariate-conditional share of reliers among subsequent reliers is then $p(x) = r(x)/r^+(x)$.

The covariate-conditional quantile function of the early motherhood outcome distribution among subsequent reliers is identified from the outcome distribution of women whose first IUI succeeded and who have only IUI children:

$$q(u, x) = \inf \left\{ q : u \leq \Pr(Y \leq q \mid X_1 = x, Z_1 = 1, D^+ = 0) \right\}. \tag{7}$$

Finally, $q(p(x), x)$ and $q(1 - p(x), x)$ identify the covariate-conditional $p(x)$-th and $1 - p(x)$-th quantiles of the early motherhood outcome distribution among subsequent reliers. These quantiles are used to trim the outcome distribution and select reliers in scenarios where they have either the lowest or highest early motherhood outcomes.

The nuisance functions are combined with the data to construct the moments:

$$m^L(G, \eta^0) = Y(1 - D^+)1_{\{Y < q(p(X_1), X_1)\}} \frac{Z_1}{e_1(X_1)} - Y(1 - D^+)\Pi_{j=1}^{A} \frac{(1 - Z_j)}{(1 - e_j(X_j))} \tag{8}$$

$$m^U(G, \eta^0) = Y(1 - D^+)1_{\{Y > q(1 - p(X_1), X_1)\}} \frac{Z_1}{e_1(X_1)} - Y(1 - D^+)\Pi_{j=1}^{A} \frac{(1 - Z_j)}{(1 - e_j(X_j))}, \tag{9}$$

where vector G contains observed variables and η^0 contains the nuisance functions:

$$\eta^0(x_1, \dots, x_A) = \{r^+(x_1), r(x_1), q(p(x_1), x_1), q(1 - p(x_1), x_1), e_1(x_1), \dots, e_{\overline{w}}(x_{\overline{w}})\}. \tag{10}$$

The first term in $m^L(G, \eta^0)$ is used to bound the relier average early motherhood outcome. It assigns positive weights to women whose first IUI succeeded, who have only IUI children, and whose outcomes fall below the covariate-conditional trimming threshold $q(p(x), x)$. The second term, used to identify the relier average childless outcome, assigns positive weights to outcomes of childless women. Larger weights are given to women who underwent more IUIs to account for the fact that reliers willing to undergo more IUIs are less likely to not experience IUI success, making them underrepresented in this group. $m^U(G, \eta^0)$ mirrors this for the scenario where reliers have the highest early motherhood outcomes. The moments are then scaled by the relier share.

Theorem 1 (Bounds on the Treatment Effect)**.** *Under Assumptions 2, 4, and 5, sharp lower and upper bounds on τ_{ATR} are given by $\theta_L = \mathbb{E}[m^L(G, \eta^0)]/\mathbb{E}[r(X_1)]$ and $\theta_U = \mathbb{E}[m^U(G, \eta^0)]/\mathbb{E}[r(X_1)]$.*

Proof. See the Appendix.

The next moment is used to identify the sequential reduced-form estimand, which captures the average effect of the first IUI succeeding relative to the entire sequence of IUIs failing. This estimand is then used both to back out timing effects from the treatment-effect bounds and to identify the sequential IV estimand, which identifies the relier average treatment effect under the assumption of no timing effects.

$$m^{RF}(G, \eta^0) = Y \frac{Z_1}{e_1(X_1)} - Y\Pi_{j=1}^{A} \frac{(1 - Z_j)}{(1 - e_j(X_j))} \tag{11}$$

Corollary 1 (Identification in the Absence of Timing Effect)**.** *Under Assumptions 4 and 5, and if* $\delta = 0$ *a.s., then* $\tau_{ATR} = \mathbb{E}[m^{RF}(G, \eta^0)]/\mathbb{E}[r(X_1)]$*.*

Proof. See the Appendix.

Since the sequential reduced-form estimand is a weighted average of the parenthood effect for reliers and the timing effect for non-reliers, the moments used to bound the average timing effect for non-reliers combine the moment identifying the sequential reduced form with those used to bound the relier average treatment effect:

Theorem 2 (Bounds on the Timing Effect)**.** *Under Assumptions 2, 4, and 5, sharp lower and upper bounds on* δ_{ANR} *are given by* $\gamma_L = \mathbb{E}[m^L(G, \eta^0) - m^{RF}(G, \eta^0)]/\mathbb{E}[1 - r(X_1)]$ *and* $\gamma_U = \mathbb{E}[m^U(G, \eta^0) - m^{RF}(G, \eta^0)]/\mathbb{E}[1 - r(X_1)]$*.*

Proof. See the Appendix.

The next assumption yields point identification when non-IUI conception selection is limited:

Assumption 6 (Conditionally random reliance). $\mathbb{E}[Y^E | R = 1, X_1] = \mathbb{E}[Y^E | R^+ = 1, X_1]$ *a.s.*

The assumption states that, among subsequent reliers with the same covariates, reliance carries no information about average outcomes under early motherhood. In the IUI context, it formalizes the idea that, conditional on pre-IUI labor market outcomes and other observables, whether a couple would remain reliant on IUI after a failed sequence may be unrelated to career potential.

$$m^{RND}(G, \eta^0) = Y(1 - D^+)\, p(X_1)\, \frac{Z_1}{e_1(X_1)} - Y(1 - D^+)\, \Pi_{j=1}^{A} \frac{(1 - Z_j)}{(1 - e_j(X_j))}. \tag{12}$$

This moment replaces the trimming indicator in m^L and m^U with the covariate-conditional relier share $p(X_1)$. Rather than assigning reliers the lowest or highest early-motherhood outcomes among subsequent reliers, it assigns them the average.

The following theorem establishes identification and shows that the identified parameters are guaranteed to lie within the bounds.

Theorem 3 (Identification under Random Reliance)**.** *Under Assumptions 4, 5, and 6,* $\tau_{ATR} = \mathbb{E}[m^{RND}(G, \eta^0)]/\,\mathbb{E}[r(X_1)]$ *and* $\delta_{ANR} = \mathbb{E}[m^{RND}(G, \eta^0) - m^{RF}(G, \eta^0)]/\,\mathbb{E}[1 - r(X_1)]$*. Moreover,* $\theta_L \leq \tau_{ATR} \leq \theta_U$ *and* $\gamma_L \leq \delta_{ANR} \leq \gamma_U$*.*

Proof. See the Appendix.

3.5 Estimation

The bounds can be estimated by replacing the expectations and nuisance functions in the theorem with their empirical counterparts. However, inference for such estimators is complicated by their sensitivity to first-stage estimation error in the conditional quantile function, which itself must be estimated nonparametrically. If overly conservative—i.e., non-sharp—bounds are acceptable, one can overcome this challenge by restricting attention to discrete covariates or by partitioning continuous covariates into bins. In that case, covariate-conditional bounds can be estimated separately within each covariate cell and then aggregated. Asymptotic normality and a consistent estimator of the asymptotic variance then follow from conventional GMM results.

Ensuring sharp bounds, however, requires using all available covariates, including continuous ones, making such an approach infeasible. To address this challenge, I draw on insights from Semenova (2025), who develops an estimator for the Lee (2009) bounds, which also rely on conditional quantiles. The method combines orthogonalization with sample splitting. I adapt this approach by constructing new orthogonal moment functions for my bounds.

The moment functions given in Table 1 identify the same parameters as the baseline moments:

$$\mathbb{E}[\psi^L(G,\xi^0)] = \mathbb{E}[m^L(G,\eta^0)], \qquad \mathbb{E}[\psi^U(G,\xi^0)] = \mathbb{E}[m^U(G,\eta^0)], \tag{13}$$

$$\mathbb{E}[\psi^{RF}(G,\xi^0)] = \mathbb{E}[m^{RF}(G,\eta^0)], \qquad \mathbb{E}[\psi^{RND}(G,\xi^0)] = \mathbb{E}[m^{RND}(G,\eta^0)]. \tag{14}$$

However, the original moments are sensitive to small errors in the nuisance parameters, whereas the new moments are not.[7] I use this property together with sample splitting to justify asymptotic inference as if the nuisance parameters were known, following the argument in Semenova (2025).

The moments in this paper generalize those in Semenova (2025) by incorporating the sequential structure of my procedure. This shifts the focus from compliers to reliers and requires additional correction terms for sequential propensity scores. When only the first IUI attempt is used, rather than the full sequence, variables can be mapped directly between the two frameworks, and the moments ψ^L and ψ^U exactly match their counterparts in Semenova (2025).

The estimator for θ_b, for $b \in \{L, U\}$, is given by:

$$\widehat{\theta_b} = \left(\Sigma_i \left(\psi^b(G_i,\widehat{\xi_i})1_{\{p_i(X_1)\leq 1\}} + \psi^{RND}(G_i,\widehat{\xi_i})1_{\{p_i(X_1)>1\}}\right)\right) / \left(\Sigma_i \psi^R(G_i,\widehat{\xi_i})\right) \tag{15}$$

where G_i is the data for observation i, $\widehat{\xi_i}$ is the nuisance parameter for observation i, estimated on a fold that excludes observation i. ψ^b is the orthogonal counterpart of m^b, and ψ^R is the orthogonal counterpart of r. ψ^{RND} is used to obtain point estimates under conditionally random reliance and also serves as a fallback in constructing the bounds when the estimated relier share exceeds the subsequent relier share. Appendix SA5 discusses monotonicity further and shows that the estimates are insensitive to this choice. The moment for the sequential reduced form is ψ^{RF}, and the estimator for γ_b, for $b \in \{L, U\}$, follows Theorem 2.

I use 4-fold cross-fitting, estimating the nuisance functions for each fourth of the sample using the remaining three-fourths.[8] Propensity scores are estimated using logistic regressions. The key covariates are women's and their partners' ages at the time of the procedure, each included as a second-order polynomial. Because success rates vary smoothly with age, I model age parametrically

[7]For example, consider a perturbation of the propensity score at attempt j, $e^*_j(X^*_j, r) = e^*_j(X^*_j) + r(\widehat{e}^*_j(X^*_j) - e^*_j(X^*_j))$. For the orthogonal upper-bound moment, $\partial_r \, \mathbb{E}[\psi^U(G,\xi_r) \mid X^*_j]|_{r=0} = 0$ almost surely, whereas the corresponding derivative for the original moment is generally nonzero. Orthogonality is established by repeated application of Lemma 1.

[8]The main results are insensitive to this choice (see Appendix SA6).

Table 1: Orthogonal Moments

	Moment functions
$\psi^L(G,\xi^0)$	$Y(1-D^+)1_{\{Y<q(p(X_1^*),X_1^*)\}}\frac{Z_1}{e_1^*(X_1^*)} - Y(1-D^+)\Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $+q(p(X_1^*),X_1^*)\Big[\Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}(1-D^+-r_1(X_1^*))$ $-\frac{Z_1}{e_1^*(X_1^*)}p(X_1^*)(1-D^+-r^+(X_1^*)) - \frac{Z_1}{e_1^*(X_1^*)}(1-D^+)(1_{\{Y<q(p(X_1^*),X_1^*)\}}-p(X_1^*))\Big]$ $-\frac{Z_1-e_1^*(X_1^*)}{e_1^*(X_1^*)}z^{L+}(X_1^*)r_1(X_1^*) + \sum_{k=1}^{\overline{w}} 1_{\{A\geq k\}}\Pi_{j=1}^{k-1}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}\frac{e_k^*(X_k^*)-Z_k}{1-e_k^*(X_k^*)}[r_k(X_k^*)\beta_k(X_k^*)$ $+q(p(X_1^*),X_1^*)(r_1(X_1^*)-r_k(X_k^*))]$
$\psi^U(G,\xi^0)$	$Y(1-D^+)1_{\{Y>q(1-p(X_1^*),X_1^*)\}}\frac{Z_1}{e_1^*(X_1^*)} - Y(1-D^+)\Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $+q(1-p(X_1^*),X_1^*)\Big[\Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}(1-D^+-r_1(X_1^*))$ $-\frac{Z_1}{e_1^*(X_1^*)}p(X_1^*)(1-D^+-r^+(X_1^*)) - \frac{Z_1}{e_1^*(X_1^*)}(1-D^+)(1_{\{Y>q(1-p(X_1^*),X_1^*)\}}-p(X_1^*))\Big]$ $-\frac{Z_1-e_1^*(X_1^*)}{e_1^*(X_1^*)}z^{U+}(X_1^*)r_1(X_1^*) + \sum_{k=1}^{\overline{w}} 1_{\{A\geq k\}}\Pi_{j=1}^{k-1}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}\frac{e_k^*(X_k^*)-Z_k}{1-e_k^*(X_k^*)}[r_k(X_k^*)\beta_k(X_k^*)$ $+q(1-p(X_1^*),X_1^*)(r_1(X_1^*)-r_k(X_k^*))]$
$\psi^{RND}(G,\xi^0)$	$Y(1-D^+)\frac{Z_1}{e_1^*(X_1^*)}p(X_1^*) - Y(1-D^+)\Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $-\beta^+(X_1^*)\Big[\frac{Z_1}{e_1^*(X_1^*)}\frac{(1-D^+-r^+(X_1^*))}{r^+(X_1^*)}r_1(X_1^*) - \Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}(1-D^+-r_1(X_1^*))\Big]$ $-\frac{Z_1-e_1^*(X_1^*)}{e_1^*(X_1^*)}\beta^+(X_1^*)r_1(X_1^*) + \sum_{k=1}^{\overline{w}} 1_{\{A\geq k\}}\Pi_{j=1}^{k-1}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}\frac{e_k^*(X_k^*)-Z_k}{1-e_k^*(X_k^*)}\big[r_k(X_k^*)\beta_k(X_k^*)$ $+\beta^+(X_1^*)(r_1(X_1^*)-r_k(X_k^*))\big]$
$\psi^{RF}(G,\xi^0)$	$Y\frac{Z_1}{e_1^*(X_1^*)} - Y\Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $-\gamma^1(X_1^*)\frac{Z_1-e_1^*(X_1^*)}{e_1^*(X_1^*)} + \sum_{k=1}^{\overline{w}}\gamma_k^0(X_k^*)1_{\{A\geq k\}}\Pi_{j=1}^{k-1}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}\frac{e_k^*(X_k^*)-Z_k}{1-e_k^*(X_k^*)}$
$\psi^R(G,\xi^0)$	$r_1(X_1^*) + (1-D^+-r_1(X_1^*))\Pi_{j=1}^{\overline{w}}\frac{(1-Z_j)}{(1-e_j^*(X_j^*))}$ $+\Sigma_{k=1}^{\overline{w}}1_{\{A\geq k\}}\Pi_{j=1}^{k-1}\frac{(1-Z_j)}{1-e_j^*(X_j^*)}\frac{(e_k^*(X_k^*)-Z_k)}{1-e_k^*(X_k^*)}[r_1(X_1^*)-r_k(X_k^*)]$
	Nuisance functions
$\xi^0(x_1^*,\ldots,x_{\overline{w}}^*)$	$\{e_1^*(x_1^*),\ldots,e_{\overline{w}}^*(x_{\overline{w}}^*), r_1(x_1^*),\ldots,r_{\overline{w}}(x_{\overline{w}}^*), r^+(x_1^*), q(p(x_1^*),x_1^*), q(1-p(x_1^*),x_1^*),$ $\beta_1(x_1^*),\ldots,\beta_{\overline{w}}(x_{\overline{w}}^*), \beta^+(x_1^*), z^{U+}(x_1^*), z^{L+}(x_1^*), \gamma_1^0(x_1^*),\ldots,\gamma_{\overline{w}}^0(x_{\overline{w}}^*), \gamma^1(x_1^*)\}$
$r_k(x)$	$\frac{\mathbb{E}[(1-D^+)/(\Pi_{j=k+1}^{A}(1-e_j^*(X_j^*))) \mid X_k^*=x, Z_A=0]}{\mathbb{E}[\Pi_{j=k+1}^{A}(1-e_j^*(X_j^*)) \mid X_k^*=x, Z_A=0]}$
$\beta_k(x)$	$\frac{\mathbb{E}[Y/(\Pi_{j=k+1}^{A}(1-e_j^*(X_j^*))) \mid X_k^*=x, D=0]}{\mathbb{E}[\Pi_{j=k+1}^{A}(1-e_j^*(X_j^*)) \mid X_k^*=x, D=0]}$
$\beta^+(x)$	$\mathbb{E}[Y \mid X_1^*=x, Z_1=1, D^+=0]$
$z^{U+}(x)$	$\mathbb{E}[Y \mid X_1^*=x, Z_1=1, D^+=0, Y\geq q(1-p(x),x)]$
$z^{L+}(x)$	$\mathbb{E}[Y \mid X_1^*=x, Z_1=1, D^+=0, Y\leq q(p(x),x)]$
$\gamma_k^0(x)$	$\frac{\mathbb{E}[Y/(\Pi_{j=k+1}^{A}(1-e_j^*(X_j^*))) \mid X_k^*=x]}{\mathbb{E}[\Pi_{j=k+1}^{A}(1-e_j^*(X_j^*)) \mid X_k^*=x]}$
$\gamma^1(x)$	$\mathbb{E}[Y \mid X_1^*=x, Z_1=1]$
	Covariate vectors
X_k^*	$\big(X_k, 1_{\{A\geq k\}}\big)$

to avoid sparse cells at later attempts; other nuisance functions control for age nonparametrically.

Following Heiler (2024), I estimate the remaining nuisance functions using Generalized Random Forests (Athey et al., 2019).[9] For attempt k, covariates include all propensity-score covariates from attempts 1 through k and both partners' work hours and earnings in the year before the first IUI. The pre-IUI outcomes are highly predictive of post-IUI outcomes and thus key for obtaining narrow bounds. Including age at the current and previous attempts controls for the time between attempts. Confidence intervals for the partially identified parameters follow Heiler (2024), based on Stoye (2020), and account for uncertainty in the bounds when covering the underlying parameter.

I use data on the first ten procedures, which makes the bounds only negligibly wider than if additional procedures were included, while avoiding the need to estimate nuisance functions on small subsamples of women who undergo more than ten (fewer than 10% of the sample). This means that reliers are women who would not have a child if their first ten observed procedures failed. Varying the number of procedures used between 9 and 13 has little impact on the estimates (Appendix SA7).

3.6 Characterizing Potential Selection on Unobservables

This section develops a procedure for bounding the potential extent of selection on unobservables. The results characterize how such selection may bias the estimates and, in particular, how much the bias may be amplified by using the full sequence of IUI attempts rather than only the first attempt. The procedure also provides an empirical benchmark for the degree of selection consistent with the observed data, which I use in the sensitivity analysis to assess how the estimates change under selection models that cannot be rejected.

The procedure formalizes the idea that women may differ in latent health, unobserved in the data, that is correlated with both potential outcomes and IUI success rates. For simplicity, suppose women belong to one of two latent health groups, $H = 1$ and $H = 0$, although the approach extends straightforwardly to multiple types or a continuous distribution. Women with $H = 1$ succeed at the first attempt with probability p^H and women with $H = 0$ with probability p^L, where $p^H > p^L > 0$.[10]

Although the procedure can assess selection across all three potential outcomes and the entire IUI sequence, it is easiest to illustrate using women whose first IUI succeeds, since all of them realize the same potential outcome, Y^E. Let $\pi = \Pr(H = 1)$ and $p = \mathbb{E}[Z_1]$ denote the share of high-health women and the observed first-attempt success rate. Since $p = \pi p^H + (1 - \pi)p^L$, the sample share of high-health women is pinned down by the two success rates: $\pi = (p - p^L)/(p^H - p^L)$. Let $\mu_h = \mathbb{E}[Y^E \mid H = h]$ denote the average early-motherhood potential outcome in group h, and $\Delta = \mu_1 - \mu_0 > 0$ the difference between groups.

[9] The baseline specification uses 250 trees; results are insensitive to this choice (Appendix SA6). I add $u \sim U(0, 0.001)$ noise to outcomes to avoid ties in the trimming step. I estimate z_t^{U+} and z_t^{L+} by trimming above or below the estimated quantiles and estimating the resulting conditional expectations. Theoretical results for nonparametric estimation of truncated expectations exist (Olma, 2021), but no implementation is currently available.

[10] The exercise therefore abstracts from sterility and instead captures latent health factors or behaviors that may jointly affect IUI success and labor market outcomes.

The outcomes of women whose first IUI succeeds differ from those of the overall sample by

$$\mathbb{E}[Y^E \mid Z_1 = 1] - \mathbb{E}[Y^E] = \frac{\pi(1-\pi)\left(p^H - p^L\right)}{p}\Delta.^{11} \tag{16}$$

The difference is increasing in the health gap in outcomes, Δ, in the health gap in success probabilities, $p^H - p^L$, and in the extent to which both health groups are represented in the sample.

The next step is to quantify how large this imbalance could be. The first term in (16), the average outcome of women whose first IUI succeeds, is directly observed because Y^E is realized for all of these women. The second term, the average early-motherhood outcome, is not. If latent health were observed, however, it could be recovered using a standard reweighting argument. Conditional on H, first-attempt success does not select women on their potential outcomes, so

$$\mathbb{E}[Y^E \mid H = 1, Z_1 = 1]\pi + \mathbb{E}[Y^E \mid H = 0, Z_1 = 1](1-\pi) = \mathbb{E}[Y^E]. \tag{17}$$

Thus, the outcomes of successful women can be made representative of the sample by reweighting the two health groups back to their sample shares.

Neither the two type-specific success rates nor H is observed. The procedure therefore fixes a candidate pair $\left(p^H, p^L\right)$, which determines π, and assigns H among women whose first IUI succeeds so as to generate the strongest possible relationship between latent health and the outcome. By Bayes' rule, the share of successful women who must belong to the high-health group is $\pi p^H/p$. I therefore rank women whose first IUI succeeds by their realized outcome and assign the fraction $\pi p^H/p$ with the highest outcomes to $H = 1$, with the remainder assigned to $H = 0$. For a given pair $\left(p^H, p^L\right)$, this is the most adverse assignment because it maximizes the relationship between latent health and Y^E. Evaluating (17) under this assignment yields the bias implied by this selection model and, equivalently, the largest proportional gap $\mathbb{E}[Y^E \mid Z_1 = 1]/\,\mathbb{E}[Y^E]$ between the outcomes of first-attempt successes and the mean.

The second part of the procedure asks whether a candidate selection model is consistent with the observed data. The key observation is that the same reweighting argument can be applied to characteristics measured before IUI. Let X denote a pre-IUI characteristic, or a vector of characteristics. Under the candidate model,

$$\mathbb{E}[X \mid H = 1, Z_1 = 1]\pi + \mathbb{E}[X \mid H = 0, Z_1 = 1](1-\pi) = \mathbb{E}[X]. \tag{18}$$

Unlike (17), the right hand side of (18) is directly observed. The candidate pair $\left(p^H, p^L\right)$ and the associated worst-case assignment of H therefore imply a testable restriction on pre-IUI characteristics. If the reweighted characteristics of women whose first IUI succeeds differ significantly from those of the full sample, the corresponding selection model is inconsistent with the observed data.

Notably, failure to reject (18) does not provide support for the selection model. Even large imbalance in Y^E may generate little or no imbalance in X, failing to reject (18). This may occur, for instance, when X is only weakly related to Y^E. I therefore use the largest degree of selection

[11] By Bayes' rule, $\Pr(H = 1 \mid Z_1 = 1) = \pi p^H/p$ and $\Pr(H = 0 \mid Z_1 = 1) = (1-\pi)p^L/p$, so $\mathbb{E}[Y^E \mid Z_1 = 1] = [\pi p^H \mu_1 + (1-\pi)p^L \mu_0]/p$. The result follows by subtracting the corresponding average outcome, $\mathbb{E}[Y^E]$.

among the models that cannot be rejected as a conservative sensitivity benchmark.

The framework also clarifies how much selection can be amplified by focusing on women who experience a sequence of unsuccessful IUIs. A potential concern is that even modest selection at each attempt could accumulate across repeated failures and generate substantial imbalance among women who remain in the failure group.

By an argument symmetric to (16), the potential outcomes of women whose first IUI fails differ from the average by

$$\mathbb{E}[Y^E] - \mathbb{E}[Y^E \mid Z_1 = 0] = \frac{\pi(1-\pi)\left(p^H - p^L\right)}{1-p}\Delta. \tag{19}$$

Taking the ratio of imbalance in the failure group (19) and in the success group (16) gives

$$\frac{\mathbb{E}[Y^E] - \mathbb{E}[Y^E \mid Z_1 = 0]}{\mathbb{E}[Y^E \mid Z_1 = 1] - \mathbb{E}[Y^E]} = \frac{p}{1-p}. \tag{20}$$

The ratio in (20) has two important implications. First, because it depends only on the observed success probability p, and not on π, Δ, p^H, or p^L, selection among failures follows directly from selection among successes and the observed success rate. This makes it possible to quantify imbalance in their potential early-motherhood outcomes without observing those outcomes directly.

Second, because the ratio is increasing in p, it shows that when the success rate is low, even a strong relationship between success probabilities and potential outcomes generates only limited selection among women whose IUI fails. The intuition is straightforward: when success is uncommon, most women remain in the failure group, so its average potential outcome cannot move far from the population mean.

Because the per-procedure IUI success rate remains around 10% throughout the sequence, most of any imbalance is concentrated among women whose attempt succeeds. For example, if the relevant potential outcomes of women whose first IUI succeeds are 1% above the mean, then the corresponding potential outcomes of women whose first IUI fails are only 0.11% below the mean.

This substantially limits the extent to which selection can accumulate across a sequence of unsuccessful attempts. If the same imbalance—women whose IUI succeeds having relevant potential outcomes 1% above the mean—arose at each of ten consecutive attempts, the maximum number considered in the analysis, the cumulative selection factor among women experiencing ten failures would be $(1 - 0.01/9)^{10} \approx 0.989$. Thus, even after ten failures, the group's average potential outcomes would lie only about 1.1% below the mean, barely more than the 1% imbalance among women whose first attempt succeeded.

These calculations also provide a natural way to correct the main estimates for potential selection. Suppose the sensitivity exercise implies that the ratio of the potential outcomes of women whose first IUI succeeds to the mean is x. Their realized outcomes can then be adjusted by dividing by x. Similarly, if the ratio for women who experience a particular sequence of unsuccessful attempts to the corresponding type's mean is y, their outcomes can be divided by y. For repeated failures, the relevant ratio can be constructed by multiplying the attempt-specific failure-selection ratios implied by (20). The main analysis can then be repeated using the adjusted outcomes.

The procedure cannot rule out every form of selection on unobservables, since it imposes a particular model of latent heterogeneity. It nevertheless provides a transparent benchmark for how much selection is consistent with the data and how strongly it could accumulate over repeated attempts. In the sensitivity analysis, the main estimates change little even under the most extreme models I cannot reject.

Appendix SA8 explains how the procedure incorporates covariates. In short, I consider selection on potential outcomes conditional on the baseline propensity scores, thereby netting out selection already captured by the characteristics entering these scores, most notably the woman's and her partner's ages. I then test balance in pre-IUI labor market characteristics and quantify imbalances in post-IUI outcomes after additionally controlling for these characteristics. These two steps mirror the baseline balance tests and the specification used for the main estimates, respectively.

4 Institutions, Intrauterine Insemination, and Data

Section 4.1 describes Dutch family policies and the labor market context. Section 4.2 discusses IUI. Section 4.3 describes the data. Section 4.4 presents sample descriptives, compares the IUI and representative samples, and characterizes pre-IUI differences between reliers and non-reliers. Section 4.5 documents procedure profiles and pre-IUI differences across women with different levels of willingness. Section 4.6 provides empirical support for the unconfoundedness assumption.

4.1 Family Policies in the Netherlands

Dutch women are entitled to 4 to 6 weeks of pregnancy leave before the due date and at least 10 weeks of maternity leave following childbirth, totaling a minimum of 16 weeks. In the case of multiple births, the total entitlement increases to 20 weeks. During this period, mothers receive full wage replacement from the unemployment insurance agency, subject to a daily maximum. In the sample period, fathers are entitled to one week of fully paid leave within the first four weeks after childbirth, financed by the employer.

Children can enroll in private daycare at three months old. In 2022, 72% of children under two attended formal child care, averaging 20 hours per week (OECD, 2023a). After turning four and starting elementary school, they become eligible for out-of-school care. In 2023, families using child care paid an average of 8,950 euros, of which 64% was reimbursed by the government, resulting in a net cost equivalent to 10% of median disposable household income.[12]

The Netherlands has average family policies compared to other OECD countries. Paternity and maternity leave durations are slightly below the OECD averages of 2.5 and 21 weeks, respectively (OECD, 2023c). While formal child care enrollment for children under two is the highest among OECD countries, average time spent in care is the lowest (OECD, 2023a). After age four, enrollment rates and out-of-school care hours align with OECD averages (OECD, 2022).

[12] www.cbs.nl/nl-nl/nieuws/2024/30/ouders-betaalden-gemiddeld-3-210-euro-aan-kinderopvang-in-2023, longreads.cbs.nl/materiele-welvaart-in-nederland-2024/inkomen-van-huishoudens/.

While employment rates for mothers, fathers, and non-parents in the Netherlands exceed their respective OECD averages, part-time work is far more common, making average hours worked comparable to the OECD average (OECD, 2023b). In 2021, the maternal employment rate was 80%, compared to the OECD average of 71%. However, in 2023, 52% of women and 18% of men worked part-time (less than 30 hours per week), more than twice the respective OECD averages (OECD, 2023d). Among two-parent families, only 14% had both parents working full-time, 52% had a full-time working father and a part-time working mother, and 12% had both parents working part-time.[13]

4.2 Assisted Conception Procedures

My analysis focuses on Dutch couples who undergo IUI for their first child. As in most countries, the procedure is usually a first-line treatment for couples who are unable to conceive naturally within a year, especially in cases of male-factor or unexplained infertility. Prior to the procedure, women typically undergo cycle tracking and may receive hormonal stimulation to enhance egg production. During IUI, sperm is injected directly into the uterus via a catheter, mimicking natural conception by facilitating fertilization within the body. With a success rate of about 10% per attempt, the procedure is minimally invasive, typically lasting five minutes and causing little or no discomfort.

Some couples who do not succeed with IUI eventually proceed to IVF, which has previously been used to study the career impact of parenthood in Denmark, Norway, and Sweden (Lundborg et al., 2017, 2024; Gallen et al., 2024; Bensnes et al., 2025). IVF is a surgical procedure in which eggs are retrieved through the vaginal wall, fertilized in the laboratory, and transferred as embryos into the uterus. It is more invasive than IUI, performed under sedation or anesthesia, and in most countries it is used for women with severe fertility problems, such as tubal damage or advanced endometriosis. It also has a higher success rate than IUI, of about 25% per embryo transfer. Because IVF also generates as-good-as-random fertility variation, I include eventual IVF attempts in the analysis. To account for differences in success rates between IUI and IVF, all terms in each propensity score are interacted with a procedure-type indicator.

In the Netherlands, couples without a specific infertility diagnosis are typically required to undergo six IUI cycles before becoming eligible for IVF. Compulsory health insurance covers an unlimited number of IUI cycles and up to three IVF procedures. In 2022, each additional IVF cycle cost approximately 4,000 euros, but because multiple embryos can be frozen per cycle, subsequent transfers could cost 1,000 euros or less.

4.3 Data

I use administrative data from Statistics Netherlands, covering all residents. Medical records from 2012–2017 are drawn from the Diagnosis-Treatment Combination system, which Dutch hospitals are required to maintain. The main variables are the procedure type—IUI or IVF—and the date of sperm or embryo insertion.

[13] `www.cbs.nl/en-gb/news/2024/10/fewer-and-fewer-families-in-which-only-the-father-works`

Importantly, these records indicate procedures that were actually carried out: a cycle that a woman cancels or chooses not to undergo does not appear. An attempt is therefore defined as an actual insertion, corresponds directly to the conceptual experiment, and the key variation is whether a completed insertion results in a birth. Procedure success is defined as having a child born within ten months of insertion with no subsequent procedures, a definition validated against medical records by Lundborg et al. (2017).

The medical registry also contains information on the treatment protocol surrounding each attempt, including semen testing and preparation, ovulation monitoring, and hormone monitoring. I use these variables as additional controls in sensitivity analyses.

Labor market data span 2011–2024 and include annual work hours and gross earnings derived from tax records. Gross earnings are expressed in 2024 euros using the consumer price index. Annual work hours and gross earnings are censored at zero and the 99th percentile of their respective population distributions. Reported work hours include maternity leave, and earnings include maternity pay. This means that earnings are closer to labor-related income, while hours need not map one-to-one to actual time spent working (see Adams et al., 2024, for a detailed discussion). In Appendix SA9, I repeat the main analyses using adjusted hours that account for the maximum potential leave duration. This adjustment affects estimates only in the first year after childbirth. I also use several demographic variables, including an indicator for higher education attainment, number of children, year and month of birth, marital status, and cohabitation status. Fertility is measured based on legal parenthood and is independent of coresidence. Adoptions in the Netherlands are extremely rare, with fewer than 40 occurring annually.

My main sample includes 12,892 Dutch-born women who underwent IUI to conceive their first child and were cohabiting with or in a registered partnership with a Dutch-born man at the time of the procedure. Throughout the analysis, I refer to these men as the partners. I address potential complications related to separation in Appendix SA4. Following Lundborg et al. (2017), I exclude women whose first observed procedure occurred in the first data year to reduce the risk of including women with unobserved prior procedures, and those whose first observed procedure occurred in the last data year to avoid misclassifying failed procedures as successful because subsequent procedures are unobserved. Thus, the sample consists of women whose first observed fertility procedure is an IUI between 2013 and 2016.

For comparison, I use 84,337 Dutch-born women who conceived their first child between 2013 and 2016 without prior assisted conception procedures and were cohabiting with or in a registered partnership with the child's Dutch-born father. To assess the role of the intensive fertility margin, I also use a sample of 11,426 women who underwent IUI or IVF to conceive a second child, imposing the same restrictions as in the main sample. All analyses use the full samples, with hours and earnings set to zero for individuals not in paid employment.

4.4 Sample Descriptives and Representativeness

Table 2 compares pre-IUI characteristics of couples whose first procedure succeeded (column 1) and failed (column 2). Because age strongly predicts IUI success, these comparisons are descriptive

Table 2: Pre-Parenthood Descriptives by First IUI Outcome, Latent Type, and Sample

	First IUI outcome			Latent type			Representative sample		
	Success (1)	Fail (2)	Dif. (1)-(2)	Relier (4)	Non-rel. (5)	Dif. (4)-(5)	Rep. (7)	vs relier (7)-(4)	vs non-rel. (7)-(5)
Panel A: Woman									
Work	0.910	0.916	-0.006	0.891	0.928	-0.037	0.935	0.043	0.007
	[0.287]	[0.278]	(0.008)	(0.008)	(0.005)	(0.010)	[0.248]	(0.004)	(0.003)
Hours	1299.447	1296.947	2.500	1227.727	1314.270	-86.543	1307.439	79.712	-6.831
	[548.456]	[556.609]	(15.872)	(15.532)	(11.627)	(18.933)	[543.748]	(7.758)	(6.632)
Earn. 1000s EUR	52.651	52.111	0.539	49.257	52.033	-2.776	46.264	-2.993	-5.768
	[30.414]	[30.804]	(0.879)	(0.771)	(0.559)	(0.985)	[27.116]	(0.388)	(0.333)
Bachelor deg.	0.517	0.502	0.016	0.470	0.526	-0.056	0.533	0.063	0.007
	[0.500]	[0.500]	(0.014)	(0.011)	(0.009)	(0.017)	[0.499]	(0.007)	(0.006)
Age	31.351	32.061	-0.710	33.078	31.096	1.982	28.871	-4.207	-2.225
	[3.883]	[4.264]	(0.121)	(0.099)	(0.059)	(0.135)	[3.908]	(0.056)	(0.048)
Panel B: Partner									
Work	0.891	0.884	0.007	0.866	0.889	-0.023	0.894	0.028	0.004
	[0.311]	[0.320]	(0.009)	(0.008)	(0.007)	(0.011)	[0.308]	(0.004)	(0.004)
Hours	1501.993	1485.385	16.608	1444.074	1502.663	-58.588	1484.782	40.707	-17.881
	[631.601]	[643.639]	(18.346)	(15.694)	(13.059)	(22.055)	[639.634]	(9.096)	(7.799)
Earn. 1000s EUR	68.391	69.063	-0.672	66.739	69.290	-2.551	61.744	-4.994	-7.545
	[35.450]	[36.557]	(1.041)	(0.891)	(0.701)	(1.174)	[32.606]	(0.466)	(0.401)
Bachelor deg.	0.416	0.406	0.010	0.351	0.439	-0.088	0.431	0.079	-0.009
	[0.493]	[0.491]	(0.014)	(0.011)	(0.009)	(0.017)	[0.495]	(0.007)	(0.006)
Age	34.059	34.855	-0.796	36.056	33.868	2.188	31.429	-4.628	-2.439
	[4.988]	[5.493]	(0.155)	(0.128)	(0.086)	(0.178)	[4.809]	(0.069)	(0.059)
Observations	1,372	11,520		2,348	3,141		84,337		

Note: Success – average among women whose first IUI succeeded; *Fail* – average among women whose first IUI failed; *Relier* – average among women who would remain childless if all IUIs failed; *Non-rel.* – average among women who would have a child even if all IUIs failed; *Rep.* – average in the representative sample of women who conceived their first child without assisted conception procedures. Relier and non-relier averages are estimated using the baseline specification. Earnings, hours, and employment are measured in the year preceding the first IUI for the IUI sample and in the year preceding conception for the representative sample; age and education are measured in the reference year. *Bachelor deg.* – indicator for completing a bachelor's degree. Standard deviations are in brackets. Standard errors are in parentheses; relier and non-relier statistics use bootstrap standard errors based on 300 replications.

and not intended to test unconfoundedness. Despite this, in the year before the procedure, the groups are similar in earnings, employment rates, and education. The only notable difference is age: both women and their partners whose first IUI succeeded were around nine months younger, consistent with age being the key determinant of procedure success.

Table 2 also compares pre-IUI characteristics of reliers (column 4) and non-reliers (column 5), estimated using the baseline specification. Non-relier women have 5–6% higher employment, earnings, and hours worked and are more likely to have a bachelor's degree, indicating positive selection on labor market outcomes among women who conceive after IUI failure. Reliers are also about two years older on average, consistent with age being an important determinant of natural conception success. Given these baseline differences, comparisons between relier and non-relier effects should be interpreted cautiously; when making such comparisons in the discussion, I therefore compare effects in relative terms.

Column (7) reports pre-pregnancy characteristics for the representative sample, while the final two columns report differences relative to reliers and non-reliers. The pattern is mixed: reliers have higher employment and hours worked, while these outcomes are similar for non-reliers; both groups have higher earnings, with a larger difference for non-reliers. Women in the IUI sample are also about three years older, which can largely be explained by eligibility requiring at least a year of unsuccessful natural conception attempts and treatment typically beginning about six months later, as well as by the representative sample additionally including unplanned births, which tend to occur at younger ages.

Six years after the first birth, women whose first IUI succeeded have about as many children as mothers in the representative sample: 1.88 compared with 1.96. They are, however, considerably more likely to have a multiple birth at their first delivery (7.1% vs. 1.4%), which could limit the external validity of the estimated effects. In Section 5.5, I estimate the direct effect of a multiple birth in both the IUI and representative samples and re-estimate the main specification excluding couples with a multiple birth; both exercises are justified if multiple births are as good as random. A multiple birth raises the number of children six years later by about 0.45 in both samples, yet has little effect on women's earnings or work hours, and the main estimates are largely unchanged when multiple births are excluded. Since a multiple birth represents a large shift in the intensive fertility margin, these results also suggest that this margin plays a limited role in the effects estimated here.

Beyond observable characteristics, IUI mothers may differ systematically in unobservables, limiting the generalizability of the estimates. For example, they may have a stronger desire for children and therefore respond differently to motherhood. Since only about 5% of Dutch mothers undergo IUI before their first child, such selection could in principle be substantial.

However, conditional on age, fertility problems are largely unpredictable. IUI mothers can therefore be viewed as drawn from a broader group of women with similar preferences who were trying to conceive and would have initiated treatment had they not conceived earlier. Within this group, reaching IUI depends on whether natural conception occurs first, which may itself be largely a matter of chance, making the IUI sample representative of this broader group.

The size of this broader group can be approximated using medical evidence on fecundity and the institutional setting. Medical guidelines typically require couples to try to conceive naturally for at least a year before becoming eligible for treatment, and IUI usually begins about six months later. With a mean IUI initiation age of 32, most women therefore began trying by about age 30.5. At this age, about 80% conceive naturally within 1.5 years (Leridon, 2004). If natural conception within this group is largely a matter of chance, each woman who reaches IUI therefore corresponds to about four otherwise similar women who conceived naturally beforehand. Combined with the 5% of mothers who undergo IUI, this implies that the broader group accounts for roughly a quarter of all mothers.

This approximation is likely conservative because many women who would eventually use IUI may conceive well before the age at which they would otherwise begin treatment, making their cumulative probability of natural conception higher than 80%. The magnitude is also plausible given that most couples who want children and fail to conceive naturally would eventually seek medical assistance, with IUI among the first treatments they receive. The estimates may therefore be relevant for a substantially broader population than IUI mothers alone.

4.5 Procedure Profiles

Figure 2 describes women's procedure histories. Panel A shows the distribution of the number of procedures women undergo after the first failure and the number they would undergo if the first ten failed, estimated using the baseline specification. Women undergo on average 4.1 additional procedures, while average willingness is 6.8. Among women whose first IUI fails, 43% eventually undergo IVF, which accounts for 20% of all procedures. Panel B reports the relier share six years after the first IUI by willingness and shows little relationship between the two, although this relationship is irrelevant for identification.

Panels C and D report average work hours and earnings in the year before the first IUI by willingness, relative to the sample mean, estimated using the baseline specification. Both increase systematically with willingness: women who would undergo more procedures in case of failure have higher pre-IUI labor market outcomes. If this pattern extends to post-IUI outcomes, women who eventually succeed through IUI will be positively selected, because those with greater willingness undergo more procedures and are therefore more likely to experience at least one success. Using the outcome of the last IUI a woman undergoes as an instrument should therefore yield smaller estimated career costs of motherhood than conventional IV. Whether these estimates are also smaller than the motherhood effect itself additionally depends on the bias in conventional IV estimates due to timing effects.

Panels E and F report the distribution of gaps between consecutive procedures and the elapsed time from the first to the tenth procedure among women who reach the tenth, using centered three-month bins. The modal gap between consecutive procedures is one month throughout the sequence, while the average is around two months. The average time from the first to the tenth procedure is 18.4 months. Thus, women who continue treatment generally undergo subsequent procedures in quick succession. Notably, procedure spacing does not itself characterize the timing effects. These depend on the time from the first procedure to conception outside treatment after the failed sequence, which may occur well after the last procedure. Section 5.1 quantifies this delay.

4.6 Unconfoundedness

Since women have limited direct control over IUI outcomes (see Section 3.1 for discussion), the main threat to unconfoundedness is that success depends on latent health factors that also affect labor market outcomes. If such factors also affect pre-IUI outcomes, this can be assessed by comparing pre-IUI characteristics between women whose procedures succeed and fail.

I make these comparisons separately for each procedure in the sequence, using only women who reach that procedure. This holds continuation decisions—which may themselves be endogenous—fixed and directly matches the identifying assumption, which conditions on reaching a given attempt. The comparisons use inverse-probability weights from the baseline specification that account for age and education, following Lundborg et al. (2017); results are unchanged when education is

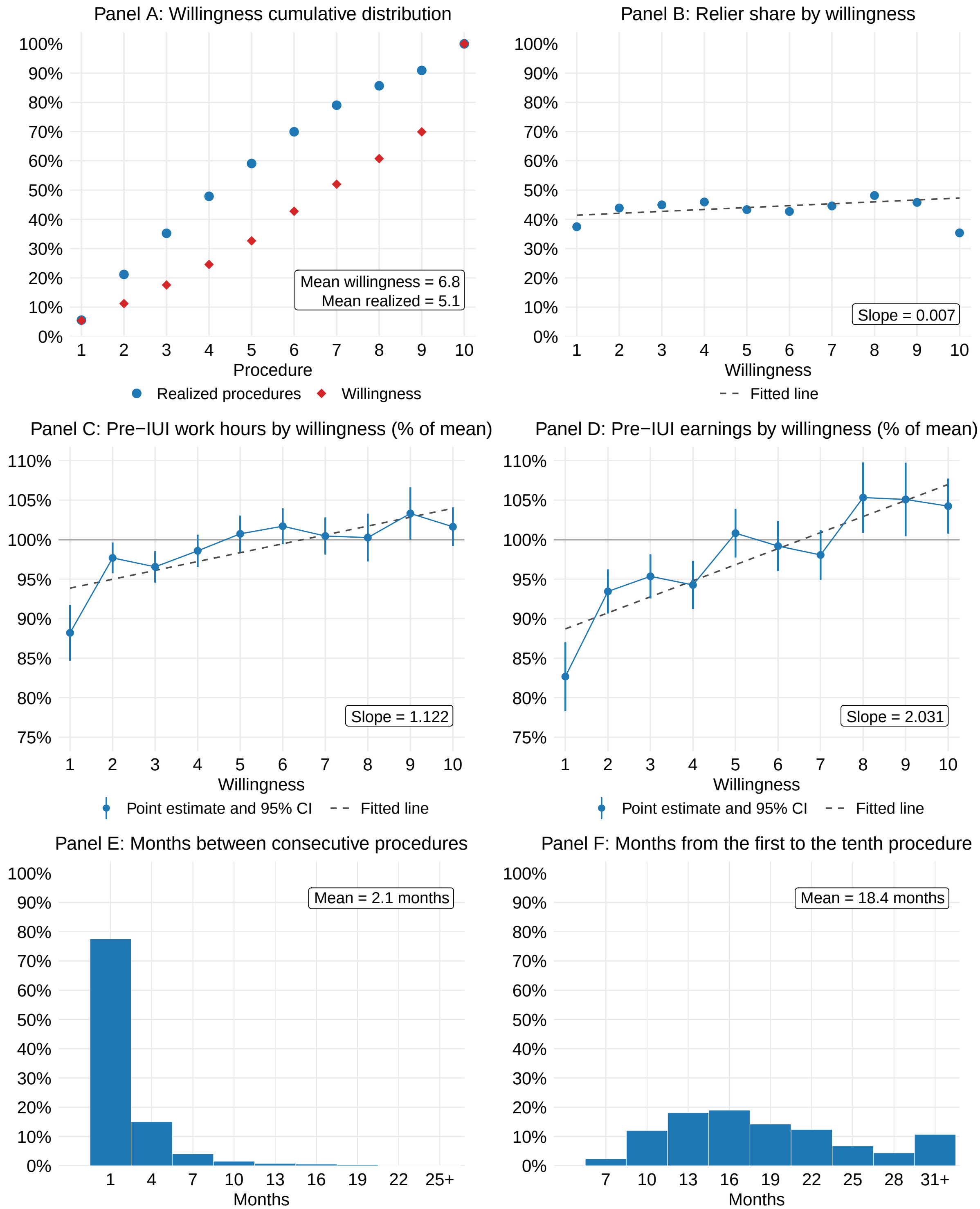

Figure 2: Willingness, Timing and Pre-IUI Characteristics

Note: Panel A: cumulative distribution of the number of assisted conception procedures pursued after the first failure, and of the number women would pursue if the first ten procedures failed. Panel B: relier shares by willingness. Panels C and D: women's work hours and earnings in the year before the first procedure, by willingness, expressed as a percentage of the overall mean; bootstrap standard errors are based on 300 replications. Panel E: distribution of the number of months between consecutive procedures. Panel F: number of months between the first and tenth procedures. Gaps are rounded down to whole months and grouped into three-month bins labeled by their midpoint. Dashed lines in Panels B, C, and D are least-squares fits. Panels A–D use the baseline specification.

Table 3: Balance Across Procedures

	Procedure number									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Woman										
First procedure										
Work	-0.011	0.008	-0.013	0.008	0.001	0.013	-0.003	0.006	0.011	0.019
	(0.009)	(0.009)	(0.010)	(0.010)	(0.011)	(0.012)	(0.017)	(0.014)	(0.016)	(0.020)
Hours	0.923	19.299	-17.682	27.530	24.955	50.190	11.717	-4.531	41.572	76.042
	(16.286)	(17.802)	(19.795)	(20.340)	(22.272)	(25.782)	(30.737)	(30.776)	(36.374)	(42.895)
Earn. 1000s EUR	1.850	1.234	0.613	0.287	2.796	2.071	-0.067	0.411	0.280	3.203
	(0.956)	(1.082)	(1.105)	(1.257)	(1.384)	(1.516)	(1.561)	(1.843)	(2.057)	(2.869)
Current procedure										
Work		0.009	-0.011	0.011	-0.000	0.021	0.005	-0.006	-0.011	0.040
		(0.009)	(0.010)	(0.010)	(0.012)	(0.013)	(0.017)	(0.017)	(0.022)	(0.022)
Hours		16.725	-19.258	14.255	18.232	58.119	14.366	-42.638	-10.830	97.769
		(17.960)	(19.924)	(21.111)	(23.176)	(25.979)	(30.531)	(31.920)	(41.028)	(47.537)
Earn. 1000s EUR		1.238	0.430	0.040	2.923	2.121	0.758	-0.667	-1.490	3.739
		(1.092)	(1.113)	(1.304)	(1.447)	(1.540)	(1.637)	(1.906)	(2.183)	(3.025)
Panel B: Partner										
First procedure										
Work	0.001	0.009	0.020	0.007	0.018	-0.002	-0.006	-0.023	0.002	0.034
	(0.010)	(0.010)	(0.010)	(0.012)	(0.012)	(0.016)	(0.016)	(0.022)	(0.022)	(0.024)
Hours	2.719	21.260	25.860	23.631	34.325	-2.022	-12.127	-45.307	-10.891	21.889
	(19.105)	(21.103)	(21.312)	(24.084)	(26.310)	(31.967)	(32.792)	(41.345)	(46.126)	(51.120)
Earn. 1000s EUR	0.028	-0.764	0.024	1.808	2.148	-1.705	0.553	1.665	-1.988	1.128
	(1.105)	(1.221)	(1.262)	(1.470)	(1.598)	(1.820)	(1.877)	(2.401)	(2.630)	(3.574)
Current procedure										
Work		0.008	0.015	0.013	0.022	0.011	-0.016	-0.030	-0.015	0.061
		(0.010)	(0.011)	(0.012)	(0.013)	(0.016)	(0.017)	(0.022)	(0.024)	(0.022)
Hours		20.906	17.521	16.404	50.694	31.633	-24.683	-25.072	-51.934	44.335
		(21.151)	(21.644)	(24.375)	(25.900)	(31.820)	(33.496)	(41.788)	(48.426)	(51.225)
Earn. 1000s EUR		-0.568	0.235	1.110	1.856	-1.954	1.612	2.524	-1.506	1.761
		(1.222)	(1.268)	(1.490)	(1.577)	(1.783)	(2.003)	(2.534)	(2.737)	(3.707)
Observations	12,892	10,880	9,082	7,462	6,006	4,713	3,463	2,418	1,652	1,048
Joint p-val.	0.415	0.932	0.508	0.297	0.287	0.088	0.747	0.335	0.344	0.546

Note: Column (k) reports the difference in average characteristics between women whose k-th procedure succeeded and those for whom it failed, among women who underwent a k-th procedure, adjusted for age and education using inverse probability weights from the baseline specification. *First procedure* measures characteristics in the year preceding the first procedure and is therefore the same variable in every column; *Current procedure* measures them in the year preceding the procedure of that column. Earn. – earnings. Standard errors in parentheses.

excluded. Since subsequent procedures also include IVF, I additionally control for each partner's age interacted with treatment type. This requires procedure success to be as good as random only among couples of the same ages undergoing the same treatment at the same point in the sequence, allowing selection into IUI or IVF to depend on women's types and potential outcomes.

Table 3 reports the results. Across the sequence, the remaining differences in pre-IUI employment, work hours, and earnings are negligible. The table also reports differences in outcomes measured in the year before each procedure. These measures are often identical for the first few procedures, which typically occur only a few months apart, but increasingly differ later in the sequence. These differences are also small. Notably, because women for whom conception is im-

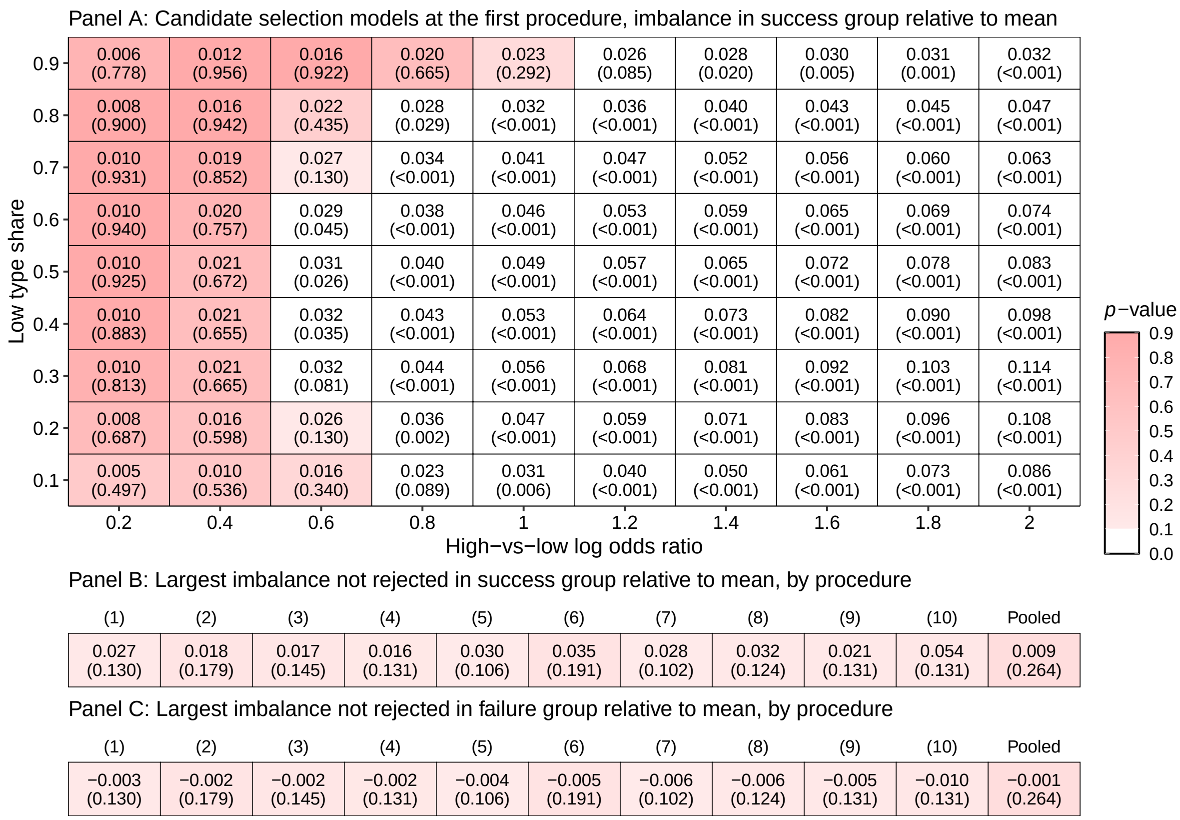

Figure 3: Selection on Unobservables in Potential Earnings

Note: Panel A: each cell represents a candidate model of selection on unobservables at the first procedure; Section 3.6 describes the test. The horizontal axis reports the log odds ratio in success propensity between the two unobserved types, beyond what is captured by the baseline specification; the vertical axis reports the share of the low type. The top number gives the implied difference in potential outcomes under IUI success between women whose procedure succeeds and the sample mean, relative to that mean. The outcome is cumulative earnings over the first six years following the first IUI. The number in parentheses is the p-value from a joint test of balance in mean pre-IUI employment, earnings, and hours for the woman and her partner. Cells not rejected at the 10% level are shaded. *Panel B*: largest non-rejected imbalance at each procedure, with the corresponding p-value below. *Panel C*: corresponding imbalance among women whose procedure fails. In Panels B and C, the final column pools procedures 1–10; the pooled p-value accounts for within-woman correlation across attempts.

possible are necessarily in the failure group, these comparisons also suggest that such underlying fertility conditions are not systematically related to labor-market outcomes. Overall, these results support conditional local sequential unconfoundedness.

The remaining concern is that latent health may affect post-IUI labor market outcomes without leaving a trace in pre-IUI employment, hours, or earnings. I use the approach introduced in Section 3.6 to assess how large such selection might be. Panel A in Figure 3 applies the exercise to the first procedure, using cumulative earnings over the full observation window under first-attempt success—the early-motherhood potential outcome. The procedure assumes two unobserved types with high and low IUI success propensities. It assigns types among women whose first IUI succeeds by sorting on the potential outcome so as to maximize differences between types. The type-specific probabilities are then used to reweight women whose first procedure succeeded, recovering both the mean potential outcome and the mean pre-IUI outcomes of all who entered the procedure. The first is compared with the average outcome observed among successes, which gives the selection the model implies; the second with the average pre-IUI outcomes observed for everyone, which tests whether the model is consistent with the data.

The horizontal axis fixes the log odds ratio in success propensity between the two types beyond what is captured by the baseline specification.[14] The vertical axis fixes the share of the low type among women whose first IUI succeeds. The top number in each cell reports the largest implied difference in potential outcomes between women whose first IUI succeeds and the sample mean, relative to that mean; the number below reports the p-value from a joint test of whether the selection model is consistent with mean pre-IUI employment, earnings, and hours for the woman and her partner.

A high top number indicates sizable selection, while a low p-value indicates that the model can be rejected. The relevant benchmark is therefore the largest imbalance among models that cannot be rejected. Extreme models are rejected: log odds ratios above one are rejected at every type share. Models with less selection generally survive but imply little imbalance; at a log odds ratio of 0.2, the implied difference in potential outcomes is at most 1.0%. At the 10% level, the largest non-rejected imbalance is 2.7%. This means that women whose first IUI succeeds have potential outcomes 2.7% above the mean. Since 10% of first procedures succeed, this implies that women whose first IUI fails then have potential outcomes at most 0.3% below the mean (see Section 3.6 for the formula). As emphasized earlier, failure to reject does not imply such selection; it provides a conservative benchmark for the largest selection consistent with the observable balance tests.

Panel B reports, for each procedure, the largest non-rejected imbalance among women whose procedure succeeds relative to the mean among all women who reach that procedure; Appendix SA8 reports the full grids by type share and log odds ratio. The largest imbalance is similar across the first nine procedures and increases at the tenth as the sample shrinks. Because the analysis of subsequent procedures uses outcomes only among women whose procedures fail, the more relevant quantity is their imbalance relative to the corresponding sample mean, reported in Panel C. These imbalances are substantially smaller because most procedures fail, so the failure-group mean remains close to the overall mean. Across procedures, their potential outcomes lie at most 1% below the corresponding sample mean. The pooled specification, which tests balance jointly across all procedures, implies an imbalance below 0.1%. Applied across ten failures, the first-procedure and pooled benchmarks accumulate to roughly 3% and 1%, respectively. Section 5.6 uses these benchmarks to assess the sensitivity of the main estimates to non-rejected selection models.

Lastly, I investigate whether women who proceed to later procedures differ systematically in success propensity. For instance, if women who are generally healthier are both more likely to succeed at each procedure and more likely to continue treatment, success rates might be expected

[14] I use log odds because baseline propensities are heterogeneous and level shifts can imply probabilities outside the unit interval. To translate the magnitudes into levels, consider a baseline success probability of 10% and equally common high and low types. A log odds ratio of 0.2 corresponds to success probabilities of about 11% and 9% for the high and low types, respectively; log odds ratios of one and two correspond to about 14% and 6%, and 17% and 3%.

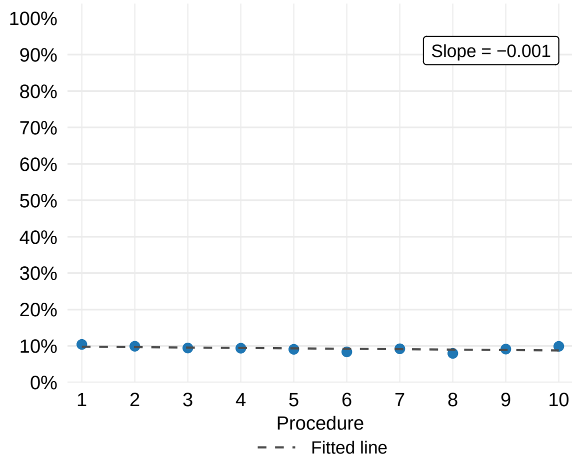

Figure 4: Success Rate by Procedure

Note: IUI success rates by procedure among women reaching each procedure, estimated using the baseline specification and holding covariates at their mean values at the first procedure. The dashed line shows the least-squares fit.

to change across the sequence as the sample becomes increasingly selected. Because success rates decline with age, I estimate adjusted rates holding age and other covariates fixed at their first-procedure means. Figure 4 shows little variation across procedures, providing little evidence of such sorting. Any remaining sorting would threaten identification only if this variation in success propensity were also related to potential labor market outcomes, for which the balance and selection-test results above leave little scope. Taken together, the evidence supports conditional local sequential unconfoundedness.

5 Results

Section 5.1 presents the effects on fertility. Section 5.2 quantifies the effects of parenthood and their contribution to within-couple gender gaps, and Section 5.3 the effects of parenthood timing. Section 5.4 examines how both sets of effects vary across women, and Section 5.5 investigates the role of the intensive fertility margin. Section 5.6 reports sensitivity analyses, and Section 5.7 compares the estimates with those obtained from conventional methods.

5.1 Fertility

Panel A in Figure 5 reports the share of mothers under first-IUI success and entire sequence failure, by years since the first IUI. Because the identified sets for fertility outcomes are extremely narrow, I refer to point estimates under conditionally random reliance throughout the discussion. Under first-IUI success, only around one third have given birth by the end of year 0, since not all successful conceptions result in a birth within the same calendar year. From year 1 onward, the share is mechanically 100%. Under failure, the share rises gradually from zero, begins to plateau around year 4, and reaches 59% by year 6. This 59% is the non-relier share: the remaining 41%, who are still childless by year 6, are reliers over the observation window.

Panel B reports the timing of motherhood among non-reliers. Under first-IUI failure, around 12% become mothers in the following year, 29% in year 2, 28% in year 3, 16% in year 4, and the remaining 15% in years 5 and 6. IUI failure therefore delays motherhood by 2.3 years on average among non-reliers. The estimated timing effects on labor market outcomes thus capture

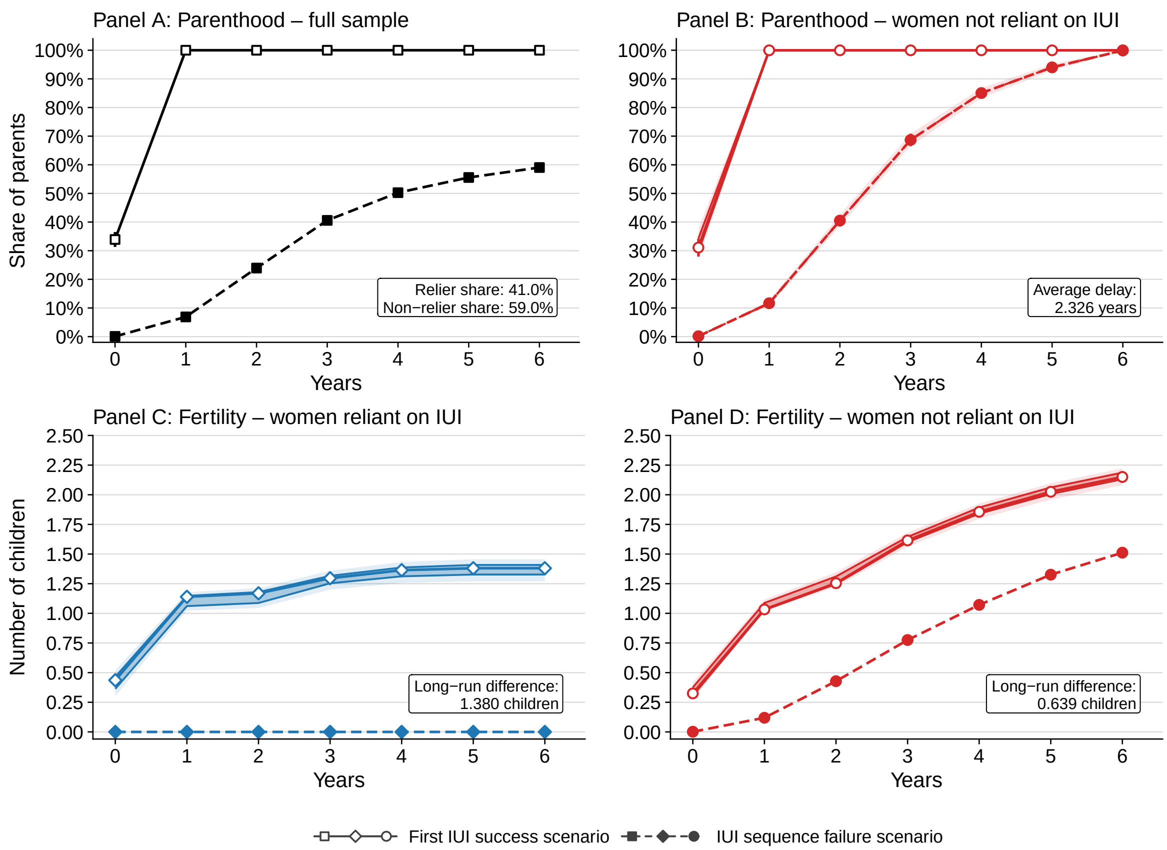

Figure 5: Parenthood and Fertility under Alternative IUI Sequence Scenarios

Note: Outcomes by year relative to the first intrauterine insemination. Panels A and B report the share of women who are parents under each scenario, with Panel B restricted to women who would become parents within the window irrespective of IUI success. Panels C and D report the corresponding number of children born among women who rely on IUI to become parents and women who do not, respectively. The translucent band gives the estimated identified set for the partially identified parameters, and the fainter outer band the corresponding 95% confidence interval for the parameters; markers give point estimates with their 95% confidence intervals.

the consequences of an average delay of 2.3 years.

Panels C and D turn to the intensive fertility margin. Panel C shows that, among reliers, average fertility under first-IUI success is slightly above one child in year 1 and rises gradually to around 1.4 by year 6. Panel D reports the corresponding outcomes for non-reliers. Under success, average fertility rises from slightly above one child in year 1 to around 2.15 by year 6; under failure, it reaches around 1.5. The resulting gap of about 0.64 children in year 6 implies that the estimated timing effects reflect not only later motherhood but also differences in total fertility over the observation window. I investigate the role of this intensive fertility margin in Section 5.5.

5.2 Effects of Parenthood

Panels A and B in Figure 6 present the effects of parenthood on women's annual work hours and earnings. The estimates apply to reliers and are reported in percentage terms relative to their average childless potential outcome in each year. Years −5 through −1 provide placebo tests of

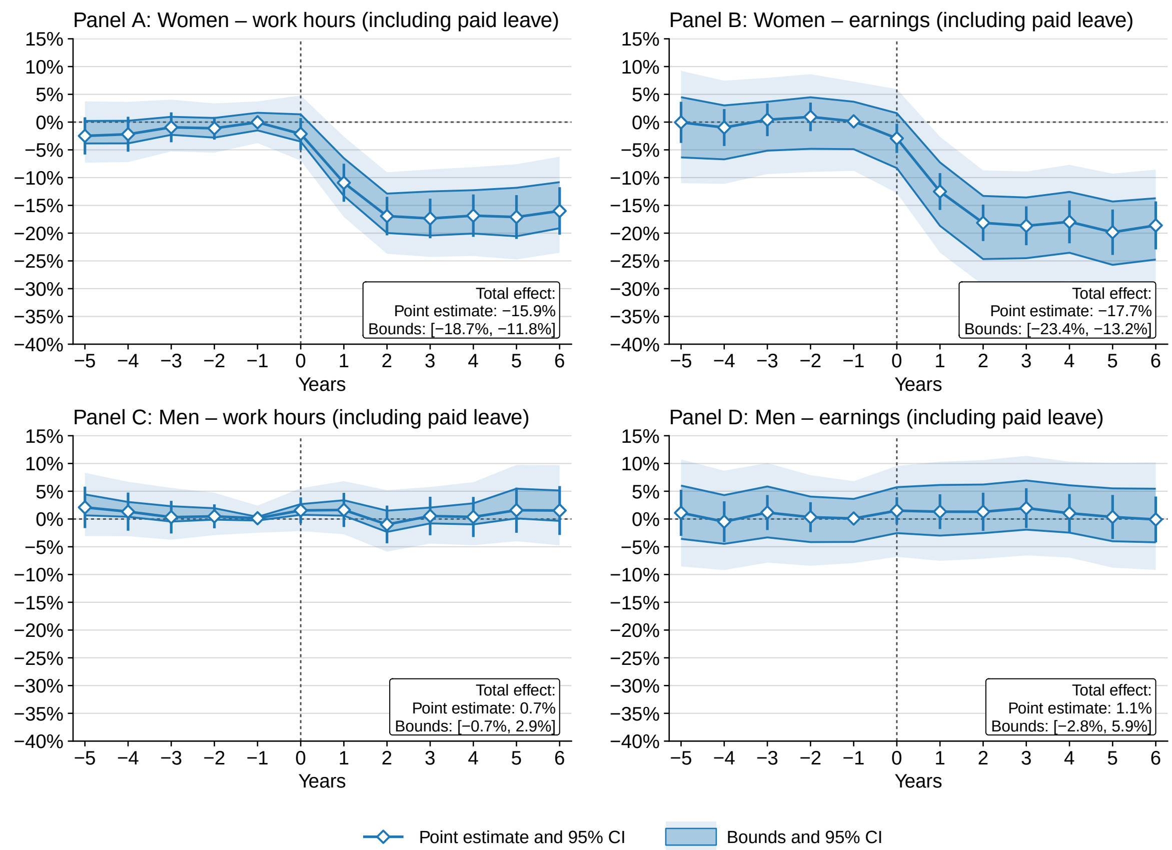

Figure 6: Effects of Parenthood on Work Hours and Earnings

Note: Estimates by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level without children. Panels A and B report estimates for women, and Panels C and D for their partners. The boxes report effects totalled over years 1–6, expressed as a percentage of the totalled counterfactual level.

sequential unconfoundedness, monotonicity, and conditionally random reliance (see Section 4.6 and Appendix SA5). The bounds include zero throughout the pre-IUI period, and the point estimates are close to zero, supporting these assumptions.

In the conception year, the bounds indicate negligible effects on either outcome. Effects become only slightly more negative in the following year, in part because maternity leave is included in measured hours and earnings[15]. Between the second and sixth years of motherhood, the average bounds imply reductions of 12%–20% in annual work hours and 13%–25% in annual earnings. The point estimates under conditionally random reliance lie approximately midway between the bounds. Over the first six years of parenthood, the bounds imply reductions of 12%–19% in work hours and 13%–23% in earnings, with corresponding point estimates of 16% and 18%.

Figure 7 relates these effects to gender gaps in labor market outcomes, measured as the percentage by which women's outcomes are lower than men's. In the year before the first IUI, women work 14.5% fewer hours and earn 25.2% less than their partners. If the couples remain childless, these gaps, which are point identified under the childless scenario, increase modestly over the next

[15]See Appendix SA9 for leave-adjusted estimates; only first-year estimates change meaningfully.

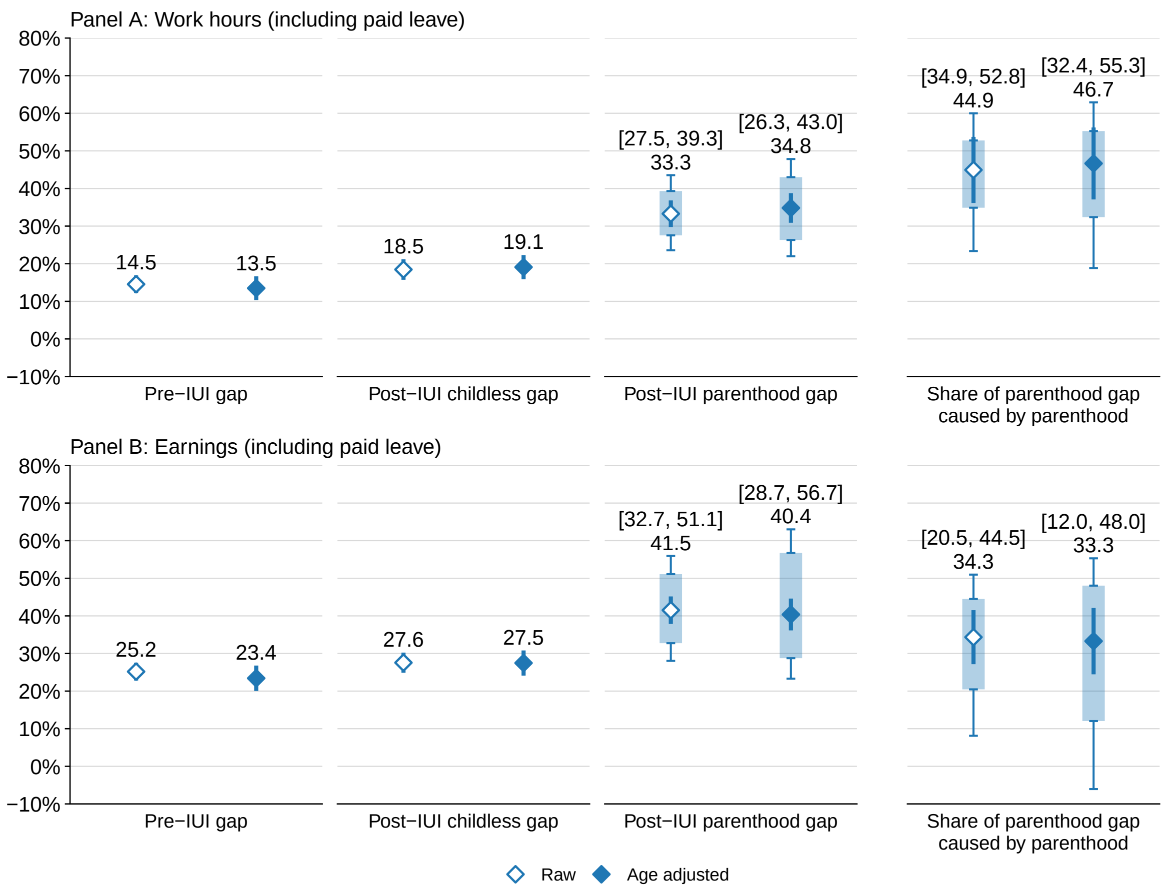

Figure 7: Parenthood and Within-Couple Gender Gaps

Note: Average within-couple gender gaps in work hours and earnings, defined as the percentage by which the woman's outcome is below her partner's. *Pre-IUI* is the gap in the year before the first procedure; *childlessness* and *parenthood* are the gaps over the six years following it under each scenario; the final block gives the share of the post-parenthood gap attributable to parenthood. Translucent bars give the estimated identified set for the partially identified parameters, with whiskers showing the corresponding 95% confidence interval; markers give point estimates with their 95% confidence intervals. Bracketed numbers give the identified sets; unbracketed numbers give the point estimates. *Age-adjusted* estimates use partner outcomes aligned to the woman's age.

six years, to 18.5% for work hours and 27.6% for earnings. If the first IUI succeeds and the couples become parents, the bounds imply corresponding gaps of 27.5%–39.3% for work hours and 32.7%–51.1% for earnings, with point estimates of 33.3% and 41.5%. The final block shows the share of the post-parenthood gender gap caused by parenthood. The bounds imply shares of 34.9%–52.8% for work hours and 20.5%–44.5% for earnings, with point estimates of 44.9% and 34.3%, respectively.

Male partners are, on average, 2.5 years older, so outcome gaps measured at the same time relative to the first IUI also reflect age differences. Given the limited effects of parenthood on men, a natural adjustment is to keep women's outcomes indexed by years since the first IUI and compare them with their partners' outcomes at the same age. For example, if a woman is 35 three years after the first IUI, the gender gap is measured relative to her partner's outcome at age 35, regardless of when that age falls relative to the first IUI. This reduces the sample somewhat because

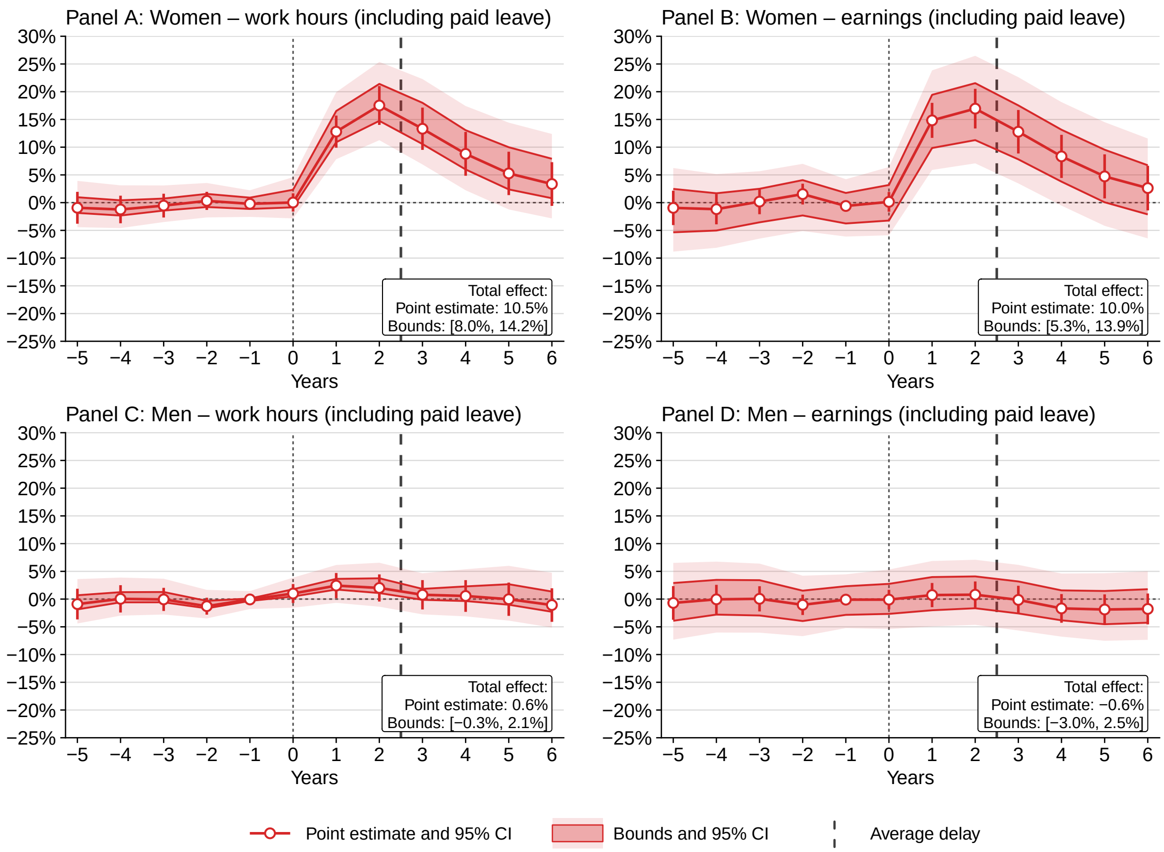

Figure 8: Effects of Later Parenthood on Work Hours and Earnings

Note: Estimates by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level under early parenthood. Panels A and B report estimates for women, and Panels C and D for their partners. The boxes report effects totalled over years 1–6, expressed as a percentage of the totalled counterfactual level. The dashed line marks the average later conception moment, 2.3 years after the first procedure.

the corresponding age is not observed for all couples.[16] Figure 7 also reports these age-adjusted estimates. The adjustment has little effect: the parenthood gap point estimate moves from 33.3% to 34.8% for work hours and from 41.5% to 40.4% for earnings, while the share of the post-parenthood gap attributable to parenthood moves from 44.9% to 46.7% and from 34.3% to 33.3%, respectively.

5.3 Effects of Parenthood Timing

Figure 8 presents the effects of parenthood timing. The estimates are reported in percentage terms and compare outcomes when the first IUI fails and motherhood occurs later with outcomes when the first IUI succeeds and motherhood begins earlier. Panels A and B report effects on women's work hours and earnings, respectively. As before, the pre-IUI years provide placebo tests of the identifying assumptions: the bounds and the point estimates are close to zero.

After the first IUI, the benefits of delay increase through year 2, when the bounds imply gains of 15%–21% in work hours and 11%–22% in earnings, with point estimates under conditionally

[16]The sample falls from 12,892 to 8,798 post-IUI and 8,762 pre-IUI.

random reliance of around 17% for both outcomes. The effects then decline gradually, with point estimates of around 3% by year 6. Over years 1–6 after the first IUI, the bounds imply gains of 8%–14% in work hours and 5%–14% in earnings, with point estimates of 10% for both outcomes.

Next, I consider whether the benefits of postponement arise only during the delay, before the later birth, or persist once both groups of women are mothers. The vertical dashed line approximately marks the average delay in motherhood among women who become mothers after IUI failure, 2.3 years. If all of them became mothers exactly 2.3 years later, estimates before this point would capture the benefit of remaining childless relative to having already become a mother, while estimates afterward would compare recent with earlier motherhood. In practice, the timing of later births varies: some occur within 2.3 years of the first IUI and others later. If parenthood weakly reduces women's labor-market outcomes, as the previous section suggests, this shifts the first-period effect downward and the second-period effect upward.[17]

To assess the importance of this timing heterogeneity, I construct alternative estimates for the two periods. For the first three years, I use the negative of the parenthood effect estimated in Section 5.2, scaled by the corresponding average outcome under first-IUI success. This directly measures the effect of remaining childless relative to early motherhood. From year 3 onward, I instead re-estimate the timing effect using parenthood status in each period rather than status at the end of the observation window, effectively treating each period as the end of the window. This ensures that each estimate compares early mothers with late mothers who have already become mothers by that period, so the effect reflects later rather than earlier motherhood. Appendix SA1 presents the formal identification results.

Figure 9 reports the resulting point estimates under conditionally random reliance; Appendix SA2 reports the corresponding bounds, which lead to the same conclusions. The estimates change little overall and move in the expected direction. Before later motherhood, the average estimated gain increases from 9.4% to 11.0% for work hours and from 10.1% to 12.7% for earnings. After later motherhood, it falls from 7.7% to 5.7% for work hours and from 7.1% to 5.4% for earnings.

Nearly all of the difference is concentrated in years 2 and 3, which surround the average later birth and are therefore the most contaminated years of their respective windows. As Figure 5 shows, only about 12% of later births have occurred by year 1, so the estimate for that year is largely uncontaminated. By year 2, about 40% have occurred, so the group meant to be childless already includes mothers, understating the benefit of remaining childless. By contrast, by year 3 about 30% of later births have yet to occur, so the group meant to consist of mothers still includes childless women, overstating the benefit of postponement. By year 4, about 85% have occurred, and the remaining contamination is limited.

Taken together, the results suggest that delaying parenthood substantially increases women's work hours and earnings. Importantly, these gains are not limited to eliminating the losses before

[17] In the first period, women who have already become mothers have worse outcomes than if they had remained childless, attenuating the estimated benefit. In the second, women who remain childless have better outcomes than if they had already become mothers, inflating it.

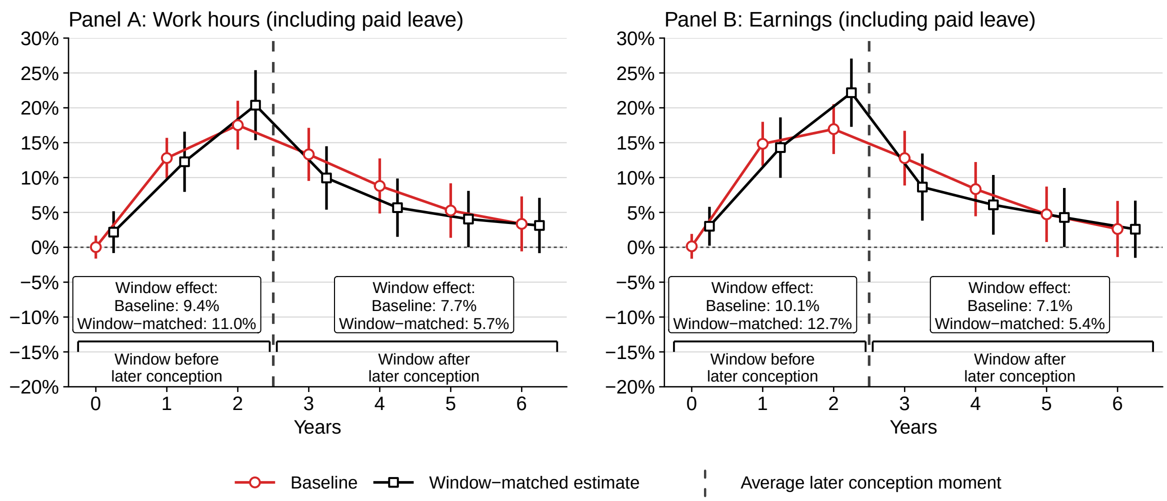

Figure 9: Window-Matched Estimates of the Effect of Later Parenthood on Women's Outcomes

Note: Estimates by year relative to the first intrauterine insemination, expressed as percentages of the corresponding counterfactual levels. *Baseline* uses parenthood status at the end of the observation window. *Window-matched* uses the negative of the parenthood effect from Section 5.2, scaled by the average outcome under first-IUI success, for years 0–2, and re-estimates the timing effect using parenthood status in each period from year 3 onward. Boxes report effects totaled over each window, expressed as percentages of the corresponding totaled counterfactual levels. Markers show point estimates with their 95% confidence intervals.

delayed birth: women who become mothers later continue to have better labor market outcomes even after motherhood begins. This occurs even though they have become mothers more recently and impacts are often thought to be largest at the onset of motherhood (see Lundborg et al., 2017, for discussion). It also occurs even though these delays are unintended, arising from IUI failure, which should lead to larger losses if fertility is timed to minimize career costs (see Bensnes et al., 2025, for discussion). Overall, the evidence points to early childbearing as an important driver of career setbacks. It suggests that postponement helps mitigate the losses not only by eliminating them before the later birth but also by reducing them afterward. I present indicative calculations of how much postponement may offset the career cost of motherhood in Section 6.

5.4 Heterogeneity

Figure 10 reports heterogeneity in the effects on women's outcomes, with the sample split at the median of three pre-IUI characteristics: the woman's earnings, the female–male earnings gap, and her age at the first procedure. The woman's earnings are residualized on her age, while the earnings gap is residualized on both her earnings and age. Outcomes are summed over the six years following the first IUI, and effects are expressed as a percentage of the corresponding counterfactual levels. The estimation approach follows Heiler (2024): nuisance functions are estimated jointly in the full sample, and the resulting moments are then split by the relevant pre-IUI characteristic to maximize precision. The dashed line indicates the pooled-sample point estimate.

The point estimates under conditionally random reliance show a consistent pattern: losses from parenthood and gains from postponement are larger for women with lower pre-IUI earnings, women

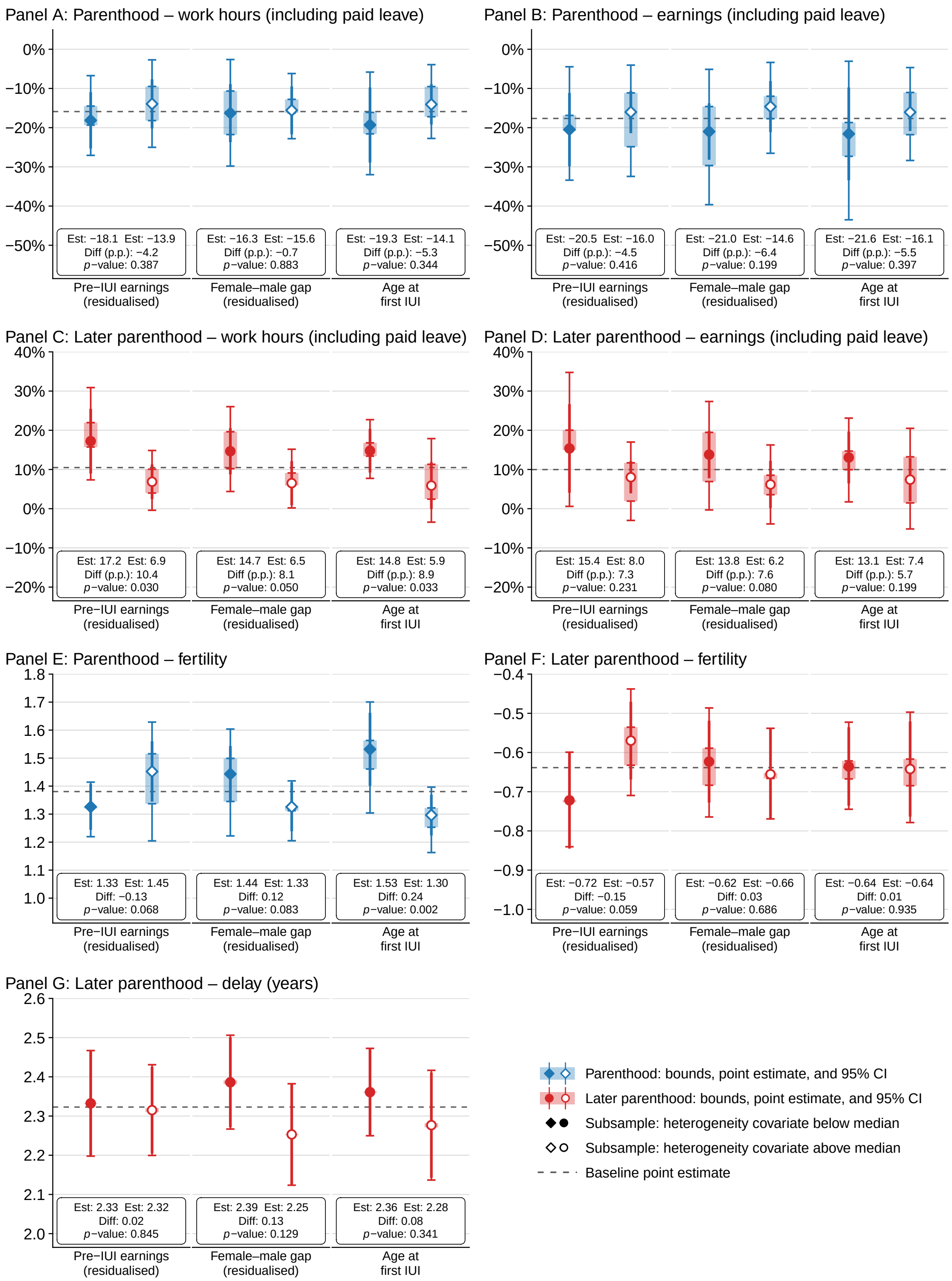

Figure 10: Heterogeneity in the Effects of Parenthood and Its Timing on Women's Outcomes

Note: Heterogeneity in effects by above- versus below-median values of three characteristics measured in the year before the first IUI: (1) earnings residualized on age, (2) the female–male earnings gap residualized on age and female earnings, and (3) age. Panels A–D report estimated effects on outcomes totaled over years 1–6 relative to the first intrauterine insemination, expressed as percentages of the corresponding totaled counterfactual levels; Panels E and F report estimated effects on the number of children, while Panel G reports estimated effects on timing in years. Bars give the estimated identified set for the partially identified parameters, with whiskers showing the corresponding 95% confidence interval; markers give point estimates with their 95% confidence intervals. The dashed line reports the full-sample estimate. Estimation follows Heiler (2024).

who earn less relative to their partner, and younger women. Although the parenthood and timing effects are estimated for distinct groups, this pattern is consistent with postponement being most valuable for women who face the greatest costs of early motherhood.

Statistical evidence is more limited. Only the effect of later parenthood on work hours in Panel C differs significantly across groups: 17.2% versus 6.9% by pre-IUI earnings, 14.7% versus 6.5% by the within-couple earnings difference, and 14.8% versus 5.9% by age, all significant at the 5% level. Elsewhere, the groups cannot be statistically distinguished and the identified sets overlap substantially, making the remaining heterogeneity only suggestive.

Panels E and F examine heterogeneity in the intensive fertility margin. Fertility responses vary across groups but do not consistently align with heterogeneity in labor-market effects. For parenthood, younger women have more children despite similar labor-market effects by age. For timing, lower-earning women experience both larger fertility reductions and labor-market gains, whereas fertility responses vary little by the within-couple earnings gap or age despite differences in labor-market effects. Panel G also shows little heterogeneity in delay length. Thus, total fertility may explain some heterogeneity in labor-market effects but does not account for it systematically; I investigate this margin further in the next section.

Overall, the results in this section are consistent with the benefits of postponement being largest when careers are least established.

5.5 The Intensive Fertility Margin

The timing estimates compare women who become mothers earlier with those who become mothers later. As shown in Section 5.1, the latter group also has fewer children within the observation window: about 0.64 fewer by year 6. Differences in subsequent fertility may therefore be one mechanism behind the labor market returns to postponement. Because subsequent births may themselves be selective, the intensive margin cannot be fully separated. This section therefore provides indirect evidence on its importance.

Figure 11 reports three sets of results. The first, in Panels A and B, uses couples who already have one child and undergo IUI or IVF for a second, and reports sequential reduced-form estimates comparing success at the first procedure with failure of the whole sequence. This estimand mirrors the intensive-margin component of the main analysis, where delay both reduces subsequent fertility and postpones subsequent births that still occur. Panel A shows that a successful first procedure raises the number of children by 0.76 by the end of the window, from 1.48 to 2.24, a larger shift than the 0.64 difference in the main analysis. Panel B reports the corresponding labor-market effects. Cumulated over years 1–3, the point estimates are −3.2% for work hours and −1.0% for earnings; over years 1–6, they are −3.1% and −0.4%. These effects are small relative to the timing effects.

The second exercise, in Panels C and D, uses a multiple first birth as a shock to subsequent fertility, as in Rosenzweig & Wolpin (1980). In the IUI sample, I restrict attention to couples whose first procedure succeeds and compare those with multiple and singleton births; I make the same comparison in a representative sample of couples having a first child between 2013 and 2016. In both samples, I control for the baseline covariates. Panel C confirms that a multiple birth

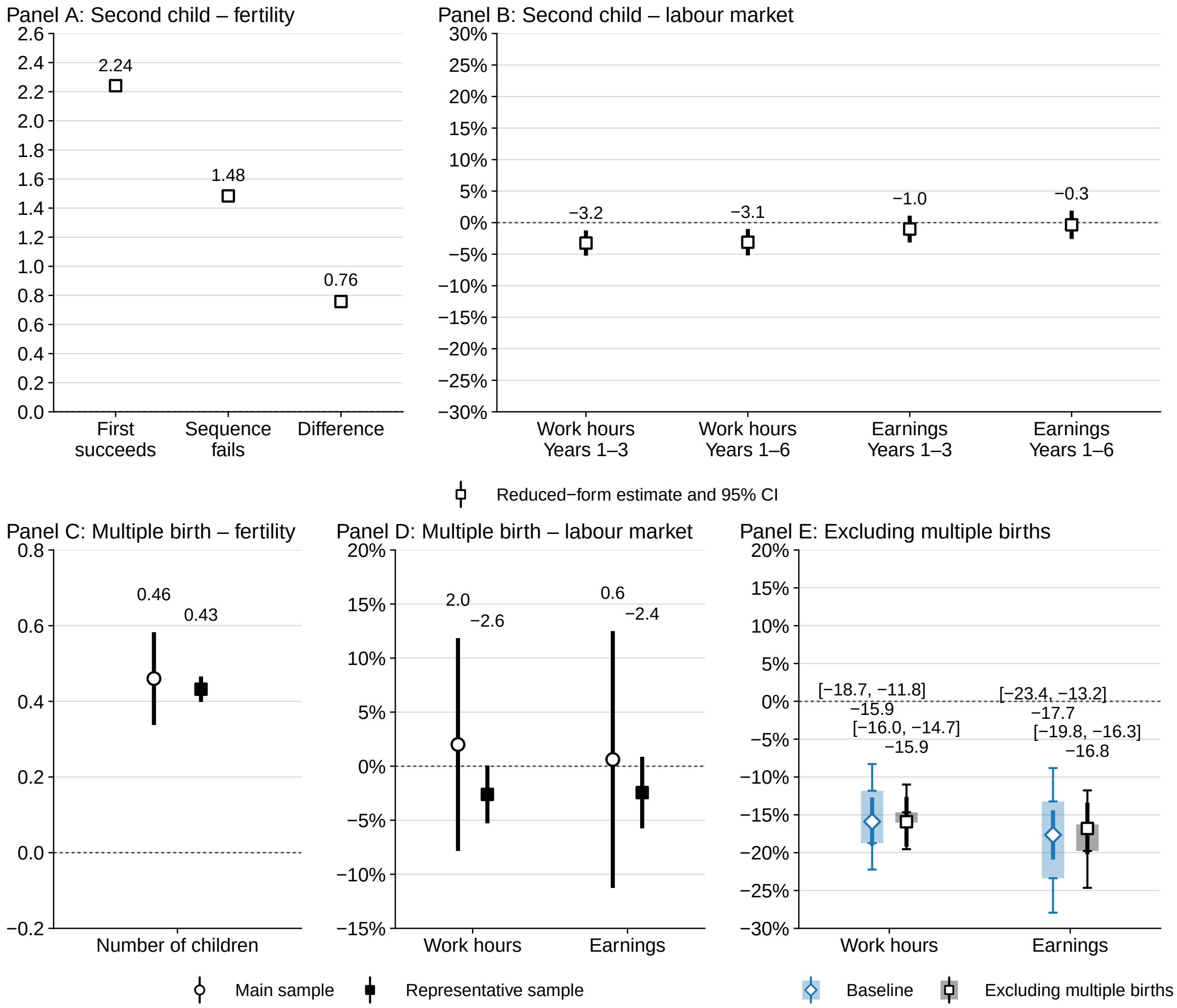

Figure 11: The Intensive Fertility Margin

Note: Panels A and B use couples who already have one child and undergo IUI or IVF for a second, and report point-identified reduced-form estimates. Panel A reports the number of children under first-procedure success and sequence-failure scenarios, six years after the first procedure; Panel B reports the effects of first-procedure success relative to sequence failure on work hours and earnings totalled over years 1–3 and 1–6, expressed as a percentage of the corresponding level under sequence failure. Panels C and D compare couples with multiple and singleton first births, controlling for the baseline covariates. In the IUI sample, these panels restrict to couples whose first procedure succeeds; the representative sample includes couples having a first child between 2013 and 2016. Work hours and earnings are totalled over six years from first conception, while the number of children is measured six years after first conception. Panel C reports effects on the number of children, and Panel D reports labor market effects as a percentage of the mean among singleton births. Panel E reports effects of parenthood in the main sample on women's outcomes totalled over years 1–6, under the baseline specification and the same specification excluding couples with a multiple first birth. Translucent bars give the estimated identified set for the partially identified parameters, with whiskers showing the corresponding 95% confidence interval; markers give point estimates with their 95% confidence intervals. Bracketed numbers give the identified sets; unbracketed numbers give the point estimates. Work hours and earnings include paid leave throughout.

substantially increases total fertility over the following six years, by 0.46 children in the IUI sample and 0.43 in the representative sample. Panel D shows that the corresponding labor-market effects over the same six-year window are small: 2.0% for work hours and 0.6% for earnings in the IUI sample, and −2.6% and −2.4% in the representative sample.

Finally, I assess the sensitivity of the main estimates of the effect of parenthood to excluding multiple births, which are more common following IUI and may therefore limit generalizability to non-IUI parents. Panel E re-estimates the effect of parenthood dropping couples with a multiple birth at the first IUI. The point estimates are essentially unchanged: −15.9% for work hours, against −15.9% in the baseline, and −16.8% for earnings, against −17.7%.[18]

Together with the previous section, where the benefits of postponement are larger for younger women with less established careers despite similar fertility responses, these results suggest that the intensive margin may contribute to labor-market effects but is unlikely to be their primary driver.

5.6 Sensitivity

The baseline specification predicts IUI success using both partners' age, education, and procedure type. Age is the single most important clinical determinant of IUI success, while education may absorb behavioral or health differences associated with success probabilities, for instance through smoking. One may nevertheless worry that women who are similar along these observed dimensions differ in their behavior during treatment, and that these differences are related to potential labor-market outcomes. To investigate this further, I additionally control for detailed features of the procedure protocol. These controls capture differences in treatment behavior directly and may also proxy for remaining differences in the behavior or circumstances surrounding conception.

I construct six indicators for whether a given activity occurs within a fixed window around the insemination date: a semen analysis in the preceding year, sperm preparation on the day of insemination, ovulation ultrasound or hormonal monitoring in the preceding fortnight, and progesterone administration in the fortnight before or after. Each is measured at the level of the individual procedure and included among the controls for the corresponding procedure.

Figure 12 re-estimates the effects on women's outcomes adding three progressively richer sets of these controls to the propensity scores and other nuisance functions: a single indicator for whether any of the six activities is recorded; three indicators grouping them into male-factor testing and preparation, cycle monitoring, and progesterone; and the six separately. The point estimates are virtually unchanged from the baseline; Appendix SA10 reports the corresponding bounds, which are also virtually unchanged. This suggests that protocol differences play little role in the results.

I next assess sensitivity to selection on unobservables. Section 3.6 quantifies how much imbalance in post-IUI potential outcomes may be consistent with balance in pre-IUI outcomes under a model in which women with higher success probabilities have systematically higher potential outcomes. I use these magnitudes to adjust observed outcomes. For example, if the test implies that women who succeed at the first procedure may have potential outcomes up to $x\%$ above the mean among women reaching it, I scale their observed outcomes down accordingly. Likewise, if women who fail a procedure may have potential outcomes a given percentage below the corresponding mean, I scale their outcomes up. Failure adjustments compound along the realized treatment sequence: a woman who fails three procedures has her outcome scaled up three times in succession.

[18] The bounds also narrow because excluding multiple births makes subsequent non-IUI conception relatively more common, lowering the subsequent relier share and hence the trimming share.

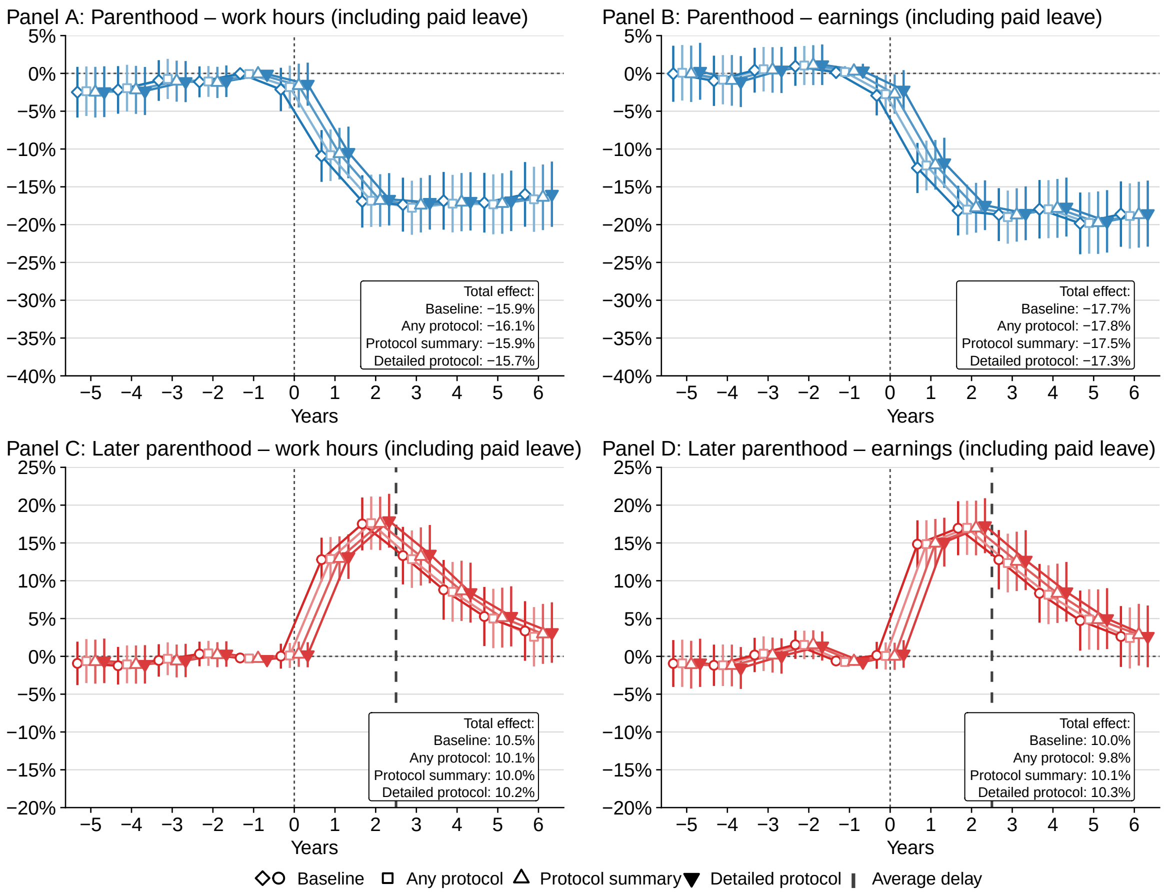

Figure 12: Sensitivity of Effects on Women's Outcomes to Controls for the Clinical Protocol

Note: Estimates by year relative to the first intrauterine insemination, expressed as percentages of the corresponding counterfactual levels. *Baseline* is the specification used throughout the paper, which predicts procedure success using both partners' age and education and procedure type. Relative to the baseline, *Any protocol* adds an indicator for whether any of six procedure-level care activities is recorded within its relevant window: semen analysis, sperm preparation, ovulation ultrasound, hormonal monitoring, and progesterone administration before or after insemination. *Protocol summary* adds three indicators grouping these activities into male-factor testing and preparation, cycle monitoring, and progesterone; *Detailed protocol* adds all six separately. Markers show point estimates with their 95% confidence intervals.

The key additional assumption is that selection operates proportionally across potential outcomes. For instance, if women with higher success probabilities have early-motherhood outcomes that are 1% higher, their late-motherhood and childlessness outcomes are also assumed to be 1% higher. This seems natural if the main concern is latent health or other persistent characteristics that affect labor-market outcomes similarly across fertility scenarios. Appendix SA8 discusses how this assumption can be relaxed, though doing so makes the exercise more elaborate.

Importantly, the exercise does not require committing to the particular latent-type selection model used in Section 3.6. It accommodates any selection process in which women who conceive at a given procedure have proportionally higher potential outcomes than the mean among women reaching it, while women who fail have proportionally lower potential outcomes; the model in Section 3.6 is used only to calibrate the benchmark magnitudes.

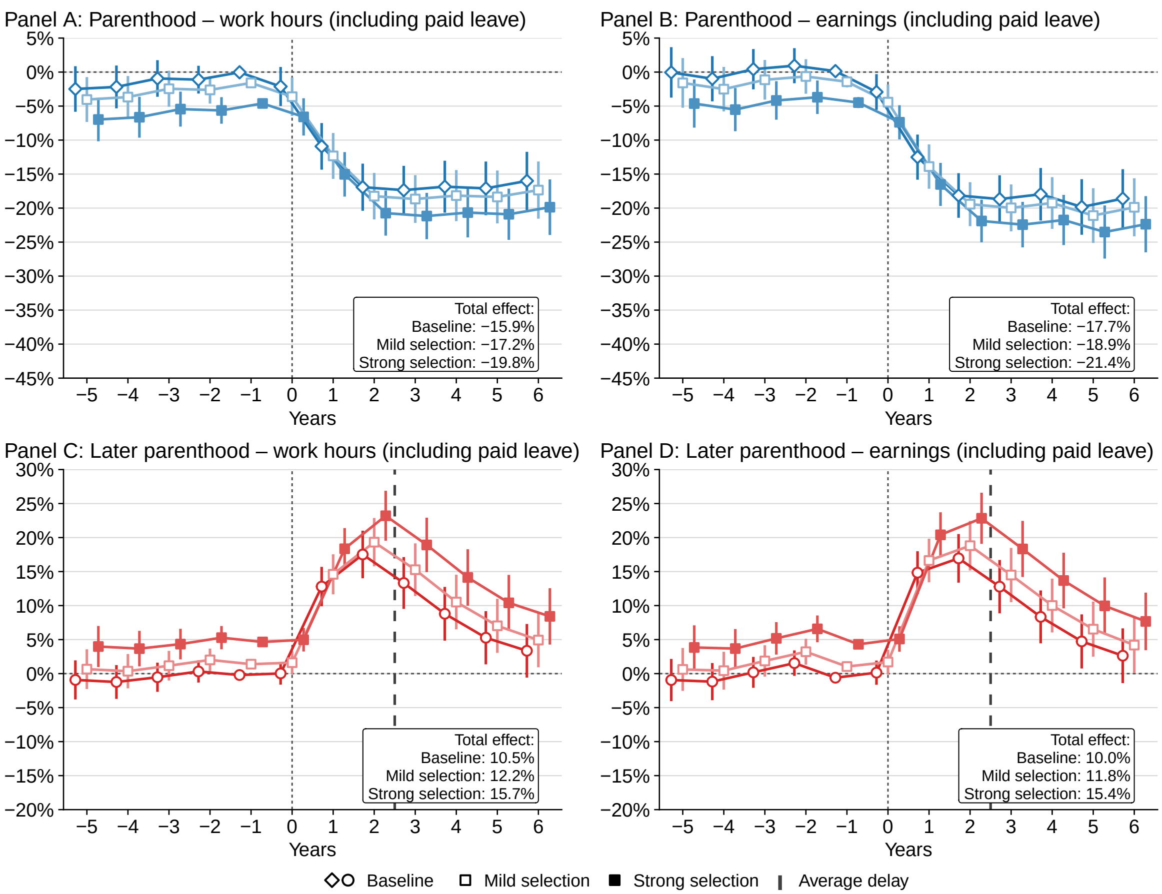

Figure 13: Sensitivity of Effects on Women's Outcomes to Selection on Unobservables

Note: Estimates by year relative to the first intrauterine insemination, expressed as percentages of the corresponding counterfactual levels. Prior to estimation, outcomes are rescaled according to each woman's number of procedures. Under *strong selection*, outcomes are scaled by 1.027^{-1} for women who conceive at the first procedure and by 0.997^{-A} for women with no IUI success; under *mild selection*, the corresponding factors are 1.009^{-A} and 0.999^{-A}, where A denotes the number of procedures. The mild adjustment uses the largest imbalance not rejected by the pooled balance test in Section 3.6; the strong adjustment uses the largest imbalance not rejected at the first procedure. Markers show point estimates with their 95% confidence intervals.

I use two adjustment benchmarks. The first—mild-selection—scenario uses the largest differences not jointly rejected over the full procedure sequence, implying potential outcomes 0.9% above the mean among women entering a given procedure for those who succeed and 0.1% below it for those who fail. The second—strong-selection—scenario uses the largest difference not rejected at the first procedure, where the corresponding gaps are 2.7% and 0.3%. Although the test suggests that selection this strong may arise at individual procedures but cannot persist over the full sequence, I conservatively apply it to every procedure rather than take a stance on where it arises.

Figure 13 reports point estimates based on the adjusted outcomes. Under strong selection, parenthood effects over the window move from −15.9% to −19.8% for work hours and from −17.7% to −21.4% for earnings, while the timing effects move from 10.5% to 15.7% and from 10.0% to 15.4%, respectively. Notably, this adjustment also moves pre-IUI effects away from zero, providing evidence against selection this strong: accounting for selection should instead bring them closer to zero. Under milder selection, estimates change little relative to baseline: the parenthood effects move to −17.2% and −18.9%, and the timing effects to 12.2% and 11.8%, for work hours and earnings, respectively. Appendix SA11 reports the corresponding bounds, which move proportionally. Overall, the conclusions change little, with both adjustments mechanically implying larger career costs of motherhood and larger benefits of postponement.

The Appendix reports further sensitivity analyses. Appendix SA4 assesses the role of relationship stability and mental health, measured by antidepressant use. Appendix SA5 tests and relaxes monotonicity. Appendix SA6 varies the number of cross-fitting folds and trees, and Appendix SA7 the maximum number of procedures used for estimation. Appendix SA9 adjusts work hours for maximum unobserved parental leave. The main results are generally insensitive to these changes.

5.7 Comparison with Conventional Methods

Next, I compare my estimates with those from conventional methods. Because these methods target parenthood effects and typically rule out timing effects by assumption, the comparison is limited to the parenthood margin. Figure 14 reports the comparison, summing effects over the six years following the first IUI and expressing them as a percentage of the corresponding counterfactual level. Appendix SA3 reports the per-period estimates and provides estimation details.

Panels A1 and B1 report my baseline estimates, and Panels A2 and B2 place them alongside bounds that use only the first IUI and do not impose monotonicity, but otherwise exploit the same covariate information as the baseline estimation. These bounds are equivalent to Zhang & Rubin (2003) and Lee (2009). They range from −59.4% to 20.2% for work hours and from −57.3% to 27.2% for earnings, and therefore do not rule out large effects of either sign. My bounds are roughly ten times narrower. This improvement follows directly from exploiting sequential variation in IUI success and is guaranteed by the method: the full sequence identifies the childless outcome for the larger group of reliers, rather than only for the compliers targeted by these bounds, so bounding their early-motherhood outcome requires trimming fewer observations from first-attempt successes, making the retained group means more similar whether trimming from the upper or lower tail.

Panels A3 and B3 report conventional instrumental-variable estimates. Scaling the cumulative reduced form by the first stage at the end of the window—the *endline* estimator—gives −33.1% for work hours and −34.9% for earnings, well outside the identified set. Scaling each period's reduced form by its contemporaneous first stage before cumulating gives −21.5% and −22.8%, closer to the lower bound of my identified set, though still outside it for work hours.

Panels A4 and B4 report the sequential IV estimates introduced in Section 3.3.1, which cover reliers rather than compliers but, like conventional IV, identify parenthood effect only under the assumption of no timing effects. The endline estimates are −28.6% and −29.9% for work hours and earnings, respectively, while the corresponding per-period estimates are −20.2% and −21.3%. The fact that the sequential estimates remain close to the conventional IV estimates indicates that the difference from my estimates is not due to heterogeneity in treatment effects between reliers and compliers alone. Instead, the remaining difference is accounted for by contamination from timing effects: the sequential IV estimand and the parenthood and timing effects are linked by identity,

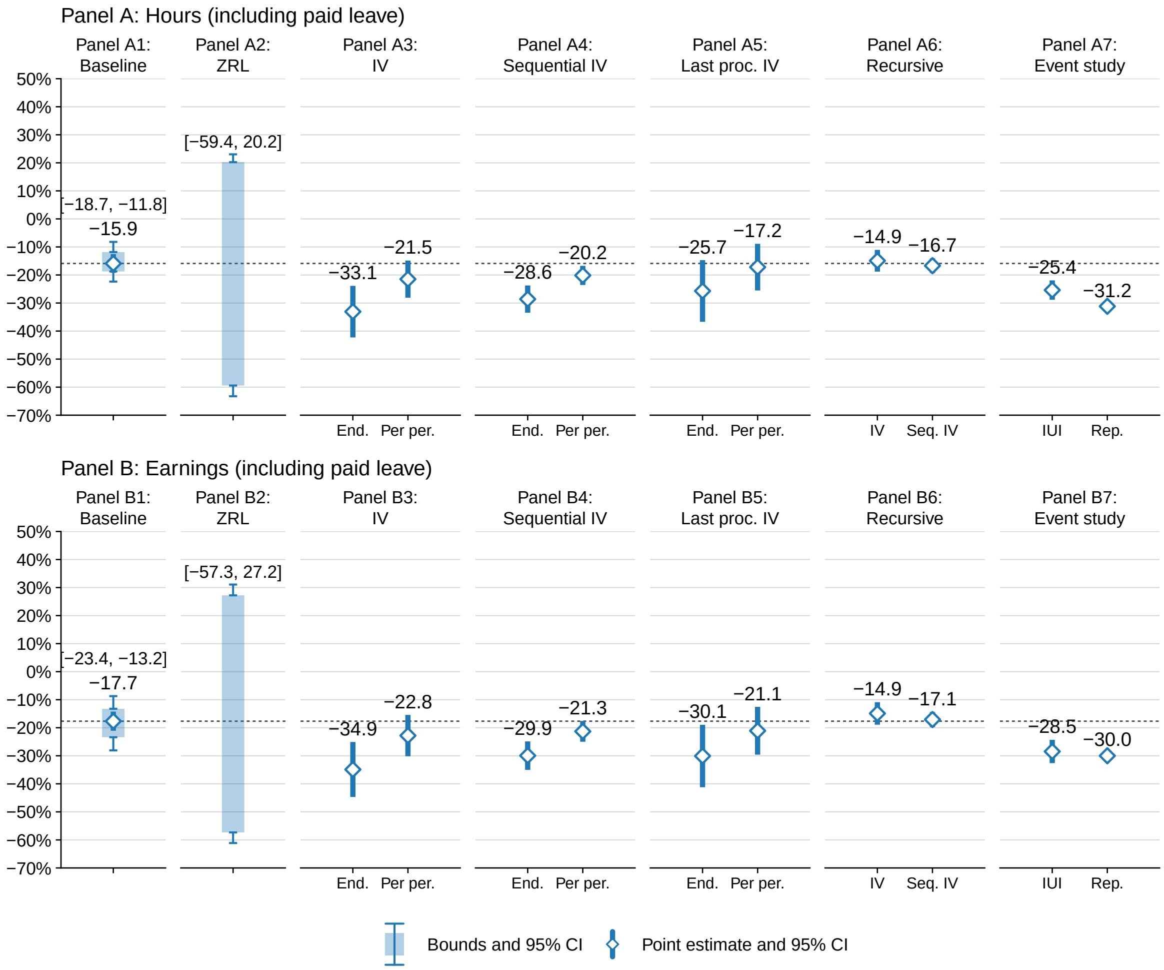

Figure 14: Comparison of Estimators of the Effect of Parenthood on Women's Hours and Earnings

Note: Effects of parenthood on women's work hours and earnings, totalled over years 1–6 relative to the first intrauterine insemination, expressed as a percentage of the totalled counterfactual level without children. Panels A1 and B1 report the baseline estimates. Panels A2 and B2 report bounds using only the first procedure and imposing no monotonicity, following Zhang & Rubin (2003) and Lee (2009). Panels A3–A5 and B3–B5 report instrumental-variable estimates: first-procedure success versus failure; first-procedure success versus sequence failure; and success at the last procedure undergone. For each instrument, *endline* scales the cumulative reduced form by the first stage at the end of the window, while *per period* scales each period's reduced form by its contemporaneous first stage before cumulating. Panels A6 and B6 report recursively adjusted estimates, following Gallen et al. (2024) and Bensnes et al. (2025), applied to the first-procedure-success versus first-procedure-failure and first-procedure-success versus sequence-failure reduced forms. Panels A7 and B7 report event-study estimates for the intrauterine-insemination sample and a representative sample of women having a first child between 2013 and 2016, following Kleven, Landais, & Søgaard (2019). The horizontal dotted line gives the baseline point estimate from Panels A1 and B1. Bracketed numbers give the identified sets; unbracketed numbers give the point estimates. See Appendix SA3 for further details.

so the positive timing effects above imply that sequential IV overstates the cost of parenthood.

Panels A5 and B5 instead define the instrument as success at the last procedure a woman undergoes. Relative to conventional IV, the endline estimates fall to −25.7% and −30.1% for work hours and earnings, respectively, and the per-period estimates to −17.2% and −21.1%. This is consistent with the positive selection into additional procedures documented in Section 4.5, which attenuates estimated career costs. Because the timing bias operates in the opposite direction in this setting, this selection moves the estimates closer to the baseline estimates.

Recursive IV estimates, reported in Panels A6 and B6, adjust the outcomes of women who conceive in later periods using earlier-period estimates of the effect of conceiving at the first IUI, following Gallen et al. (2024) and Bensnes et al. (2025). The conventional recursive estimator gives −14.9% for both outcomes and the sequential recursive estimator −16.7% for work hours and −17.1% for earnings. All lie inside the identified set and close to my point estimates. Unlike in Bensnes et al. (2025), however, the recursive adjustment moves the estimates toward smaller rather than larger losses. The difference reflects the time profile of estimated losses in the two settings. Both estimators start from the reduced-form comparison between women whose first IUI succeeds and fails. Conventional IV scales it by a first stage below one, pushing the estimate away from zero. Recursive IV instead adjusts the outcomes of control-group women who become mothers later by subtracting earlier-period parenthood-effect estimates, which, when negative, raises their adjusted outcomes and likewise pushes the estimated effect downward. Which adjustment is larger depends on the size of earlier-period losses. Because these are relatively small in my setting, the recursive adjustment is limited and the resulting estimates remain closer to zero than conventional IV.

Panels A7 and B7 report event-study estimates following Kleven, Landais, & Søgaard (2019), which compare mothers with similarly aged women who are about to have their first child. The estimates are −25.4% for work hours and −28.5% for earnings in the IUI sample, compared with −31.2% and −30.0% in a representative sample entering parenthood over the same period. These estimates are larger than my baseline and closer to the IV estimates. Several factors may explain this discrepancy, but one particularly important one is the relevant counterfactual. In estimates leveraging IUI success, women in the comparison group try to conceive but do not, whereas in event studies women may not yet have even tried. If labor-market outcomes differ between these states, for instance because trying and failing to conceive itself affects labor-market behavior, the resulting estimates need not coincide. I discuss these differences further in the next section.

The results in this section have an important implication for how my estimates compare with the existing literature. Conventional estimators applied to these data generally imply similar or substantially larger parenthood losses than my baseline estimates. The relatively large effects I find compared with IVF designs in Scandinavian settings, and their similarity to estimates from contraceptive-failure designs—which might be expected to be larger because unintended births are more disruptive—therefore do not appear to be driven by methodological differences. Instead, the difference is consistent with substantially more generous family policies in Scandinavia helping to mitigate the career cost of motherhood.

6 Discussion

In this section, I discuss how my estimates may generalize to scenarios in which women make different fertility and career plans, and I present suggestive calculations on how much delaying motherhood may mitigate its career costs.

The focal point of my analysis is the moment of the first IUI procedure, whose timing is chosen by the woman. My estimates for the effect of parenthood therefore reflect the consequences of failing to conceive when intended, while my estimates for the effect of timing reflect the consequences of conceiving later than intended. To understand the broader role of parenthood in the labor market, two other counterfactuals are especially important: planned childlessness and deliberate postponement of childbearing.

How my estimates differ from the impacts of parenthood relative to planned childlessness depends on whether women would invest more or less in their careers under that scenario. On the one hand, women may invest more if they plan to remain childless, since the absence of children may allow them to reap greater returns. In that case, my estimates would serve as a lower bound on the career cost of motherhood. Alternatively, women may invest more if they anticipate motherhood—both to accumulate resources for raising children and to develop careers compatible with family life. In that case, my estimates would instead provide an upper bound on the career cost.

The impacts of planned delayed childbearing are arguably easier to anticipate. One reason why planned and unplanned delays may differ concerns the experience of infertility itself: women's outcomes may be depressed by the failure to achieve pregnancy. Another reason is that women may plan fertility and their careers, at least in part, to minimize career costs. Both imply that planned delays would result in better labor market outcomes than unplanned delays. This suggests that the results may be seen as a lower bound on the gains from postponement.

I conclude with a calculation, based on my estimates, of how much postponing motherhood can mitigate its career cost. This requires outcomes under three scenarios: remaining childless, becoming a mother relatively early, and becoming a mother relatively late. My approach identifies the first two for women whose eventual fertility depends on IUI success and the latter two for women who eventually conceive regardless of IUI success, but does not identify all three for either group. One way forward is to compare the estimated effects in levels for the two groups directly. Another is to use one estimated path as an anchor and construct the other two by applying the estimated effects in percentage terms. I consider both approaches and alternative anchors and show in Appendix SA12 that the conclusions are invariant to these choices. The comparison is exact under the first approach if the corresponding effects are equal in levels across reliers and non-reliers, and under the second if they are proportional across the two groups; otherwise, it is suggestive.[19]

Figure 15 constructs the three paths using the point estimates from Sections 5.2 and 5.3: remaining childless, conceiving at the first IUI, and conceiving 2.3 years later—the average delay in the analysis. The childless path is the relier path and serves as the level anchor. The early-motherhood path scales it by the ratio of reliers' mean outcome under early motherhood to that under childlessness. From year 3 onward, the later-motherhood path further scales the early-motherhood path by the corresponding ratio of non-reliers' mean outcomes under later and early motherhood, using the window-matched specification.

[19]For example, anchoring on the relier path requires either equal relative timing effects, $\mathbb{E}[\delta|R = 0]/\,\mathbb{E}[Y^E|R = 0] = \mathbb{E}[\delta|R = 1]/\,\mathbb{E}[Y^E|R = 1]$, or equal effects in levels, $\mathbb{E}[\delta|R = 0] = \mathbb{E}[\delta|R = 1]$.

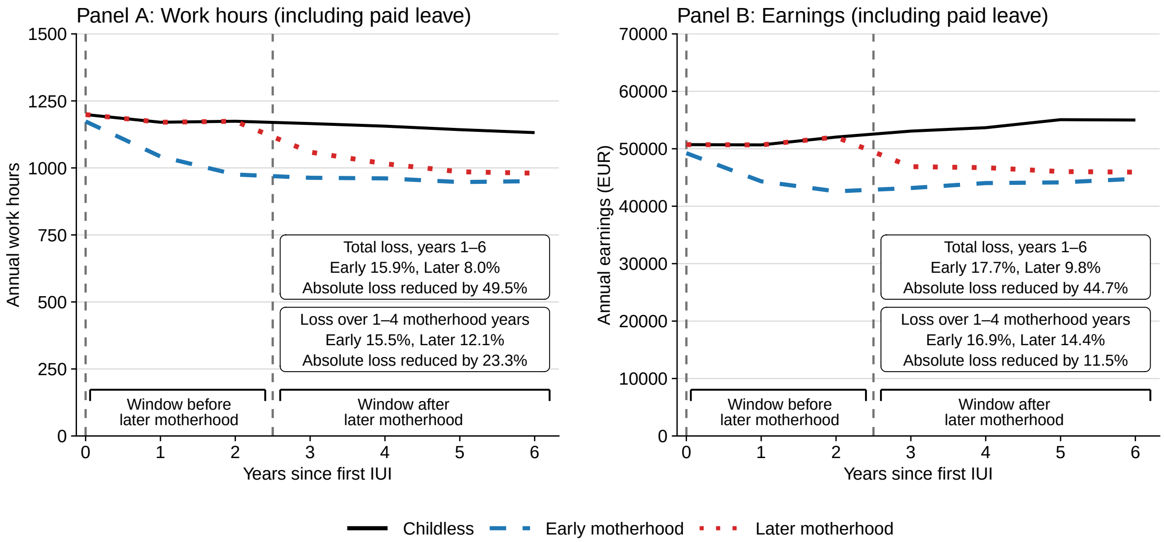

Figure 15: Childless, Early-Motherhood and Later-Motherhood Paths

Note: Work hours and earnings under alternative motherhood scenarios, by year relative to the first intrauterine insemination. The childless path is the estimated relier path; the early-motherhood path scales it by the ratio of reliers' mean outcomes under early motherhood and childlessness; from year 3 onward, the later-motherhood path additionally scales it by the corresponding ratio of non-reliers' mean outcomes under later and early motherhood from the window-matched specification. The upper box reports reductions relative to childlessness over years 1–6; the lower box reports reductions over the first four years of motherhood, years 1–4 for the early mother and years 3–6 for the later mother. The final line in each box gives the implied mitigation from postponement.

Over years 1–6, the early mother works 15.9% fewer hours and earns 17.7% less relative to the childless path, compared with reductions of 8.0% and 9.8%, respectively, for the later mother. A 2.3-year delay reduces the absolute loss in work hours by 49.5% and in earnings by about 45%.

The gain does not arise solely from postponing the reduction. Comparing the paths over the first four years of motherhood—years 1–4 for the early mother and years 3–6 for the later one—the later mother's hours loss is 12.1%, compared with 15.5% for the early mother, while the corresponding earnings losses are 14.4% and 16.9%. In levels, the total loss over the first four years is 23.3% smaller for work hours and 11.5% smaller for earnings. Thus, even at the same stage of motherhood, the later mother's career losses remain smaller.

A remaining question concerns the long-run benefits of postponement. These are difficult to assess because the data window is limited and because, as discussed above, my estimates may understate them. Nonetheless, one pattern is suggestive: parenthood losses are stable and persistent, whereas the gains from postponement are initially large but fade and are small by the end of the data window. Taken at face value, this suggests that postponement reduces career costs mainly during the delay and in the first few years after childbirth, with limited benefits thereafter.

7 Conclusion

This paper develops a method for estimating treatment and treatment-timing effects in quasi-experiments where individuals not initially assigned to treatment may obtain it later through repeated assignments. Beyond the setting of this paper, where I exploit variation in IUI success to estimate the labor-market effects of parenthood and its timing, similar structures arise in education admission lotteries, job-training programs with partially randomized access over time, and sequential clinical trials with imperfect compliance.

Using Dutch administrative data, I find that motherhood reduces women's annual work hours by 12%–19% and earnings by 13%–23% over the first six years relative to remaining childless, while the corresponding bounds for men are centered near zero. Parenthood accounts for 35%–53% of the gender gap in work hours and 21%–45% of the gap in earnings among parents. Conceiving 2.3 years later raises women's work hours by 8%–14% and earnings by 5%–14% over the following six years, with gains accruing both before and after the delayed birth. A suggestive calculation implies that, over the six-year window, a delay of this length removes roughly half of the loss in work hours and 45% of the loss in earnings. Over the first four years after childbirth, the corresponding losses are 23% smaller for work hours and 12% smaller for earnings.

I also investigate whether the gains from postponement can be explained by additional time to establish a career or by reduced fertility. The average delay of 2.3 years leaves women with 0.64 fewer children within six years. However, exercises using couples undergoing treatment for a second child and variation from multiple births at first parity suggest that reduced fertility explains little of the gains. I also find that the gains are largest for women with lower pre-IUI earnings, women who earn less than their partner, and younger women, while fertility responses vary little along the same dimensions. Taken together, these patterns are consistent with postponement benefiting women primarily by giving them additional time to establish a career before motherhood, rather than through reduced fertility.

My results inform debates about policies affecting the costs and timing of parenthood, including work-family supports and access to contraception and reproductive technologies. They point to substantial labor-market returns to delaying childbearing, suggesting a role for policies that support women who choose to postpone motherhood. This does not imply that later childbearing is welfare-improving: postponement may carry health costs outside the labor market, and the fertility reductions I observe within the data window may extend to completed fertility. If they do, policies that mitigate the career costs of early childbearing could be especially important, both for protecting women's labor-market outcomes and for supporting overall fertility.

Appendix

To simplify exposition, let $X_j^* \in \mathcal{X}_j^*$ contain X_j and $1_{A\geq j}$, and define $\mathcal{X}_j^{*1} = \{x \in \mathcal{X}_j^* : 1_{A\geq j} = 1\}$. Further, define $e_j^*(x) = \Pr(Z_j = 1|X_j^*)$ and $Z_l^* = (1-Z_l)/(1-e^*(X_l))$. Note that $e_j^*(x) = 0$ and $Z_j = 0$ if $1_{A\geq j} = 0$. Hence, Assumption 5 implies $\mathbb{E}[Z_l^*|X_l^*] = 1$. Moreover, by definition, $\Pi_j^A \frac{(1-Z_j)}{(1-e_j(X_j))} = \Pi_j^{\overline{w}} Z_j^*$.

Lemma 1. *Under Assumption 4:*

$$(Y^E, Y^N, R^+, R, W, Z_1, \ldots, Z_{j-1}, X_1^*, \ldots, X_{j-1}^*) \perp\!\!\!\perp Z_j^* | X_j^* \text{ for all } j > 1. \tag{21}$$

Proof of Lemma 1. X_j^* includes $1_{\{A\geq j\}}$, and when $A<j$, we have $Z_j=0$, which covers the cases when $X_j^* \in \mathcal{X}_j^* \setminus \mathcal{X}_j^{*1}$. The remainder follows from Assumption 4, since $1_{\{A\geq j\}}$, $Z_1,\ldots,Z_{j-1}$, $X_1^*,\ldots,X_{j-1}^*$, and $e_j^*(X_j^*)$ are fixed given X_j^*. ☐

Lemma 2. *For any l s.t. $1 \leq l \leq \overline{w}$ and any measurable function $g(M_l)$, where $M_l = (Y^E, Y^N, R^+, R, W, Z_1,\ldots,Z_l, X_1^*,\ldots,X_l^*)$, under Assumptions 4 and 5:* $\mathbb{E}\left[g(M_l)\Pi_{j=l+1}^{\overline{w}} Z_j^* \middle| X_l^*\right] = \mathbb{E}\left[g(M_l)|X_l^*\right]$.

Proof of Lemma 2. For any l s.t. $l<\overline{w}$:

$$\mathbb{E}\left[g(M_l)\Pi_{j=l+1}^{\overline{w}} Z_j^*|X_l^*\right] = \mathbb{E}\left[\mathbb{E}\left[g(M_l)\Pi_{j=l+1}^{\overline{w}} Z_j^*|X_{\overline{w}}^*\right]|X_l^*\right] \tag{22}$$

$$= \mathbb{E}\left[g(M_l)\Pi_{j=l+1}^{\overline{w}-1} Z_j^*\,\mathbb{E}\left[Z_{\overline{w}}^*|X_{\overline{w}}^*\right]\middle|X_l^*\right] \tag{23}$$

$$= \mathbb{E}\left[g(M_l)\Pi_{j=l+1}^{\overline{w}-1} Z_j^*\middle|X_l^*\right] \tag{24}$$

$$= \mathbb{E}\left[g(M_l)|X_l^*\right], \tag{25}$$

where (22) holds by law of iterated expectations and because X_j^* includes X_l^* for $j\geq l$, (23) holds by the Lemma (1), (24) holds because $\mathbb{E}[Z_l^*|X_l^*]=1$ under Assumption 5, and where (25) follows from steps similar to (22) through (24) for X_j^* for j s.t. $l<j<\overline{w}$. ☐

Proof of Theorem 1. I demonstrate the result for the upper bound, the result for the lower bound is symmetric. First, I demonstrate that $\mathbb{E}\left[Y(1-D^+)\Pi_{j=1}^{\overline{w}} Z_j^*\right]/\mathbb{E}[r(X_1^*)] = \mathbb{E}[Y^N|R=1]$. Note that:

$$\mathbb{E}\left[Y(1-D^+)\Pi_{j=1}^{\overline{w}} Z_j^*\right] = \mathbb{E}[Y^N R\Pi_{j=1}^{\overline{w}} Z_j^*] \tag{26}$$

$$= \mathbb{E}\left[\mathbb{E}\left[Y^N R\Pi_{j=1}^{\overline{w}} Z_j^*|X_1^*\right]\right] \tag{27}$$

$$= \mathbb{E}\left[\mathbb{E}\left[Y^N R Z_1^*|X_1^*\right]\right] \tag{28}$$

$$= \mathbb{E}\left[\mathbb{E}\left[Y^N R|X_1^*\right]\mathbb{E}\left[Z_1^*|X_1^*\right]\right] \tag{29}$$

$$= \mathbb{E}\left[Y^N R\right] \tag{30}$$

$$= \mathbb{E}\left[Y^N|R=1\right]\Pr(R=1), \tag{31}$$

where (26) follows from the definitions, (27) holds by law of iterated expectations, (28) holds by Lemma (2), (29) holds by Assumption 4, and (30) holds because $\mathbb{E}[Z_l^*|X_l^*]=1$ under Assumption 5. Next, note that:

$$\mathbb{E}\left[r(X_1)|X_1\right] = \mathbb{E}\left[R\Pi_{j=1}^{\overline{w}} Z_j^*|X_1^*\right] \tag{32}$$

$$= \mathbb{E}\left[RZ_1^*|X_1^*\right] \tag{33}$$

$$= \Pr(R=1|X_1^*), \tag{34}$$

where (32) follows from definitions, (33) holds by Lemma (2), and where (34) is obtained using that $\mathbb{E}[Z_l^*|X_l^*]=1$ under Assumption 5 and applying Assumption 4. Since $\mathbb{E}[\Pr(R=1|X_1^*=x)] = \Pr(R=1)$, the result holds.

It remains to show that $\mathbb{E}[Y(1-D^+)1_{\{Y>q(1-p(X_1^*),X_1^*)\}}Z_1/e_1^*(X_1^*)]/\mathbb{E}[r(X_1^*)]$ is a sharp upper bound for $\mathbb{E}[Y^E|R=1]$. I first demonstrate that $p(x) = \Pr(R=1|D^+=0, Z_1=1, X_1^*=x)$. Assumption 4 together with $D^+ = 1-R^+|Z_1=1$ implies that $r^+(x) = \Pr(R^+=1|X_1^*=x)$. Under Assumption 2, $\Pr(R=1|X_1^*=x) = \Pr(R=1, R^+=1|X_1^*=x)$. Applying the definition of conditional probability gives $p(x) = \Pr(R=1|R^+=1, X_1^*=x)$. Assumption 4 together with $D^+ = 1-R^+|Z_1=1$ gives $\Pr(R=1|D^+=0, Z_1=1, X_1^*=x) = \Pr(R=1|R^+=1, X_1^*=x)$, which implies the result.

The remainder of the proof is similar to Lee (2009). Let $\gamma_x = \mathbb{E}[Y|Z_1=1, D^+=0, Y \geq q(1-p(x), x), X_1^*=x]$. I next demonstrate that γ_x is a sharp upper bound for $\mathbb{E}[Y^E|X_1^*=x, R=1]$. Using that $p(x) = \Pr(R=1|D^+=0, Z_1=1, X_1^*=x)$, Corollary 4.1 in Horowitz & Manski (1995) gives $\gamma_x \geq \mathbb{E}[Y|Z_1=1, D^+=0, R=1, X_1^*=x]$. Using that $D^+=0|R=1$ and $Y=Y^E|Z_1=1$ and by Assumption 4, $\mathbb{E}[Y|Z_1=1, D^+=0, R=1, X_1^*=x] = \mathbb{E}[Y^E|X_1^*=x, R=1]$, meaning that γ_x is an upper bound for $\mathbb{E}[Y^E|X_1^*=x, R=1]$. Since $p(x)$ is identified and Y^E is observed only among those initially assigned treatment, Corollary 4.1 in Horowitz & Manski (1995) implies sharpness.

Let $f_{x|R=1}(x)$ be the p.d.f. of X_1^* conditional on $R=1$. Applying Bayes rule for densities to $\Pr(R=1|X_1^*=x)$ identified by $r(x)$ and p.d.f. of X_1^* identified directly identifies $f_{x|R=1}(x)$, making $\int_{\mathcal{X}_1^*} \gamma_x f_{x|R=1}(x)dx$ the sharp upper bound for $\mathbb{E}[Y^E|R=1]$.

The last step is to show that:

$$\int_{\mathcal{X}_1^*} \gamma_x f_{x|R=1}(x)dx = \mathbb{E}[Y(1-D^+)1_{\{Y>q(1-p(X_1^*),X_1^*)\}} Z_1/e_1^*(X_1^*)]/\mathbb{E}[r(X_1^*)]. \tag{35}$$

Using the law of iterated expectations and the definition of conditional probability:

$$\mathbb{E}[Y(1-D^+)1_{\{Y>q(1-p(X_1^*),X_1^*)\}} Z_1/e_1^*(X_1^*)] \tag{36}$$

$$= \mathbb{E}[\mathbb{E}[\gamma_{X_1^*}|X_1^*] \Pr(D^+=0, Z_1=1, Y>q(1-p(X_1^*), X_1^*)|X_1^*)/e_1^*(X_1^*)]. \tag{37}$$

Applying the definition of conditional probability twice and the definition of $p(X_1^*)$:

$$\Pr(D^+=0, Z_1=1, Y>q(1-p(X_1^*), X_1^*)) = r(X_1^*)e_1^*(X_1^*) \tag{38}$$

Thus:

$$\mathbb{E}[Y(1-D^+)1_{\{Y>q(1-p(X_1^*),X_1^*)\}} Z_1/e_1^*(X_1^*)] = \mathbb{E}[\gamma_{X_1^*} r(X_1^*)]. \tag{39}$$

Applying Bayes rule for densities: $\mathbb{E}[\gamma_{X_1^*} r(X_1^*)] = \int_{\mathcal{X}_1^*} \gamma_x f_{x|R=1}(x)dx \Pr(R=1)$. Since $\mathbb{E}[r(X_1^*)] = \Pr(R=1)$, the statement holds.

□

Proof of Corollary 1. Following reasoning analogous to Theorem 1 yields:

$$\mathbb{E}[Y(Z_1/e_1(X_1) - \Pi_{j=1}^{\overline{w}} Z_j^*)] = \tau_{ATR} \Pr(R=1) - \delta_{ANR} \Pr(R=0), \tag{40}$$

and $\mathbb{E}[r(X_1)] = \Pr(R=1)$. Since $\delta=0$ implies $\delta_{ANR}=0$, the statement holds. □

Proof of Theorem 2. I demonstrate the result for the upper bound, the result for the lower bound is symmetric. Following reasoning analogous to Theorem 1, yields:

$$\mathbb{E}[Y(Z_1/e_1(X_1) - \Pi_{j=1}^{\overline{w}} Z_j^*)] = \tau_{ATR} \Pr(R=1) - \delta_{ANR} \Pr(R=0), \tag{41}$$

$\mathbb{E}[r(X_1)] = \Pr(R=1)$, and $\mathbb{E}[1-r(X_1)] = \Pr(R=0)$. Theorem 1 further implies that $\mathbb{E}[m^U(G,\eta^0)]$ is a sharp upper bound on $\tau_{ATR} \Pr(R=1)$. Together, these results imply that $\mathbb{E}[m^U(G,\eta^0) - m^{RF}(G,\eta^0)]$ is a sharp upper bound on $\delta_{ANR} \Pr(R=0)$. Thus, the statement holds. □

Proof of Theorem 3. As in Theorem 1, $\mathbb{E}[Y(1-D^+)\Pi_{j=1}^{\overline{w}} Z_j^*] = \mathbb{E}[Y^N|R=1]\Pr(R=1)$, $p(x)r^+(x) = r(x)$, and $\mathbb{E}[r(X_1)] = \Pr(R=1)$.

I next show that the first term of $m^{RND}(G,\eta^0)$ identifies the relier average early motherhood outcome. By the law of iterated expectations and the definition of conditional expectation:

$$\mathbb{E}\left[Y(1-D^+)p(X_1^*)\frac{Z_1}{e_1^*(X_1^*)}\right] = \mathbb{E}\left[\frac{p(X_1^*)}{e_1^*(X_1^*)}\mathbb{E}\left[Y\middle|Z_1=1, D^+=0, X_1^*\right]\Pr(D^+=0, Z_1=1|X_1^*)\right]. \tag{42}$$

Assumption 4 together with $D^+ = 1 - R^+|Z_1 = 1$ gives $\Pr(D^+=0, Z_1=1|X_1^*) = r^+(X_1^*)e_1^*(X_1^*)$. Hence (42) equals $\mathbb{E}[\mu_{X_1^*} r(X_1^*)]$, where $\mu_x = \mathbb{E}[Y|Z_1=1, D^+=0, X_1^*=x]$. Using that $Y = Y^E|Z_1 = 1$ and $D^+ = 1 - R^+|Z_1 = 1$, and by Assumption 4, $\mu_x = \mathbb{E}[Y^E|R^+=1, X_1^*=x]$, which equals $\mathbb{E}[Y^E|R=1, X_1^*=x]$ by Assumption 6. Applying Bayes rule for densities, (42) equals $\mathbb{E}[Y^E|R=1]\Pr(R=1)$, and thus $\tau_{ATR} = \mathbb{E}[m^{RND}(G,\eta^0)]/\,\mathbb{E}[r(X_1)]$. Following reasoning analogous to Theorem 2 also yields $\delta_{ANR} = \mathbb{E}[m^{RND}(G,\eta^0) - m^{RF}(G,\eta^0)]/\,\mathbb{E}[1-r(X_1)]$.

The last step is to show that the two point-identified effects fall within the corresponding bounds. I demonstrate the argument for the upper bound on τ_{ATR}, the argument for the lower bound is symmetric. $m^{RND}(G,\eta^0)$ and $m^U(G,\eta^0)$ differ only in the first term. Theorem 1 establishes that the first term of $m^U(G,\eta^0)$ has expectation $\mathbb{E}[\gamma_{X_1^*} r(X_1^*)]$, where $\gamma_x = \mathbb{E}[Y|Z_1=1, D^+=0, Y \geq q(1-p(x),x), X_1^*=x]$, whereas the present proof establishes that the first term of $m^{RND}(G,\eta^0)$ has expectation $\mathbb{E}[\mu_{X_1^*} r(X_1^*)]$. By the definition of the truncated expectation, $\mu_x \leq \gamma_x$ for every x, and $r(x) \geq 0$. Hence, the point-identified effect cannot exceed the corresponding upper bound. The same argument applied symmetrically establishes the result for the lower bound, and for δ_{ANR} the same comparison applies since both moments are differenced with $m^{RF}(G,\eta^0)$ and scaled by $\mathbb{E}[1-r(X_1)]$. Thus, the statement follows. □

References

Abadie, A. (2003). Semiparametric instrumental variable estimation of treatment response models. *Journal of econometrics*, *113*(2), 231–263.

Adams, A., Jensen, M. F., & Petrongolo, B. (2024). Birth timing and spacing: Implications for parental leave dynamics and child penalties. *IZA Discussion Paper*.

Adda, J., Dustmann, C., & Stevens, K. (2017). The career costs of children. *Journal of Political Economy*, *125*(2), 293–337.

Agüero, J. M., & Marks, M. S. (2008). Motherhood and female labor force participation: evidence from infertility shocks. *American Economic Review*, *98*(2), 500–504.

Angelov, N., Johansson, P., & Lindahl, E. (2016). Parenthood and the gender gap in pay. *Journal of Labor Economics*, *34*(3), 545–579.

Angrist, J., & Evans, W. N. (1998). Children and their parents' labor supply: Evidence from exogenous variation in family size. *American Economic Review*, *88*(3), 450–477.

Angrist, J., Ferman, B., Gao, C., Hull, P., Tecchio, O. L., & Yeh, R. W. (2024). Instrumental variables with time-varying exposure: New estimates of revascularization effects on quality of life. *National Bureau of Economic Research*.

Angrist, J., & Imbens, G. (1995). Identification and estimation of local average treatment effects. *National Bureau of Economic Research*.

ASRM Practice Committee. (2024). Tobacco or marijuana use and infertility: A committee opinion. *Fertility and Sterility*, *121*(4), 589–603.

Athey, S., Tibshirani, J., & Wager, S. (2019). Generalized random forests. *The Annals of Statistics*, *47*(2), 1148–1178.

Bensnes, S., Huitfeldt, I., & Leuven, E. (2025). Reconciling estimates of the long-term earnings effect of fertility. *Unpublished working paper*.

Bertrand, M. (2020). Gender in the twenty-first century. *AEA Papers and Proceedings*, *110*, 1–24.

Bíró, A., Dieterle, S., & Steinhauer, A. (2019). Motherhood timing and the child penalty: Bounding the returns to delay. *Unpublished working paper*.

Black, S. E., Devereux, P. J., & Salvanes, K. G. (2005). The more the merrier? the effect of family size and birth order on children's education. *The Quarterly Journal of Economics*, *120*(2), 669–700.

Bögl, S., Moshfegh, J., Persson, P., & Polyakova, M. (2024). The economics of infertility: Evidence from reproductive medicine. *National Bureau of Economic Research*.

Bronars, S. G., & Grogger, J. (1994). The economic consequences of unwed motherhood: Using twin births as a natural experiment. *American Economic Review*, *84*(5), 1141–1156.

Bütikofer, A., Jensen, S., & Salvanes, K. G. (2018). The role of parenthood on the gender gap among top earners. *European Economic Review*, *109*, 103–123.

Cellini, S. R., Ferreira, F., & Rothstein, J. (2010). The value of school facility investments: Evidence from a dynamic regression discontinuity design. *The Quarterly Journal of Economics*, *125*(1), 215–261.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). *Double/debiased machine learning for treatment and structural parameters.* Oxford University Press Oxford, UK.

Chernozhukov, V., Chetverikov, D., & Kato, K. (2019). Inference on causal and structural parameters using many moment inequalities. *The Review of Economic Studies*, *86*(5), 1867–1900.

Chung, Y., Downs, B., Sandler, D. H., Sienkiewicz, R., et al. (2017). The parental gender earnings gap in the United States. *Unpublished working paper*.

Cortés, P., & Pan, J. (2023). Children and the remaining gender gaps in the labor market. *Journal of Economic Literature*, *61*(4), 1359–1409.

Cristia, J. P. (2008). The effect of a first child on female labor supply: Evidence from women seeking fertility services. *Journal of Human Resources*, *43*(3), 487–510.

Eichmeyer, S., & Kent, C. (2022). Parenthood in poverty. *Centre for Economic Policy Research*.

Ferman, B., & Tecchio, O. (2023). Identifying dynamic lates with a static instrument. *arXiv preprint arXiv:2305.18114*.

Fitzenberger, B., Sommerfeld, K., & Steffes, S. (2013). Causal effects on employment after first birth—a dynamic treatment approach. *Labour Economics*, *25*, 49–62.

Gallen, Y., Joensen, J. S., Johansen, E. R., & Veramendi, G. F. (2024). The labor market returns to delaying pregnancy. *Unpublished working paper*.

Goldin, C. (2014). A grand gender convergence: Its last chapter. *American Economic Review*, *104*(4), 1091–1119.

Guan, H., Tang, H., Pan, L., Song, H., & Tang, L. (2021). Pregnancy predictors in unexplained infertility after intrauterine insemination. *Journal of Gynecology Obstetrics and Human Reproduction*, *50*(8), 102071.

Han, S. (2021). Identification in nonparametric models for dynamic treatment effects. *Journal of Econometrics*, *225*(2), 132–147.

Heckman, J. J., Humphries, J. E., & Veramendi, G. (2016). Dynamic treatment effects. *Journal of econometrics*, *191*(2), 276–292.

Heiler, P. (2024). Heterogeneous treatment effect bounds under sample selection with an application to the effects of social media on political polarization. *Journal of Econometrics*, *244*(1), 105856.

Hernán, M. A., & Robins, J. M. (2020). *Causal inference: What if.* Boca Raton: Chapman & Hall/CRC.

Horowitz, J. L., & Manski, C. F. (1995). Identification and robustness with contaminated and corrupted data. *Econometrica*, *63*(2), 281–302.

Hotz, V. J., McElroy, S. W., & Sanders, S. G. (2005). Teenage childbearing and its life cycle consequences: Exploiting a natural experiment. *Journal of Human Resources*, *40*(3), 683–715.

Jakobsen, K. M., Jørgensen, T. H., & Low, H. (2022). Fertility and family labor supply.

Kan, O., Gorkem, U., Baser, E., & Alkilic, A. (2020). Impact of anxiety levels on difficulty of intrauterine insemination and treatment outcomes. *Journal of Obstetrics and Gynaecology Research*, *46*(10), 2059–2065.

Kim, H., Subramanian, V., Baird, F., Beebeejaun, Y., Sarris, I., Kamath, M. S., & Sunkara, S. K. (2023). Effect of female body mass index on intrauterine insemination outcomes: a systematic review and meta-analysis. *Human Fertility*, *26*(6), 1511–1518.

Kleven, H., Landais, C., & Leite-Mariante, G. (2024). The child penalty atlas. *The Review of Economic Studies*, rdae104.

Kleven, H., Landais, C., Posch, J., Steinhauer, A., & Zweimüller, J. (2019). Child penalties across countries: Evidence and explanations. *AEA Papers and Proceedings*, *109*, 122–126.

Kleven, H., Landais, C., & Søgaard, J. E. (2019). Children and gender inequality: Evidence from Denmark. *American Economic Journal: Applied Economics*, *11*(4), 181–209.

Lee, D. S. (2009). Training, wages, and sample selection: Estimating sharp bounds on treatment effects. *Review of Economic Studies*, *76*(3), 1071–1102.

Leridon, H. (2004). Can assisted reproduction technology compensate for the natural decline in fertility with age? a model assessment. *Human reproduction*, *19*(7), 1548–1553.

Lundborg, P., Plug, E., & Rasmussen, A. W. (2017). Can women have children and a career? IV evidence from IVF treatments. *American Economic Review*, *107*(6), 1611–37.

Lundborg, P., Plug, E., & Rasmussen, A. W. (2024). Is there really a child penalty in the long run? New evidence from IVF treatments. *IZA Discussion Paper*.

Manski, C. F. (1989). Anatomy of the selection problem. *Journal of Human Resources*, *24*(3), 343–360.

Manski, C. F. (1990). Nonparametric bounds on treatment effects. *American Economic Review*, *80*(2), 319–323.

Melentyeva, V., & Riedel, L. (2023). Child penalty estimation and mothers' age at first birth. *ECONtribute Discussion Paper*.

Miller, A. R. (2011). The effects of motherhood timing on career path. *Journal of Population Economics*, *24*, 1071–1100.

OECD. (2022). *Out-of-school-hours services.*

OECD. (2023a). *Enrolment in childcare and pre-school.*

OECD. (2023b). *OECD employment database.* Retrieved from https://stats.oecd.org/Index.aspx?DatasetCode=AVE_HRS

OECD. (2023c). *Parental leave system.*

OECD. (2023d). *Part-time employment rate (indicator).* Retrieved from https://www.oecd.org/en/data/indicators/part-time-employment-rate.html

Olivetti, C., Pan, J., & Petrongolo, B. (2024). The evolution of gender in the labor market. *Handbook of Labor Economics*, *5*, 619–677.

Olma, T. (2021). Nonparametric estimation of truncated conditional expectation functions. *arXiv preprint arXiv:2109.06150*.

Rosenzweig, M. R., & Wolpin, K. I. (1980). Life-cycle labor supply and fertility: Causal inferences from household models. *Journal of Political Economy*, *88*(2), 328–348.

Semenova, V. (2025). Generalized lee bounds. *Journal of Econometrics*, *251*, 106055.

Stoye, J. (2020). A simple, short, but never-empty confidence interval for partially identified parameters. *arXiv preprint arXiv:2010.10484*.

Syam, H. H., Madjid, T. H., Effendi, J. S., Djuwantono, T., Permadi, W., & Fatma, Z. H. (2017). Anxiety effect in the success rate of intrauterine insemination (iui) and in vitro fertilization (ivf). *Open Access Library Journal*, *4*(6), 1–10.

Van den Berg, G. J., & Vikström, J. (2022). Long-run effects of dynamically assigned treatments: A new methodology and an evaluation of training effects on earnings. *Econometrica*, *90*(3), 1337–1354.

Zhang, J. L., & Rubin, D. B. (2003). Estimation of causal effects via principal stratification when some outcomes are truncated by "death". *Journal of Educational and Behavioral Statistics*, *28*(4), 353–368.

Supplementary Appendix for "Parenthood Timing and Gender Inequality"

Appendix SA1 maps the cross-sectional framework of Section 2 into a dynamic one. Appendix SA2 reports bounds for the window-matched timing estimates. Appendix SA3 compares my estimates period by period with conventional methods and provides estimation details for each. Appendix SA4 bounds the effects for women who would remain partnered and not begin antidepressant use after procedure failure, and reports effects on antidepressant use and partnership stability directly. Appendix SA5 tests the monotonicity assumption and allows its direction to vary with covariates. Appendix SA6 and Appendix SA7 vary the cross-fitting folds and forest size and the number of procedures used in estimation, respectively. Appendix SA8 gives implementation details for the selection test of Section 3.6 and presents the full grids of test estimates. Appendix SA9 adjusts work hours for potential parental leave. Appendix SA10 and Appendix SA11 report bounds under controls for the clinical protocol and under the selection adjustment of Section 5.6, respectively. Appendix SA12 shows that the paths in Figure 15 are insensitive to the level anchor.

SA1 Dynamic Framework

This section maps the cross-sectional framework used throughout the paper into a dynamic one and explains why explicitly modeling the dynamics adds little to interpretation. It then uses this framework to establish identification when reliance is defined using per-period rather than endline fertility. A corresponding estimator is used to obtain the window-matched timing estimates before and after later birth in Section 5.3.

Time starts at the moment of a woman's first IUI and ticks up to period T. Without loss of generality, childless women can choose whether to undergo one IUI each period. Define $W_t \in \{1, \ldots, \overline{w}\}$, $R_t \in \{0,1\}$, and $R_t^+ \in \{0,1\}$ as the period-specific counterparts of W, R, and R^+, measured up to t. All three are irreversible. Taken together, W_t and R_t characterize the periods in which IUI and non-IUI conceptions occur, allowing arbitrary breaks between attempts and non-IUI conception. Let $Y_t'(k)$ denote the potential outcome in period t if motherhood starts in period k, and $Y_t'(\infty)$ if the woman remains childless.

Let $Z_j' \in \{0,1\}$ indicate whether a successful IUI occurs in period j, and define the motherhood indicator in period t as $D_t = \max(\{Z_j' : j \leq t\} \cup \{1 - R_t\})$. For notational convenience, define $A_t = t$, as if all women underwent IUI in every period up to t, and let X_t' be covariates in period t, implicitly including an indicator for undergoing IUI.

Assumption 7 (Dynamic Conditional Sequential Unconfoundedness).
$(Y_t'(d), R_{t'}^+, R_{t'}, W_{t'}) \perp\!\!\!\perp Z_j' \mid X_j'$ *for all* d, j, t, t' *and* $X_j' \in \mathcal{X}_j'$.

This assumption states that among women undergoing IUI in period j, success is independent of potential outcomes and latent types conditional on observed covariates; including the indicator for undergoing IUI ensures it holds trivially for women who do not undergo IUI this period. It differs from Assumption 4, which assumed independence among women sharing the same attempt number rather than the same period, but the two can be mapped by including both attempt number and time since the first IUI in the covariates.

Let $H = \min(\{j : R_j = 0\} \cup \{\infty\})$ denote the period in which a woman becomes a non-relier, with $H = \infty$ if she remains a relier through period T. Under Assumption 7 and adapted Assumptions 2 and 5, plugging period-t outcomes but endline fertility status into the moments of Theorem 2 yields bounds on $\mathbb{E}[Y_t'(H) - Y_t'(0) \mid H \leq T]$, the average timing effect for women who become non-reliers by the end of the sample period. This is the main timing estimand throughout the paper.

Alternatively, using period-t outcomes and period-t fertility status in the corresponding moments yields bounds on $\mathbb{E}[Y_t'(H) - Y_t'(0) \mid H \leq t]$, the average timing effect for women who have become non-reliers by period t. This compares women who have already become mothers under both scenarios, but at different points in time, and is used in Section 5.3 to construct the window-adjusted timing estimates.

Since the timing of the first IUI and the timing of conception without IUI are observed, the distribution of H is identified, and hence so is the average shift in timing up to period t, $\mathbb{E}[H \mid H \leq t]$. However, among women whose first IUI succeeds, it is not possible to determine how much each would have delayed conception had the IUI failed. Consequently, the bounds are only informative about the effect of an average timing shift induced by IUI failure, highlighting the limited additional usefulness of a fully dynamic framework for interpretation.

Finally, assuming no anticipation, $Y_t'(k) = Y_t'(\infty)$ for $t < k$, bounds on the parenthood effect follow from Theorem 1 for both women who remain reliers in period t, $\mathbb{E}[Y_t'(0) - Y_t'(\infty) \mid H > t]$, and women who remain reliers through the end of the data window, $\mathbb{E}[Y_t'(0) - Y_t'(\infty) \mid H > T]$. The mapping into the cross-sectional potential outcomes in period t is $Y_t'(0) = Y^E$, $Y_t'(\infty) = Y^N$, and $Y_t'(H) = Y^L$.

SA2 Window-Matched Timing Estimates

Section 5.3 reports point estimates for the window-matched timing specification, which uses parenthood-effect estimates in periods 0–2 and re-estimates timing effects from year 3 onward using parenthood status in each period rather than at the end of the window. Figure SA1 reports the corresponding bounds, which remain relatively close to the baseline estimates.

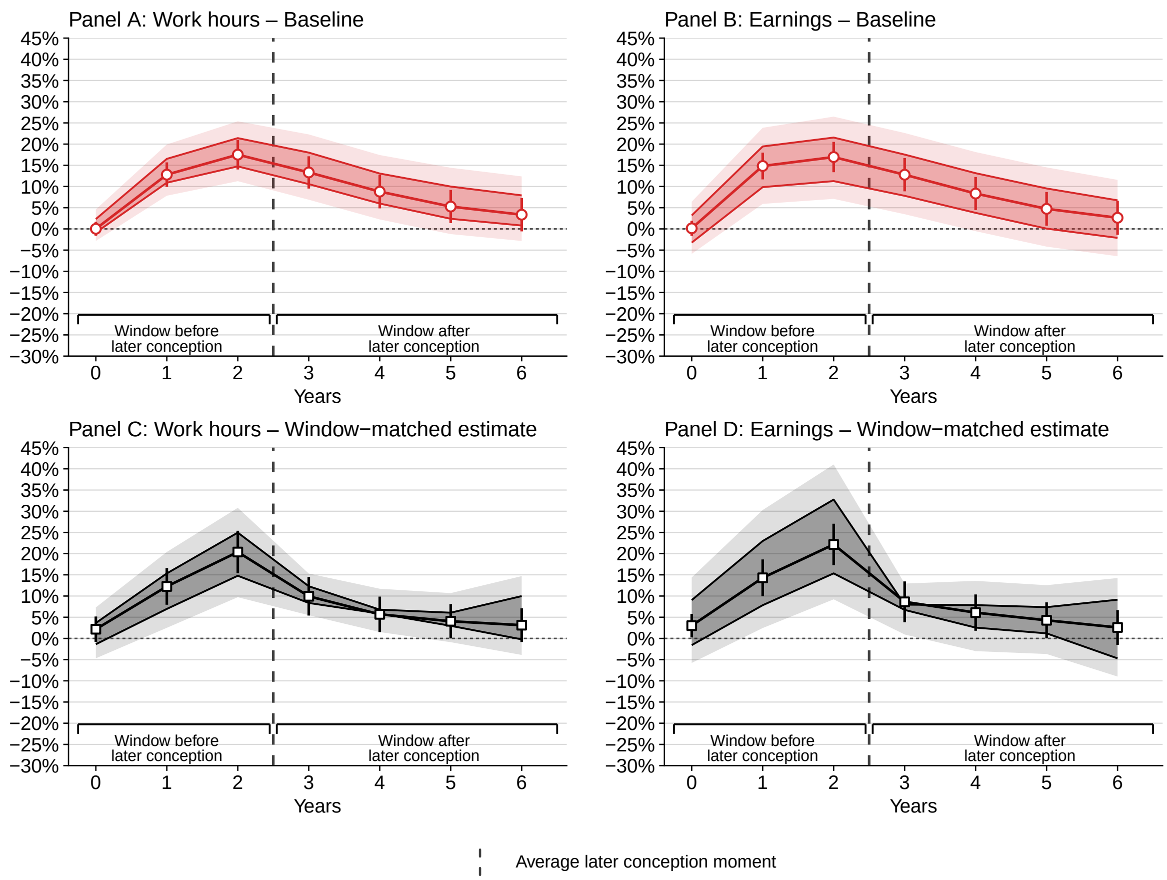

Figure SA1: Window-Matched Estimates of the Effect of Later Parenthood on Women's Outcomes

Note: Estimated effects of later parenthood on women's annual work hours and earnings, by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level under early parenthood. *Baseline* uses parenthood status at the end of the observation window; *window-matched* uses the negative of the parenthood effect for years 0–2 and re-estimates the timing effect using parenthood status in each period from year 3 onward. Darker shaded bands show the estimated identified sets for the partially identified parameters, with lighter shaded bands showing the corresponding 95% confidence intervals; markers show point estimates with their 95% confidence intervals.

SA3 Conventional Methods

The estimator comparisons use two reduced forms. The first, $\psi^{RF}(G,\xi^0)$, is the orthogonal moment for $m^{RF}(G,\eta^0)$ defined in Section 3.4 and compares first-IUI success with failure throughout the full procedure sequence. The second, $\psi^{RF,1}(G,\xi^0)$, uses only the first procedure and is the orthogonal moment for $m^{RF,1}(G,\eta^0) = YZ_1/e_1(X_1) - Y(1-Z_1)/(1-e_1(X_1))$. The *IV* rows use $\psi^{RF,1}$, whereas the *sequential IV* rows use ψ^{RF}. I denote the relevant reduced form in year t by $\widehat{\psi}_t^{RF}$ and the corresponding counterfactual level for compliers or reliers by $\widehat{C}_t$.

Each reduced form is paired with a corresponding first stage: $\pi^1(G,\xi^0)$ for the *IV* rows and $\pi(G,\xi^0)$ for the *sequential IV* rows. I denote the relevant first stage in year t by $\widehat{\pi}_t$. The *endline* estimator is $\widehat{\beta}^{end} = (\widehat{\psi}_T^{RF}/\widehat{\pi}_T)/\widehat{C}_T$, so the reduced form accumulated over the outcome window is scaled by the first stage in the final year. The *per-period* estimator instead applies the contemporaneous first stage in each year and then aggregates, $\widehat{\beta}^{pp} = \sum_t(\widehat{\psi}_t^{RF}/\widehat{\pi}_t)/\sum_t \widehat{C}_t$.

The recursive estimator adjusts for births following IUI failure, following Gallen et al. (2024) and Bensnes et al. (2025), under the restriction in equation (4). Let g_j denote the share of the control arm becoming mothers in year j. For $t = 1, \ldots, T$, the estimator recursively computes

$$\widehat{d}_t = \left(\widehat{\psi}_t^{RF} + \sum_{j=2}^{t} g_j \, \widehat{d}_{t-j+1} \right) / \widehat{\pi}_1, \tag{43}$$

$$\widehat{\mu}_t = \widehat{C}_t^{obs} - \sum_{j=1}^{t} g_j \, \widehat{d}_{t-j+1}. \tag{44}$$

and reports $\sum_t \widehat{d}_t / \sum_t \widehat{\mu}_t$. Here, $\widehat{C}_t^{obs}$ is the observed mean in the control arm, which includes women who have become mothers by year t, while $\widehat{\mu}_t$ estimates the never-treated counterfactual. I apply the recursive estimator to both reduced forms.

The event-study implementation follows Kleven, Landais, & Søgaard (2019). I define the event as the year of conception, measured as the year of first birth minus one for births occurring in the first nine months of the year, and estimate $Y_{it} = \sum_{k \neq -1} \alpha_k 1_{\{t=k\}} + \gamma_{c(it)} + \delta_{a(it)} + \varepsilon_{it}$ over the twelve years from five years before to six years after conception. The specification includes calendar-year fixed effects γ and age fixed effects δ, with the year before conception omitted, and standard errors clustered at the woman level. The counterfactual is the fitted value net of the event-time indicators, and the reported estimate is the sum of $\widehat{\alpha}_k$ for $k \geq 1$ divided by the counterfactual summed over the same years. I estimate this specification for the IUI sample restricted to conceptions between 2013 and 2016 and for the representative sample of couples conceiving a first child over the same period.

The Zhang & Rubin (2003) and Lee (2009) bounds use the same moment functions as my baseline bounds but discard all information after the first procedure and do not exploit monotonicity across the procedure sequence.

The last-procedure IV uses the last procedure observed up to and including the year in question.

Standard errors are analytic for the bounds and endline estimators. For the per-period and recursive estimators, I use 500 bootstrap draws, resampling the sample once in each draw and recomputing all components. For the event-study estimates, I use the standard error of the summed numerator while holding the denominator fixed.

Figure SA2 presents the per-period estimates. In most years, conventional IV and sequential IV estimates lie near the largest negative effects included in the bounds or are more negative. Recursive IV estimates lie closer to the smallest negative effects included in the bounds, while their sequential counterpart tracks my point estimates closely. While most estimates converge to the 10%–20% range for both work hours and earnings, the event-study estimates diverge, exceeding 30% in both the IUI and representative samples by year 6. They also display substantial pre-trends in both samples.

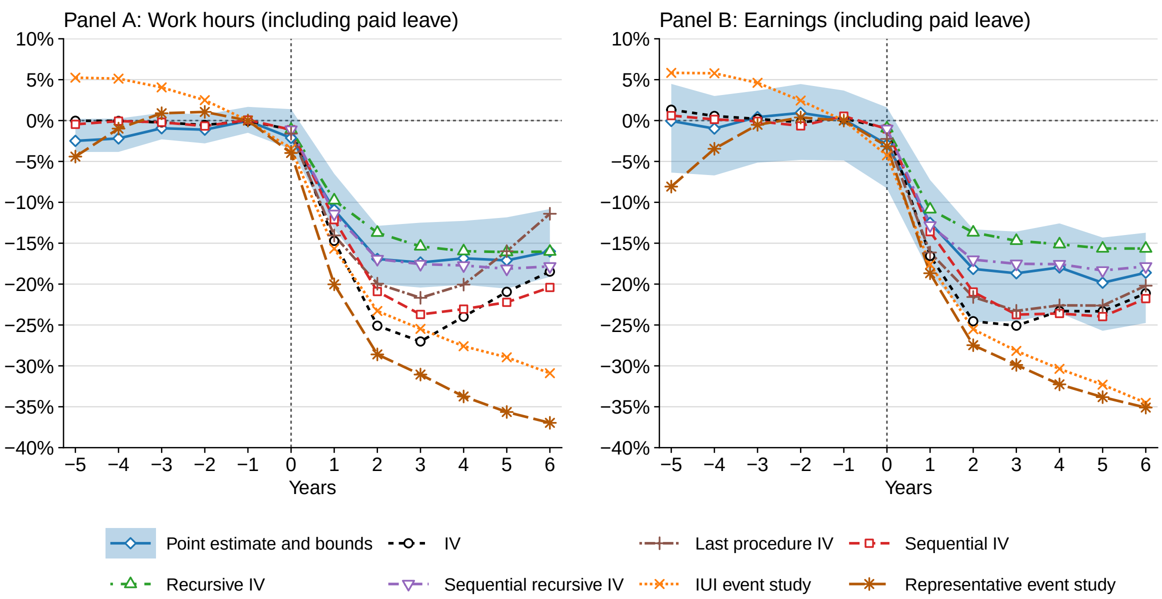

Figure SA2: Estimators of the Effect of Parenthood on Women's Hours and Earnings, by Year

Note: Estimates by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level in the absence of children. *IV* scales each period's reduced form by its contemporaneous first stage; *last procedure IV* defines the instrument as success at the last procedure a woman undergoes; *sequential IV* targets reliers rather than compliers, as described in Section 3.3.1; *recursive* variants additionally adjust for control units who subsequently become treated, following Gallen et al. (2024) and Bensnes et al. (2025). Event-study estimates follow Kleven, Landais, & Søgaard (2019) and are reported for the intrauterine-insemination sample and for a representative sample of women having a first child between 2013 and 2016.

SA4 Mental Health and Relationship Stability

Women who remain childless after trying to achieve pregnancy may experience mental health deterioration or relationship breakdowns, potentially affecting labor market outcomes. On the one hand, these consequences may be an integral part of the effect of not having children, making it important to understand the extent to which these mechanisms drive the labor market impacts. On the other hand, such issues may also stem not from childlessness itself but from fertility treatments or from the experience of trying and failing to conceive, which could limit the generalizability of the estimates to settings where women voluntarily remain childless.

Before assessing these factors, it is useful to clarify when the side effects of failing to conceive matter for generalizability. They are arguably less concerning for my estimates of timing effects: I find that women who experience delays perform better, and doing so despite potential side effects reinforces that delaying childbearing mitigates losses. By contrast, they are more important for my estimates of the effects of parenthood, where I find that large gender gaps persist even when couples fail to conceive. If these patterns reflect such side effects, the estimates may understate the role of parenthood beyond my sample.

To provide suggestive evidence, one might estimate the effect of failing to conceive, relative to successful conception, on antidepressant uptake or separation, which I do at the end of this section. However, this does not directly address the generalizability concern, since both conceiving and failing to conceive may affect mental health and relationship stability relative to not attempting conception. The contrast between failure and successful conception therefore does not reveal the effect of failure relative to not attempting.

To directly assess the role of mental health and relationship impacts, I modify the baseline approach to bound the labor-market effects of parenthood for women who, following procedure failure, would remain childless, remain partnered, and not initiate antidepressant use. If mental-health problems or partner separation worsen women's labor-market outcomes in the childless scenario, we would expect larger parenthood losses among women who do not experience these issues. Instead, a small change in the estimates would be consistent with these factors not being major drivers of the parenthood-effect estimates.

The formal procedure classifies women who meet condition (i) but not (ii) or (iii) as having conceived naturally, thereby removing them from the group used to construct the bounds. This group accounts for 10% of the sample and 15% of non-reliers. The identification argument amounts to adapting Theorem 1 by redefining R and D.

The adjustment also addresses a related potential monotonicity violation in the baseline analysis. There, women with a subsequent non-IUI birth after first-IUI success are treated as certain non-reliers, allowing them to be excluded before trimming the outcome distribution and thereby lowering the trimming share and narrowing the bounds. This can be incorrect if some would instead have remained childless after sequence failure because of mental-health problems or partner separation, in which case they are subsequent non-reliers but still reliers and should not be excluded. In the restricted analysis, however, the target population is limited to reliers who would not experience these issues after sequence failure. Thus, excluding women who would have a subsequent non-IUI birth after first-IUI success but would remain childless after failure because of these issues no longer removes women from the relevant target population.

Formally, the relaxed monotonicity assumption requires that, among women who would neither initiate antidepressant use nor separate from their partner had the sequence failed, those with a subsequent non-IUI birth after first-IUI success would also have had a non-IUI birth under sequence failure.

The adjustment procedure is conservative because it also excludes women who would have experienced such issues irrespective of parenthood, as well as those whose labor market outcomes would have remained unaffected even if the issues arose due to childlessness or procedure failure. This occurs because it is not possible to identify which women who conceived at the first IUI fall into these categories, so the trimming step must discard more observations from the tails, which in turn produces wider bounds. If such women are primarily the ones experiencing these issues, the bounds will widen without shifting systematically in either direction, much as if some childless women had been excluded at random.

Because the adjustment also addresses potential monotonicity violations, Appendix SA5 reports the resulting bounds alongside the monotonicity results. The point estimates change little, but the identified sets widen: severe mental health problems and separation do not drive the estimates, but

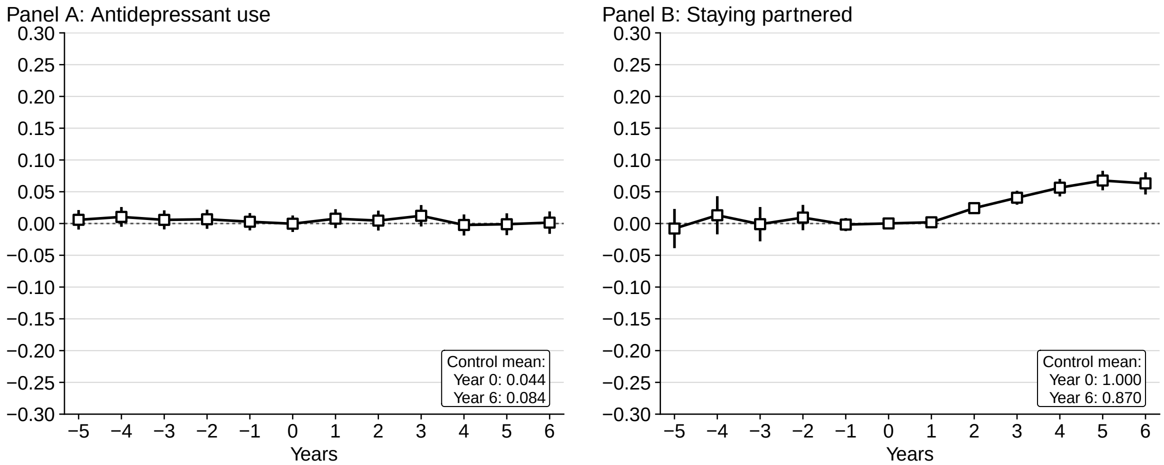

Figure SA3: Effects on Women's Antidepressant Uptake and Partnership Stability

Note: Estimates for the effect of conceiving at the first intrauterine insemination, relative to failure of the IUI sequence, on antidepressant uptake and remaining cohabiting with or registered to a partner, by year relative to the first procedure. Effects are expressed as percentages of the corresponding mean under the failure counterfactual. Markers show point estimates and their 95% confidence intervals; boxes report the corresponding failure-counterfactual means.

a conservative adjustment that holds these mechanisms fixed makes the bounds less informative.

Figure SA3 presents estimates of the effect of conceiving at the first IUI, relative to failure throughout the sequence, on antidepressant use and remaining partnered. Effects on antidepressant use are precisely estimated and indistinguishable from zero in every year. Effects on remaining partnered are zero before and immediately after the first procedure but become positive and significant from year 2, reaching 6.8 percentage points in year 5 against a control mean of 88.7%. Conceiving at the first IUI therefore has no detectable effect on antidepressant use but makes couples more likely to remain together.

SA5 Monotonicity

Monotonicity states that all reliers are subsequent reliers, implying that the subsequent relier share is at least as large as the relier share: $\Pr(R^+ = 1) \geq \Pr(R = 1)$. Monotonicity further implies that the subsequent relier share is at least as large as the relier share at each covariate value: $r^+(X_1^*) \geq r(X_1^*)$. Since the conditional shares are estimated nonparametrically, formally testing whether their differences allow rejecting monotonicity is not trivial, but comparing them offers some insight. Panel A in Figure SA4 plots the empirical distribution of the difference between estimated conditional subsequent relier and relier shares. For 34.8% of observations, the difference is below zero. While this would contradict monotonicity if observed in the true shares, such deviations can result from estimation error when the shares are close. Namely, when all subsequent reliers are reliers $\Pr(R^+ = R) = 1$, the difference should be below zero for 50% of observations. Consistent with this, the differences are generally small: the median is 0.034, and only 6.4% of observations fall below -0.1, suggesting no clear monotonicity violations.

To formally test monotonicity using covariates, I adapt the approach of Semenova (2025). I partition the sample into $J = 25$ discrete cells $\mathcal{X}_{1,j}^*$ based on quintiles of women's work hours and age in the year prior to their first IUI. Since these two covariates are highly predictive of the remaining covariates used in the analysis, including additional ones results in small cells (e.g., almost no women have extremely high work hours while having extremely low earnings). Monotonicity implies that the subsequent relier share is at least as large as the relier share in each cell, meaning that each value in the vector $\mu = (\mathbb{E}[r^+(X_1^*) - r(X_1^*) \mid X_1^* \in \mathcal{X}_{1,j}^*])_{j=1}^J$ must be non-negative. The null hypothesis is $-\mu \leq 0$, and the test statistic is $\max_{j \in J}^* \frac{-\widehat{\mu}_j}{\widehat{\sigma}_j}$. The critical value is the self-normalized critical value of Chernozhukov et al. (2019). σ_j are estimated using multiplier bootstrap with 100 draws and weights $w_i \sim \exp(1)$ to account for the uncertainty in the estimation of propensity scores (results remain unchanged when treating scores as fixed). The p-value for the test statistic is 0.93, so the differences are not statistically significant and monotonicity is not rejected. Using women's earnings instead of hours to partition the sample yields the same conclusion.

Panel B in Figure SA4 tests monotonicity among women who would neither initiate antidepressant use nor separate from their partner had the sequence failed, as discussed in Appendix SA4. The estimated difference between the subsequent relier and relier shares is below zero for only 7.0% of observations and below -0.1 for just 0.3%, with a median of 0.129 and a p-value of 0.999, providing stronger support for monotonicity in this restricted group.

To address violations of monotonicity in the Lee (2009) bounds, Semenova (2025) relaxes the assumption by allowing its direction to vary with X_1^*. To test the sensitivity of my results, I adopt an equivalent approach. If the reversal of the estimated relier and subsequent relier shares occurs because the true shares are close, this method and the baseline approach should yield similar results.

Define $\mathcal{X}_{help}^* = \{x : r^+(x) \geq r(x)\}$ and $\mathcal{X}_{hurt}^* = \mathcal{X}_1^* \backslash \mathcal{X}_{help}^*$. The relaxed monotonicity assumption is that $\forall x \in \mathcal{X}_{help}^*$ $R^+ \geq R$ $a.s.$, and $\forall x \in \mathcal{X}_{hurt}^*$ $R^+ < R$ $a.s.$. Table SA1 presents orthogonal moments for the case when $X_1^* \in \mathcal{X}_{hurt}^*$. The estimator for θ_b, for $b \in \{L, U\}$, is then given by:

$$\widehat{\theta}_b = \frac{\sum_i \left(\psi^b(G_i, \widehat{\zeta}_i) 1_{\{p(X_1^*) \leq 1\}} + \psi^{b-}(G_i, \widehat{\zeta}_i) 1_{\{p(X_1^*) > 1\}}\right)}{\sum_i \left(\psi^R(G_i, \widehat{\zeta}_i) 1_{\{p(X_1^*) \leq 1\}} + \psi^{R+}(G_i, \widehat{\zeta}_i) 1_{\{p(X_1^*) > 1\}}\right)}. \tag{45}$$

I implement it following the baseline approach. Since a weighted generalized quantile forests estimator is not available, I estimate q^0 using quantile regressions. Panels C and D of Figure SA4 report the resulting estimates under the label *direction free*.

Panels C and D report the labor-market effects of parenthood and its timing on women's work hours and earnings under three specifications: the baseline, the covariate-conditional relaxation just described, and a specification restricting the analysis to women who would neither initiate antidepressant use nor separate from their partner had the sequence failed, as discussed in Appendix SA4. The last is relevant only for the parenthood effects. The point estimates remain similar. Over years 1–6, the covariate-conditional relaxation yields estimates nearly identical to the baseline: -15.9% for the effect of parenthood on work hours and -17.7% on earnings, with corresponding timing effects of 10.5% and 10.0%. Under the restricted specification, the parenthood effect on work hours is -16.3%, compared with -15.9% at baseline, while the earnings effect remains -17.7%.

The adjustments do affect the width of the identified set. The covariate relaxation narrows

Table SA1: Moment Functions for Covariate-Conditional Monotonicity

	Moment functions
$\psi^{L-}(G,\zeta^0)$	$\frac{Z_1}{e_1^*(X_1^*)}(1-D^+)Y - \Pi_{j=1}^{\overline{w}} \frac{1-Z_j}{1-e_j^*(X_j^*)}(1-D^+)Y 1_{\{Y>q^0(1-1/p(X_1^*),X_1^*)\}}$ $-q^0(1-1/p(X_1^*),X_1^*)\Big[\frac{Z_1}{e_1^*(X_1^*)}(1-D^+-r^+(X_1^*))$ $-\Pi_{j=1}^{\overline{w}} \frac{1-Z_j}{1-e_j^*(X_j^*)} \frac{1}{p(X_1^*)}(1-D^+-r_1(X_1^*))$ $-\Pi_{j=1}^{\overline{w}} \frac{1-Z_j}{1-e_j^*(X_j^*)}(1-D^+)(1_{\{Y>q^0(1-1/p(X_1^*),X_1^*)\}}-1/p(X_1^*))\Big]$ $-\frac{Z_1-e_1^*(X_1^*)}{e_1^*(X_1^*)}\beta^+(X_1^*)r^+(X_1^*)$ $+\sum_{k=1}^{\overline{w}} 1_{\{A\geq k\}} \Pi_{j=1}^{k-1} \frac{1-Z_j}{1-e_j^*(X_j^*)} \frac{e_k^*(X_k^*)-Z_k}{1-e_k^*(X_k^*)}$ $\times\Big[\Big(r_k(X_k^*)r_k^L(X_k^*)z_k^{L-}(X_k^*)+\frac{q^0(1-1/p(X_1^*),X_1^*)}{p(X_1^*)}(r_1(X_1^*)-r_k(X_k^*))\Big)$ $+q^0(1-1/p(X_1^*),X_1^*)r_k(X_k^*)(1/p(X_1^*)-r_k^L(X_k^*))\Big]$
$\psi^{U-}(G,\zeta^0)$	$\frac{Z_1}{e_1^*(X_1^*)}(1-D^+)Y - \Pi_{j=1}^{\overline{w}} \frac{1-Z_j}{1-e_j^*(X_j^*)}(1-D^+)Y 1_{\{Y<q^0(1/p(X_1^*),X_1^*)\}}$ $-q^0(1/p(X_1^*),X_1^*)\Big[\frac{Z_1}{e_1^*(X_1^*)}(1-D^+-r^+(X_1^*))$ $-\Pi_{j=1}^{\overline{w}} \frac{1-Z_j}{1-e_j^*(X_j^*)} \frac{1}{p(X_1^*)}(1-D^+-r_1(X_1^*))$ $-\Pi_{j=1}^{\overline{w}} \frac{1-Z_j}{1-e_j^*(X_j^*)}(1-D^+)(1_{\{Y<q^0(1/p(X_1^*),X_1^*)\}}-1/p(X_1^*))\Big]$ $-\frac{Z_1-e_1^*(X_1^*)}{e_1^*(X_1^*)}\beta^+(X_1^*)r^+(X_1^*)$ $+\sum_{k=1}^{\overline{w}} 1_{\{A\geq k\}} \Pi_{j=1}^{k-1} \frac{1-Z_j}{1-e_j^*(X_j^*)} \frac{e_k^*(X_k^*)-Z_k}{1-e_k^*(X_k^*)}$ $\times\Big[\Big(r_k(X_k^*)r_k^U(X_k^*)z_k^{U-}(X_k^*)+\frac{q^0(1/p(X_1^*),X_1^*)}{p(X_1^*)}(r_1(X_1^*)-r_k(X_k^*))\Big)$ $+q^0(1/p(X_1^*),X_1^*)r_k(X_k^*)(1/p(X_1^*)-r_k^U(X_k^*))\Big]$
$\psi^{R+}(G,\zeta^0)$	$r^+(X_1^*)+(1-D^+-r^+(X_1^*))\frac{Z_1}{e_1^*(X_1^*)}$
	Nuisance functions
$\zeta^0(x_1^*,\ldots,x_{\overline{w}}^*)$	$\{e_1^*(x_1^*),\ldots,e_{\overline{w}}^*(x_{\overline{w}}^*),r_1(x_1^*),\ldots,r_{\overline{w}}(x_{\overline{w}}^*),r^+(x_1^*),q(p(x_1^*),x_1^*),q(1-p(x_1^*),x_1^*),$ $\beta_1(x_1^*),\ldots,\beta_{\overline{w}}(x_{\overline{w}}^*),\beta^+(x_1^*),z^{U+}(x_1^*),z^{L+}(x_1^*),z_1^{U-}(x_1^*),\ldots,z_{\overline{w}}^{U-}(x_{\overline{w}}^*),q^0(1/p(x_1^*),x_1^*),$ $q^0(1-1/p(x_1^*),x_1^*),z_1^{L-}(x_1^*),\ldots,z_{\overline{w}}^{L-}(x_{\overline{w}}^*),r_1^L(x_1^*),\ldots,r_{\overline{w}}^L(x_{\overline{w}}^*),r_1^U(x_1^*),\ldots,r_{\overline{w}}^U(x_{\overline{w}}^*)\}$
$q^0(u,x)$	$\inf\{q: u\leq \mathbb{E}[1_{\{Y\leq q\}}/\Pi_{j=2}^{\overline{w}}(1-e_j^*(X_j^*))\mid X_1^*=x,D=0]/$ $\mathbb{E}[\Pi_{j=2}^{\overline{w}}(1-e_j^*(X_j^*))\mid X_1^*=x,D=0]\}$
$z_k^{L-}(x)$	$\mathbb{E}[Y/(\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*)))\mid Y\geq q^0\left(1-1/p(X_1^*),X_1^*\right),D=0,X_k^*=x]$ $\mathbb{E}[\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*))\mid Y\geq q^0\left(1-1/p(X_1^*),X_1^*\right),D=0,X_k^*=x]$
$z_k^{U-}(x)$	$\mathbb{E}[Y/(\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*)))\mid Y\leq q^0\left(1/p(X_1^*),X_1^*\right),D=0,X_k^*=x]$ $\mathbb{E}[\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*))\mid Y\leq q^0\left(1/p(X_1^*),X_1^*\right),D=0,X_k^*=x]$
$r_k^L(x)$	$\mathbb{E}[1_{\{Y>q^0(1-1/p(X_1^*),X_1^*)\}}/(\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*)))\mid D=0,X_k^*=x]$ $\mathbb{E}[\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*))\mid D=0,X_k^*=x]$
$r_k^U(x)$	$\mathbb{E}[1_{\{Y<q^0(1/p(X_1^*),X_1^*)\}}/(\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*)))\mid D=0,X_k^*=x]$ $\mathbb{E}[\Pi_{j=k+1}^{\overline{w}}(1-e_j^*(X_j^*))\mid D=0,X_k^*=x]$

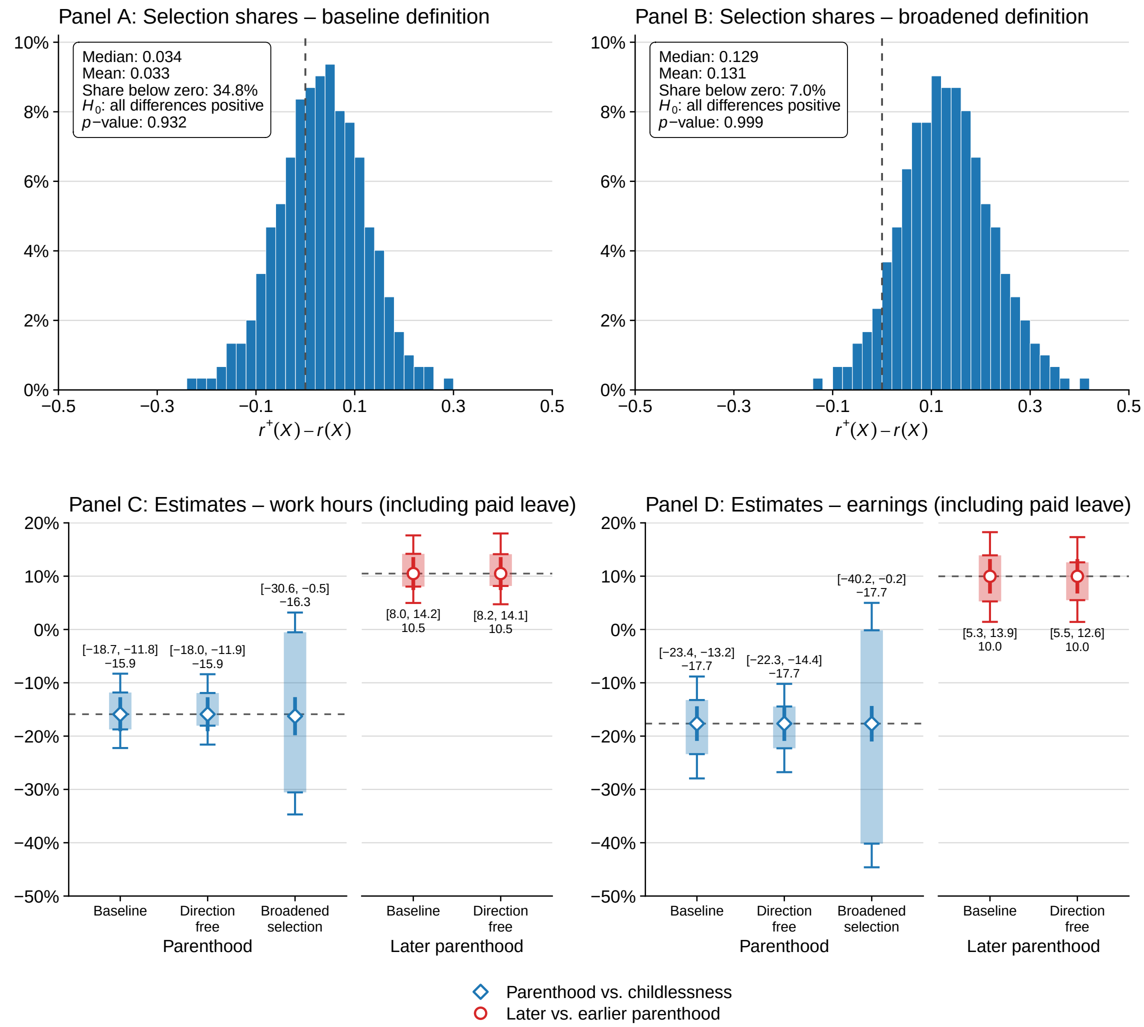

Figure SA4: Testing and Relaxing Monotonicity

Note: Panels A and B give the empirical distribution of the difference between the covariate-conditional subsequent relier and relier shares, under the baseline selection definition and under the definition described in Appendix SA4. Panels C and D give the effect of parenthood and of later parenthood on women's work hours and earnings, cumulated over years 1–6 and expressed as a percentage of the counterfactual level, under three specifications: the baseline, which imposes one direction of monotonicity; *direction free*, which lets the direction vary with covariates; and *broadened selection*, which applies the relier definition described in Appendix SA4. Translucent bars show the estimated identified sets for the partially identified parameters, with whiskers indicating the corresponding 95% confidence intervals; markers show point estimates with their 95% confidence intervals.

it slightly, by 0.8 percentage points for work hours and 2.3 points for earnings, as expected if the reversals in Panel A reflect true shares that are close rather than violations of monotonicity. Restricting attention to women who would neither initiate antidepressant use nor separate from their partner had the sequence failed instead widens the identified set substantially, from 6.9 to 30.0 percentage points for work hours and from 10.1 to 40.0 points for earnings. This widening is mechanical, as the restriction increases the share of the outcome distribution that must be trimmed. Notably, the bounds widen roughly symmetrically in both directions, as one would expect if these

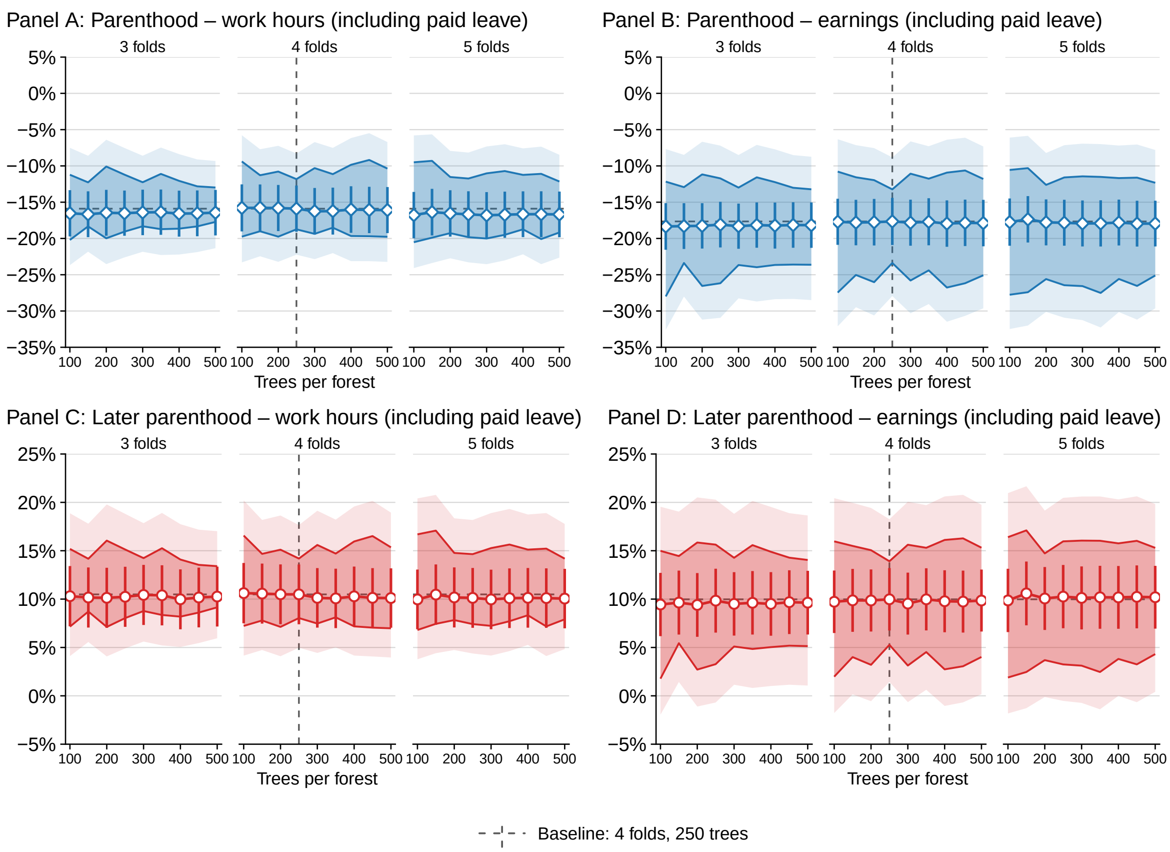

Figure SA5: Sensitivity of Estimates for Women's Outcomes to Cross-Fitting Folds and Forest Size

Note: Estimated effects on outcomes totaled over years 1–6 relative to the first intrauterine insemination, expressed as percentages of the corresponding totaled counterfactual levels. Within each panel, blocks vary the number of cross-fitting folds, while estimates within each block vary the number of trees in the Generalized Random Forest. Dashed lines indicate the baseline specification and estimate. Darker shaded bands show the estimated identified sets for the partially identified parameters, while lighter shaded bands show the corresponding 95% confidence intervals; markers show point estimates with their 95% confidence intervals.

shocks were unrelated to labor-market outcomes.

Overall, monotonicity is not rejected, allowing its direction to vary with covariates leaves the estimates essentially unchanged, and restricting the analysis to women who would not experience mental-health problems or relationship instability after procedure failure has little effect on the point estimates but does widen the bounds.

SA6 Cross-Fitting Folds and Forest Size

The baseline uses 4-fold cross-fitting and Generalized Random Forests with 250 trees. Figure SA5 assesses sensitivity to these choices by re-estimating the effects over the first six years following the first IUI using 3, 4, and 5 folds and tree counts ranging from 100 to 500. Both the point estimates and the bounds vary little across the grid.

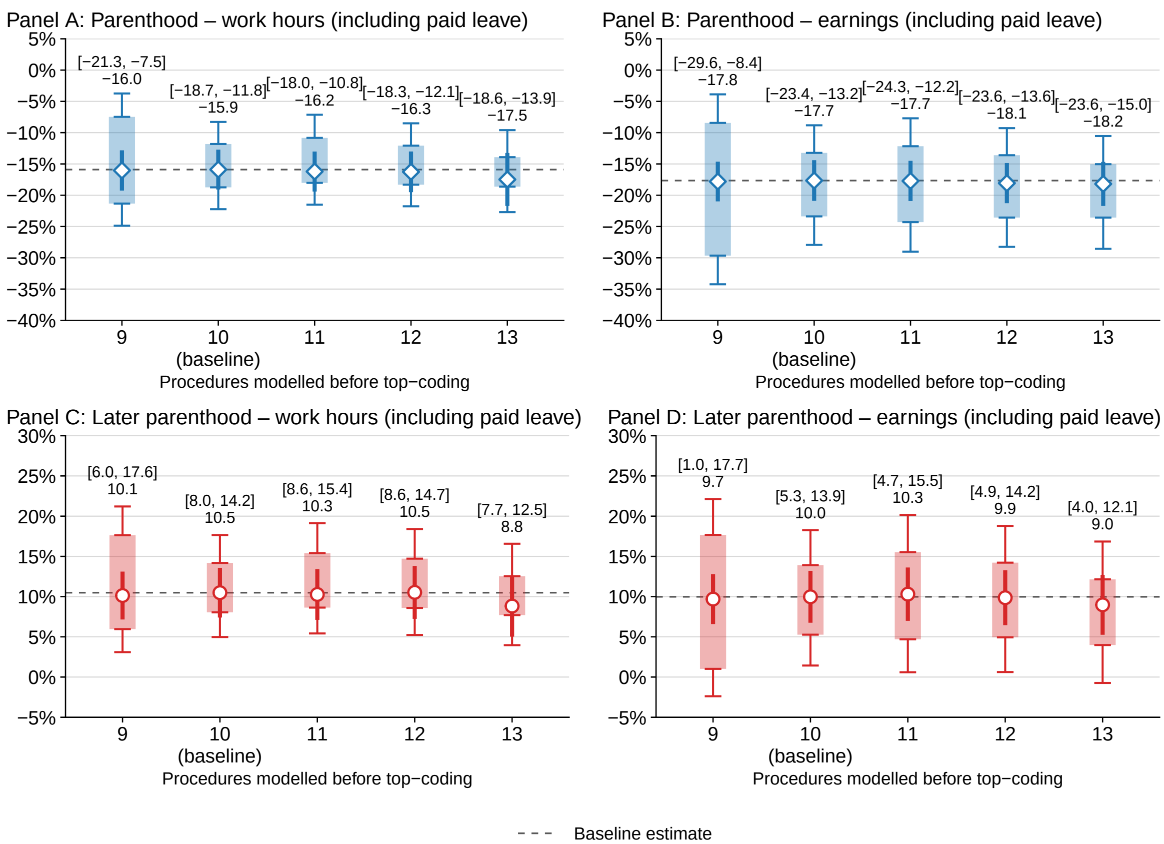

Figure SA6: Sensitivity of Effects on Women's Outcomes to the Number of Procedures Used

Note: Estimated effects on outcomes totaled over years 1–6 relative to the first intrauterine insemination, expressed as percentages of the corresponding totaled counterfactual levels. Each column within a panel varies the maximum number of procedures used for estimation; procedures beyond that number, and births following them, are treated as occurring outside the procedure sequence. Translucent bars show the estimated identified sets for the partially identified parameters, with whiskers indicating the corresponding 95% confidence intervals; markers show point estimates with their 95% confidence intervals.

SA7 Number of Procedures

The baseline estimates exploit variation across the first ten procedures women undergo. Figure SA6 re-estimates the effects capping the sequence at between nine and thirteen procedures. The estimates change little overall. Using fewer procedures widens the bounds, while using more than ten adds little because few women reach later procedures.

SA8 Selection Test

Let X_j^P denote the covariates used by the baseline specification to predict success at procedure j: both partners' age and education and the procedure type. Let $p_j(X_j^P)$ denote the fitted success probability. Let X^B denote the balance-test covariates: employment, earnings, and hours for the woman and her partner, measured in the year before the first procedure. Let H_j indicate an unobserved binary type affecting success propensity. The outcome Y is cumulative earnings over the full observation window beginning with the year of the first IUI.

I use two residualized outcomes, both defined among women whose procedure succeeds. First,

$\tilde{Y}_j^P$ is Y residualized with respect to the baseline propensity score $p_j(X_j^P)$ and determines type assignment. Selection is therefore introduced conditional on the baseline success model. Second, $\tilde{Y}_j^{PB}$ is Y residualized with respect to both X_j^P and X^B and is used to measure potential-outcome imbalances. The exercise thus asks whether selection beyond that captured by $p_j(X_j^P)$ can reproduce balance in X^B and, if so, how much imbalance in Y remains after both X_j^P and X^B are accounted for. This mirrors the observable balance test in Section 4.6 and the conditioning used in the main estimation, respectively.

A selection model is indexed by the log odds ratio δ between the two unobserved types and the share q of the low type among women whose procedure succeeds. Type membership shifts the log odds of success by $+\delta q$ for the high type and $-\delta(1-q)$ for the low type. I add a common constant to the log odds so that the implied weights reproduce the observed number of failures. This pins down the model-implied propensities $p_j(X_j^P, H_j)$. Among women whose procedure succeeds, I assign the share q with the lowest values of $\tilde{Y}_j^P$ to the low type and the remainder to the high type.

Under a candidate model, inverse-probability weighting of successful procedures must reproduce the covariate distribution among all women who reach procedure j. Hence, $\mathbb{E}[X^B(1 - Z_j/p_j(X_j^P, H_j))] = 0$. I test these six moment conditions jointly. The pooled specification clusters the covariance matrix by woman.

The implied outcome imbalance is recovered from women whose procedure succeeds. Weighting these women by $(1 - p_j(X_j^P, H_j))/p_j(X_j^P, H_j)$, normalized to mean one, gives the model-implied distribution of the potential outcome among women whose procedure fails. The difference between the unweighted and weighted means of $\tilde{Y}_j^{PB}$ is therefore the implied success–failure gap. The overall sample mean weights the success and failure means by $\mathbb{E}[p_j(X_j^P)]$ and $1 - \mathbb{E}[p_j(X_j^P)]$.

The pooled specification assigns types within each procedure, then pools all procedures to test balance in observables and quantify the implied imbalance in potential outcomes.

Because Y is measured over the full window beginning with the year of the first IUI, outcomes for women who succeed at later procedures may include some time before motherhood. Most procedures occur only a few months apart, however, so these women spend nearly the entire window as mothers. The outcome window is also defined identically for all women.

The same exercise could also be applied to the later-motherhood and childless potential outcomes, relevant for non-reliers and reliers, respectively. Extending the exercise to these outcomes would require varying success propensities jointly over the full procedure sequence. For tractability, I therefore restrict the analysis to potential outcomes observed following successful procedures.

Figure SA7 presents the full grids of test estimates corresponding to Figure 3, separately for each procedure among women who reach it and jointly across all procedures.

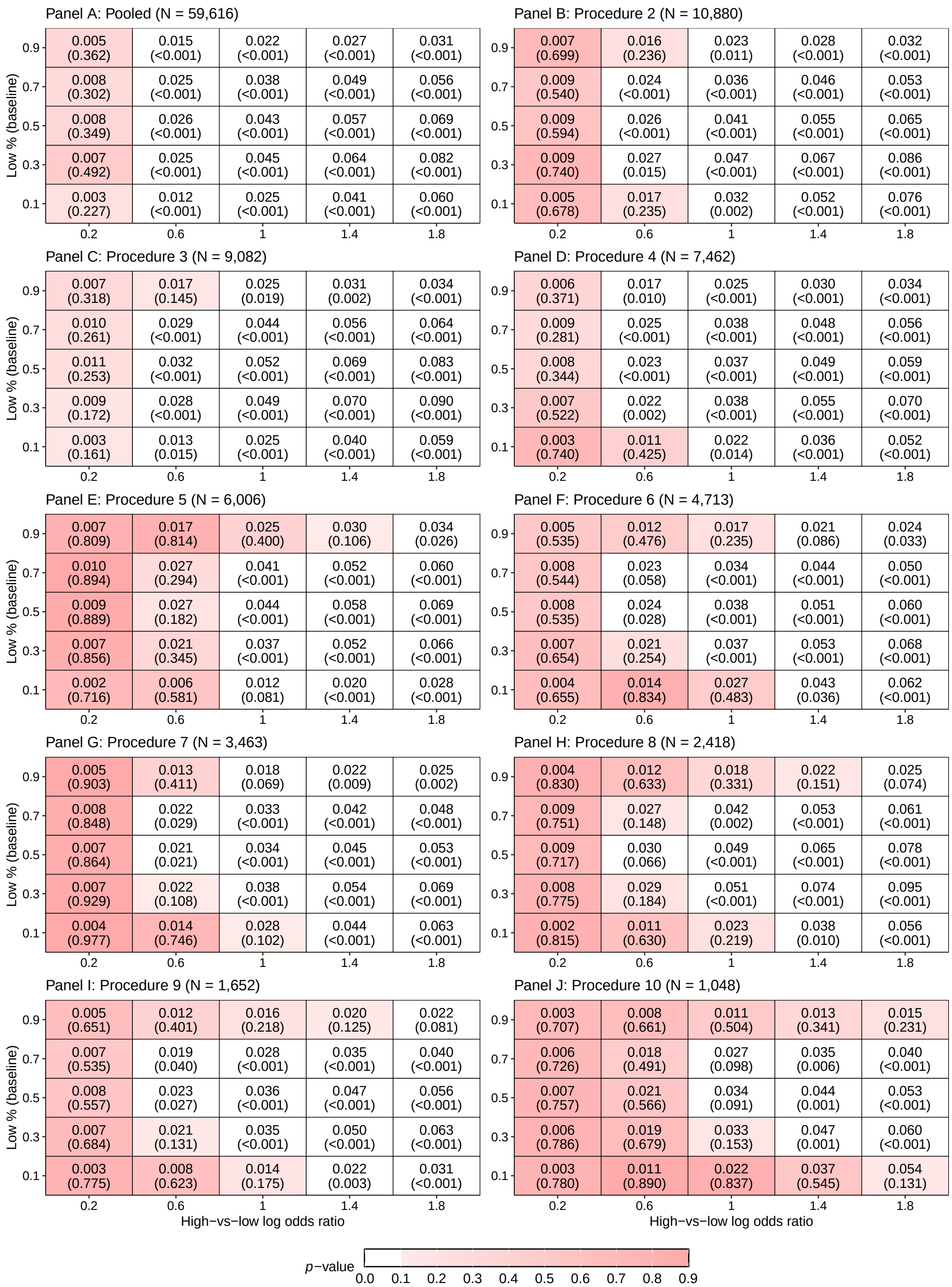

Figure SA7: Selection on Unobservables in Potential Earnings Over the Procedure Sequence

Note: Cells and axes are defined as in Figure 3, Panel A which reports estimates for the first procedure; Section 3.6 describes the test. Panel A pools all ten procedures; Panels B–J report procedures 2 through 10 separately. In each panel the top number is the implied difference in potential outcomes between women whose procedure succeeds and the sample mean at that procedure, relative to that mean. The outcome used is cumulative earnings over the first six years following the first IUI. The number in parentheses is the p-value from a joint test of whether the model is consistent with observed mean pre-IUI employment, earnings, and hours for the woman and her partner. The pooled p-value accounts for within-woman correlation across attempts.

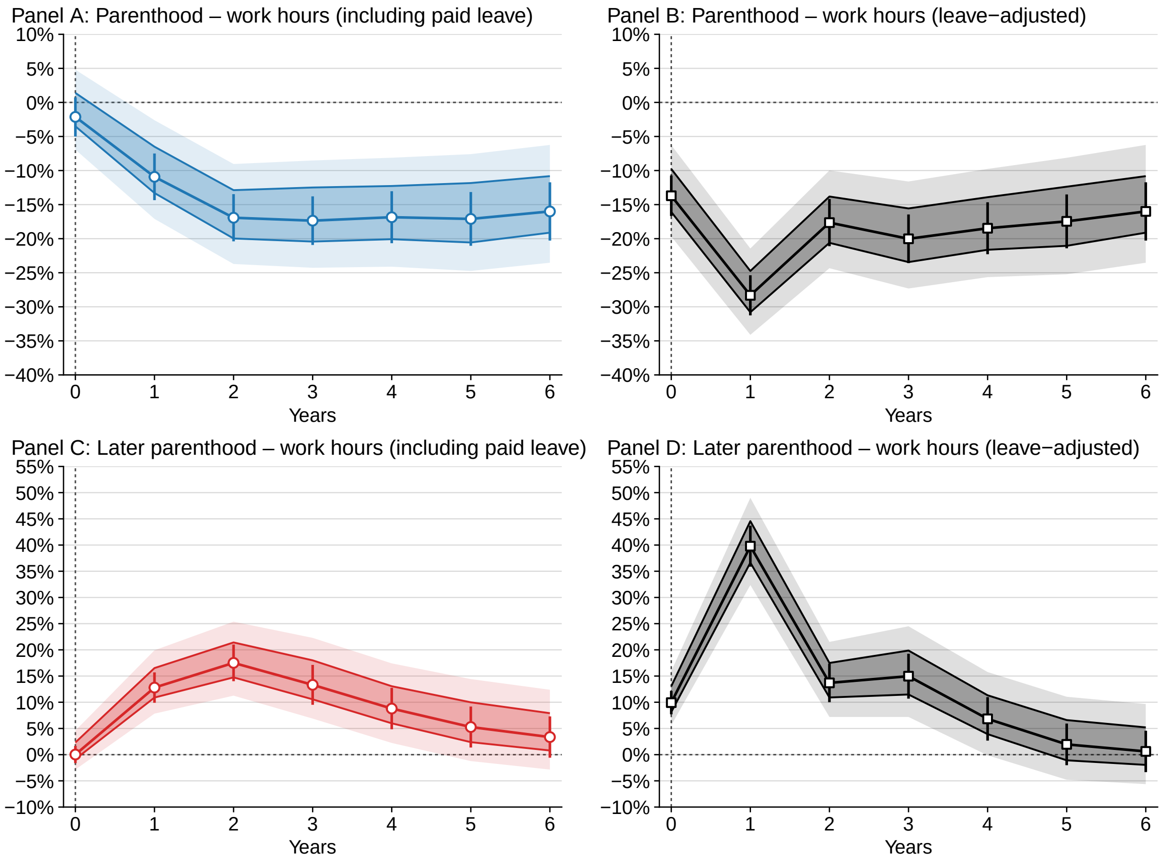

Figure SA8: Women's Work Hours Adjusted for Unobserved Parental Leave

Note: Estimates by year relative to the first intrauterine insemination, expressed as a percentage of the counterfactual level. *Reported hours* are as recorded in the data and include paid leave; *leave adjusted* scales reported hours in each childbirth year by 36/52, accounting for up to sixteen weeks of leave. Darker shaded bands show the estimated identified sets for the partially identified parameters, with lighter shaded bands showing the corresponding 95% confidence intervals; markers show point estimates with their 95% confidence intervals.

SA9 Unobserved Leave

I define maximum-leave-adjusted hours by scaling women's reported work hours in each childbirth year by 36/52, accounting for up to 16 weeks of leave. Figure SA8 shows results for female work hours. The estimates change little beyond the first year of parenthood.

SA10 Clinical Protocol Controls

Section 5.6 reports per-period point estimates under three progressively richer sets of controls for the clinical protocol. Figure SA9 reports the corresponding bounds on the window totals. The bounds vary somewhat in width, but the estimates change little overall.

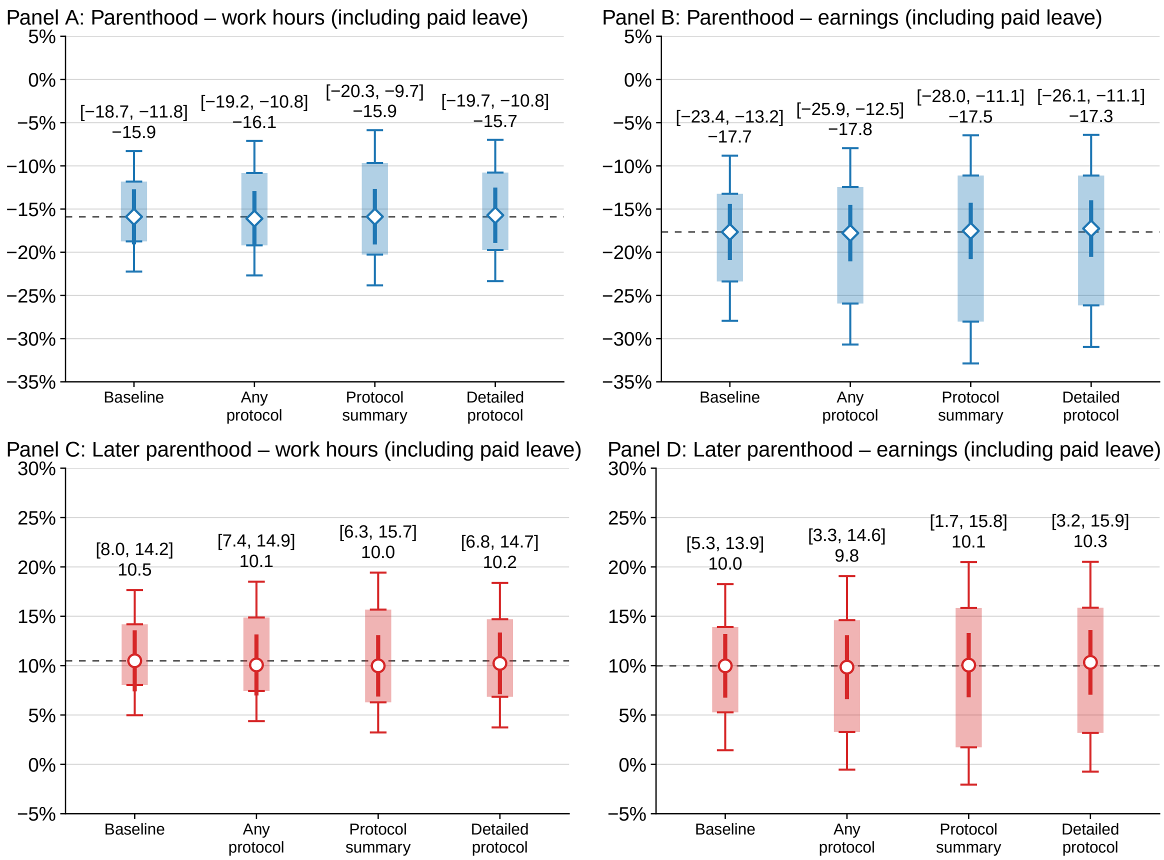

Figure SA9: Bounds for Women's Outcomes under Controls for the Clinical Protocol

Note: Estimated effects on outcomes totaled over years 1–6 relative to the first intrauterine insemination, expressed as percentages of the corresponding totaled counterfactual levels. *Baseline* is the specification used throughout the paper, which predicts procedure success using both partners' age and education and procedure type. Relative to the baseline, *Any protocol* adds an indicator for whether any of six procedure-level care activities is recorded within its relevant window: semen analysis, sperm preparation, ovulation ultrasound, hormonal monitoring, and progesterone administration before or after insemination. *Protocol summary* adds three indicators grouping these activities into male-factor testing and preparation, cycle monitoring, and progesterone; *Detailed protocol* adds all six separately. Translucent bars show the estimated identified sets for the partially identified parameters, with whiskers indicating the corresponding 95% confidence intervals; markers show point estimates with their 95% confidence intervals.

SA11 Selection Adjustment

Section 5.6 reports per-period point estimates under the selection adjustment. Figure SA10 reports the corresponding bounds on the window totals. The adjustment only shifts the identified sets and, in every case, toward larger effects.

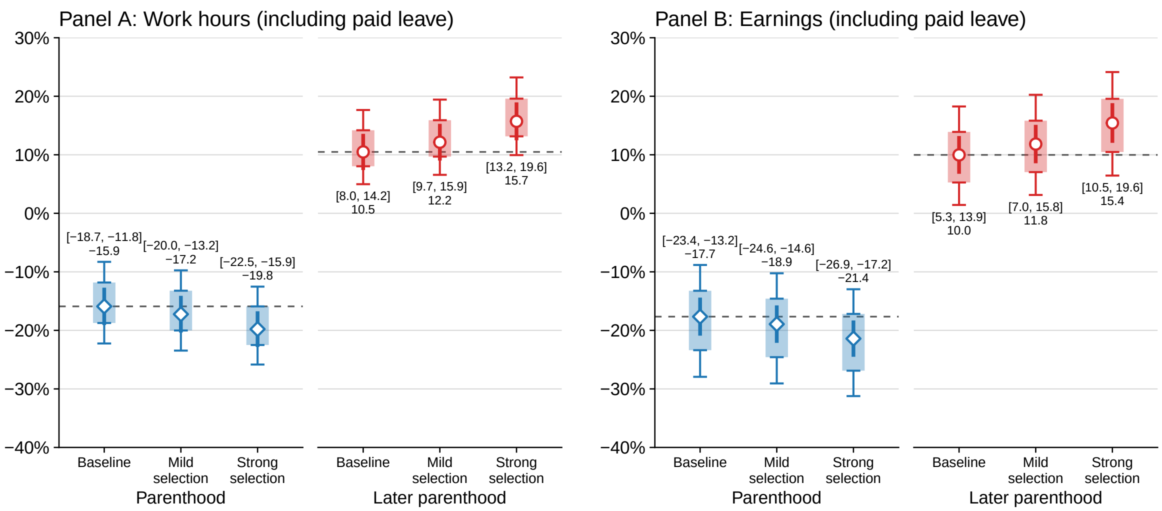

Figure SA10: Bounds for Women's Outcomes under the Selection Adjustment

Note: Estimated effects on outcomes totaled over years 1–6 relative to the first intrauterine insemination, expressed as percentages of the corresponding totaled counterfactual levels. Prior to estimation, outcomes are rescaled according to each woman's number of procedures. Under *strong selection*, outcomes are scaled by 1.027^{-1} for women who conceive at the first procedure and by 0.997^{-A} for women with no IUI success; under *mild selection*, the corresponding factors are 1.009^{-A} and 0.999^{-A}, where A denotes the number of procedures. The mild adjustment uses the largest imbalance not rejected by the pooled balance test in Section 3.6; the strong adjustment uses the largest imbalance not rejected at the first procedure. Translucent bars show the estimated identified sets for the partially identified parameters, with whiskers indicating the corresponding 95% confidence intervals; markers show point estimates with their 95% confidence intervals.

SA12 Anchoring the Estimated Paths

The three paths in Figure 15 are constructed by anchoring on the estimated childless relier level path and scaling by the estimated effects in percentage terms. Figure SA11 uses three alternative anchors: the childless relier path, the full-sample early-motherhood path, and the point-estimated non-relier early-motherhood path. The results are virtually unchanged.

Using effects estimated in levels rather than percentage terms also leaves the comparison nearly unchanged. The later mother's absolute loss is 50.0 percent smaller for hours and 46.0 percent smaller for earnings over years one to six, and 23.9 and 13.6 percent smaller over each woman's first four years of motherhood.

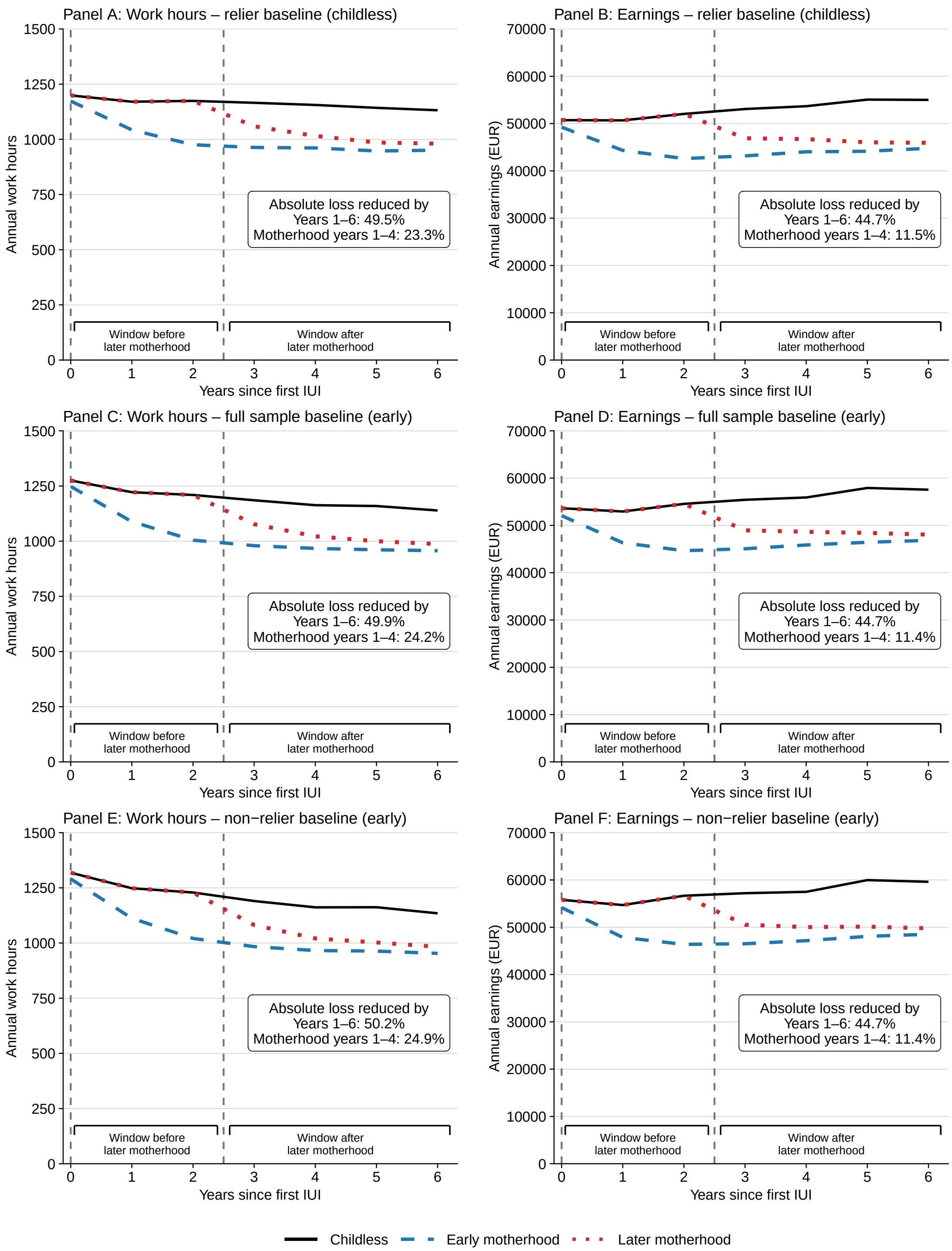

Figure SA11: Estimated Paths for Women under Alternative Level Anchors

Note: Work hours and earnings under alternative motherhood scenarios, by year relative to the first intrauterine insemination. Different panels use different paths as the level anchor. Panels A and B use the relier childless path, Panels C and D the full-sample first-procedure-success path, and Panels E and F the non-relier first-procedure-success path. Depending on the anchor, the childless or early-motherhood path is obtained by scaling the anchor by the ratio of reliers' mean outcomes under early motherhood and childlessness, or its inverse. From year 3 onward, the later-motherhood path additionally scales the early-motherhood path by the corresponding ratio of non-reliers' mean outcomes under later and early motherhood from the window-matched specification. The boxes report the implied mitigation from postponement.